Experiment Evidence

Screenshots of the test runs, with the method used for each

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About this document

This document collects the screenshots of the runs behind the project report. The projects are a quadratic equation solver and a date converter, tested with combinatorial, category-partition and metamorphic testing, then measured with JaCoCo (coverage), PIT (mutation score) and execution time. Each section states the method (the commands that were run) and shows the screenshots in the order they were taken.

The numbers in the captions come from the experiment log in the repository (results\log.md). Every screenshot shows the raw tool output. The screenshots are from my own runs.

Contents

Part A. Quadratic equation solver (Figures 1 to 47)
Part B. Date converter (Figures 48 to 89)
Part C. Execution time (Figures 90 to 91)

Commands are PowerShell, run from the repository root.

Part A. Quadratic equation solver

A.1 Baseline: one smoke test

Method. A single smoke test (quadratic.SmokeTest) gives the starting point. JaCoCo writes its report during mvn test; PIT is started separately.

```
mvn clean test "-Dtest=quadratic.SmokeTest"
start target\site\jacoco\index.html
mvn org.pitest:pitest-maven:mutationCoverage "-DtargetTests=quadratic.SmokeTest"
```

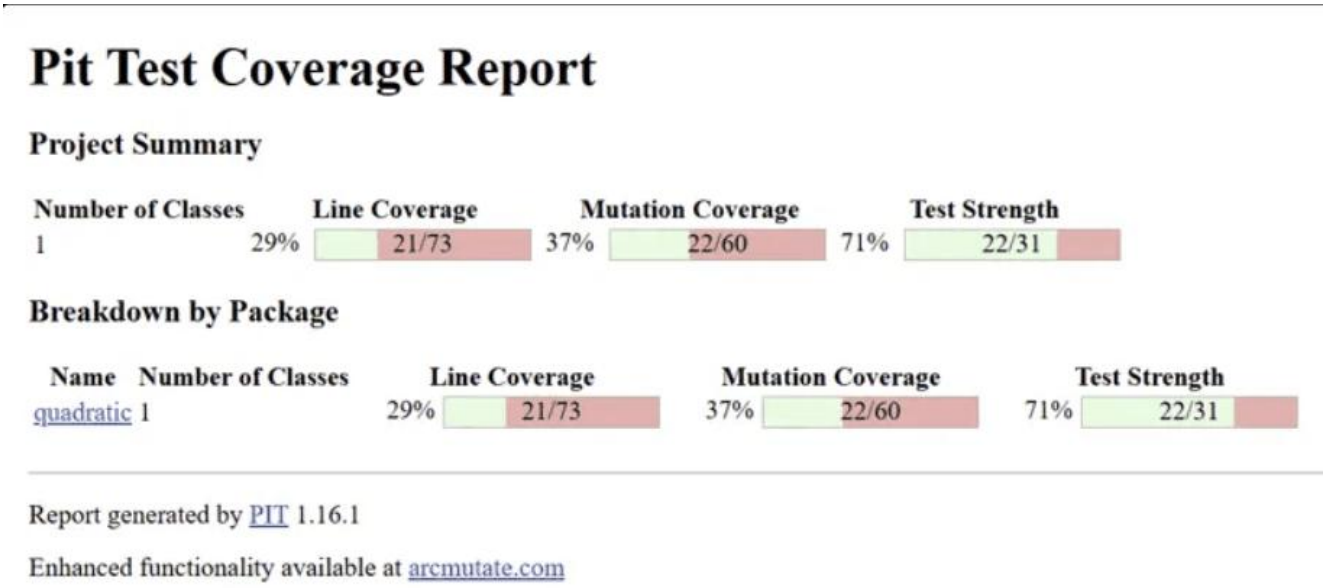


Figure 1. *PIT report, quadratic baseline: 29% line coverage, 37% mutation score (22 of 60 mutants killed).*



Figure 2. *JaCoCo project overview, baseline: 33% instruction coverage, 23% branch coverage.*

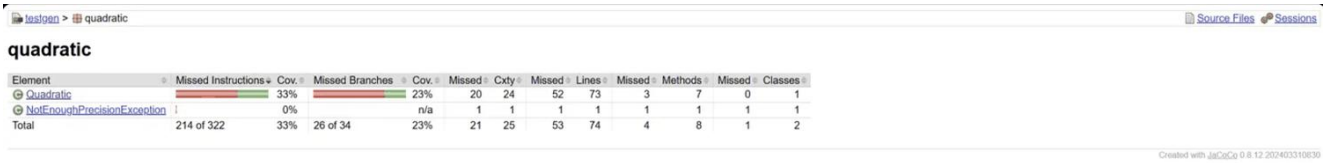


Figure 3. *JaCoCo package page: the classes Quadratic and NotEnoughPrecisionException.*

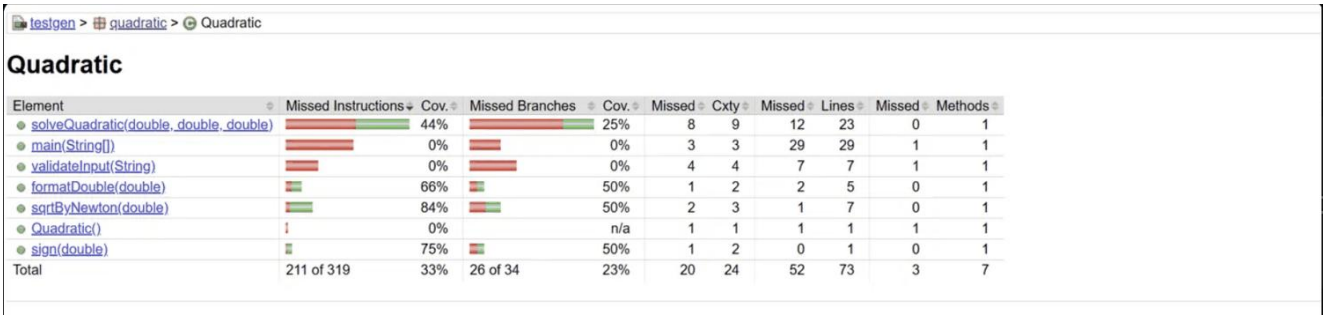


Figure 4. *JaCoCo class page for Quadratic, baseline: 23% branch coverage; main and validateInput are not covered.*

A.2 Combinatorial testing (PICT, pairwise and 3-way)

Method. Inputs were generated with Microsoft PICT from testdata\quadratic.pict (seed 42): 42 pairwise and 200 3-way tests. CombinatorialTest runs them in normal mode (known defects are recorded in results\findings_*.txt and the build stays green) and in strict mode (known defects fail the test). The oracle is a root-residual check and the exact discriminant sign. JaCoCo and PIT measure coverage and mutation score.

```
D:\pict\pict.exe testdata\quadratic.pict /o:2 /r:42 | Out-File -Encoding ascii
testdata\quad_pairwise.tsv
D:\pict\pict.exe testdata\quadratic.pict /o:3 /r:42 | Out-File -Encoding ascii testdata\quad_3way.tsv
mvn clean test "-Dtest=quadratic.CombinatorialTest"
mvn clean test "-Dtest=quadratic.CombinatorialTest" "-Dstrict=true"
start target\site\jacoco\index.html
mvn org.pitest:pitest-maven:mutationCoverage "-DtargetTests=quadratic.CombinatorialTest"
java -cp target\classes quadratic.Quadratic
```

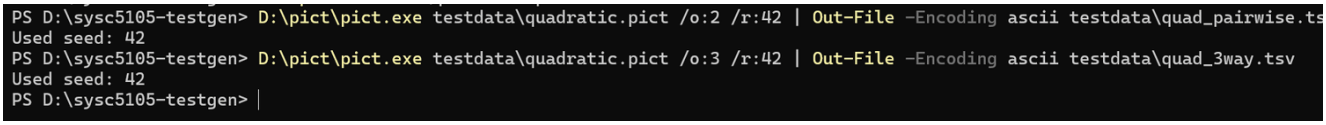


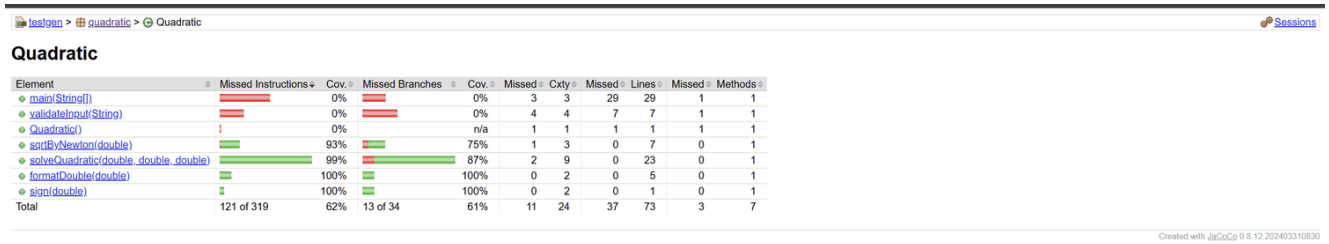
Figure 5. *PICT generation of the pairwise (/o:2) and 3-way (/o:3) suites, fixed seed 42.*

```
PS D:\sysc5105-testgen> mvn clean test "-Dtest=quadratic.CombinatorialTest"
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< sysc5105:testgen >-------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24009 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ testgen ---
[INFO] Deleting D:\sysc5105-testgen\target
[INFO]
[INFO] --- jacoco:0.8.12:prepare-agent (default) @ testgen ---
[INFO] argLine set to -javaagent:C:\\Users\\anuja\\.m2\\repository\\org\\jacoco\\org.jacoco.agent\\0.8.12\\org.jacoco.agent-0.8.12-runtime.jar=destfile=D:\\sysc5105-testgen\\targe
t\\jacoco.exec
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\main\resources
[INFO]
[INFO] --- compiler:3.16.0:compile (default-compile) @ testgen ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\test\resources
[INFO]
[INFO] --- compiler:3.16.0:testCompile (default-testCompile) @ testgen ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 4 source files with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ testgen ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running quadratic.CombinatorialTest
[INFO] Tests run: 242, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.928 s -- in quadratic.CombinatorialTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 242, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jacoco:0.8.12:report (report) @ testgen ---
[INFO] Loading execution data file D:\sysc5105-testgen\target\jacoco.exec
[INFO] Analyzed bundle 'testgen' with 4 classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  8.695 s
```

Figure 6. *Normal mode: 242 tests, 0 failures.*

```
PS D:\sysc5105-testgen> mvn clean test "-Dtest=quadratic.CombinatorialTest" "-Dstrict=true"
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< sysc5105:testgen >-------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24009 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ testgen ---
[INFO] Deleting D:\sysc5105-testgen\target
[INFO]
[INFO] --- jacoco:0.8.12:prepare-agent (default) @ testgen ---
[INFO] argLine set to -javaagent:C:\\Users\\anuja\\.m2\\repository\\org\\jacoco\\org.jacoco.agent\\0.8.12\\org.jacoco.agent-0.8.12-runtime.jar=destfile=D:\\sysc5105-testgen\\targe
t\\jacoco.exec
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\main\resources
[INFO]
[INFO] --- compiler:3.16.0:compile (default-compile) @ testgen ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\test\resources
[INFO]
[INFO] --- compiler:3.16.0:testCompile (default-testCompile) @ testgen ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 4 source files with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ testgen ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running quadratic.CombinatorialTest
[ERROR] Tests run: 242, Failures: 86, Errors: 0, Skipped: 0, Time elapsed: 1.240 s <<< FAILURE! -- in quadratic.CombinatorialTest
[ERROR] quadratic.CombinatorialTest.threeWaySuite(String, String, String)[3] -- Time elapsed: 0.015 s <<< FAILURE!
org.opentest4j.AssertionFailedError: D1: aborts although roots exist (c=0) (a=0.5, b=0, c=0)
        at org.junit.jupiter.api.AssertionUtils.fail(AssertionUtils.java:38)
        at org.junit.jupiter.api.Assertions.fail(Assertions.java:138)
        at quadratic.CombinatorialTest.known(CombinatorialTest.java:69)
        at quadratic.CombinatorialTest.checkAbort(CombinatorialTest.java:141)
        at quadratic.CombinatorialTest.runCase(CombinatorialTest.java:89)
        at quadratic.CombinatorialTest.threeWaySuite(CombinatorialTest.java:57)
        at java.base/java.lang.reflect.Method.invoke(Method.java:569)
        at java.base/java.util.stream.ForEachOps$ForEachOp$OfRef.accept(ForEachOps.java:183)
        at java.base/java.util.stream.ReferencePipeline$3$1.accept(ReferencePipeline.java:197)
        at java.base/java.util.stream.ReferencePipeline$2$1.accept(ReferencePipeline.java:179)
        at java.base/java.util.stream.ReferencePipeline$3$1.accept(ReferencePipeline.java:197)
```

Figure 7. *Strict mode: 86 of the 242 tests fail because they hit known defects.*



testgen > quadratic > Quadratic

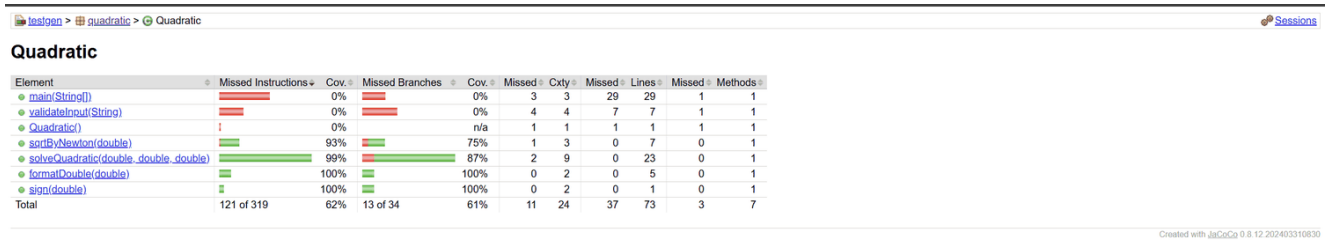
Sessions

Quadratic

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods
main(String[])		0%		0%	3	3	29	29	1	1
validateInput(String)		0%		0%	4	4	7	7	1	1
Quadratic()		0%		n/a	1	1	1	1	1	1
sqrtByNewton(double)		93%		75%	1	3	0	7	0	1
solveQuadratic(double, double, double)		99%		87%	2	9	0	23	0	1
formatDouble(double)		100%		100%	0	2	0	5	0	1
sign(double)		100%		100%	0	2	0	1	0	1
Total	121 of 319	62%	13 of 34	61%	11	24	37	73	3	7

Created with JaCoCo 0.8.12.202403310830

Figure 8. *JaCoCo class page for Quadratic, combinatorial suite (partial capture, upper rows).*



testgen > quadratic > Quadratic

Sessions

Quadratic

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods
main(String[])		0%		0%	3	3	29	29	1	1
validateInput(String)		0%		0%	4	4	7	7	1	1
Quadratic()		0%		n/a	1	1	1	1	1	1
sqrtByNewton(double)		93%		75%	1	3	0	7	0	1
solveQuadratic(double, double, double)		99%		87%	2	9	0	23	0	1
formatDouble(double)		100%		100%	0	2	0	5	0	1
sign(double)		100%		100%	0	2	0	1	0	1
Total	121 of 319	62%	13 of 34	61%	11	24	37	73	3	7

Created with JaCoCo 0.8.12.202403310830

Figure 9. *JaCoCo class page for Quadratic, combinatorial suite: 62% instruction and 61% branch coverage; main and validateInput are still not covered.*

```
PS D:\sysc5105-testgen> mvn org.pitest:pitest-maven:mutationCoverage "-DtargetTests=quadratic.CombinatorialTest"
[INFO] Scanning for projects...
[INFO] Loaded 24009 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO]
[INFO] -------------------------< sysc5105:testgen >-------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- pitest:1.16.1:mutationCoverage (default-cli) @ testgen ---
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO] Loaded 50 auto-discovered prefixes for remote repository ow2-snapshot (prefixes-ow2-snapshot.txt)
[INFO] Root dir is : D:\sysc5105-testgen
[INFO] Found plugin : Default csv report plugin
[INFO] Found plugin : Default xml report plugin
[INFO] Found plugin : Default html report plugin
[INFO] Found plugin : Static initializer code detector plugin
[INFO] Found plugin : Excluded annotations plugin
[INFO] Found plugin : Try with resources filter
[INFO] Found plugin : Inlined finally block filter plugin
[INFO] Found plugin : Implicit null check filter
[INFO] Found plugin : Method reference null check filter
[INFO] Found plugin : For each loop filter
[INFO] Found plugin : Enum junk filter
[INFO] Found plugin : Record junk mutation filter
[INFO] Found plugin : String switch filter
[INFO] Found plugin : Assertions filter
[INFO] Found plugin : Enum switch filter
[INFO] Found plugin : Logging calls filter
[INFO] Found plugin : Infinite for loop filter
[INFO] Found plugin : Long running iterator loop filter
[INFO] Found plugin : For loop counter filter
[INFO] Found plugin : Kotlin junk mutations filter
[INFO] Found plugin : Groovy junk mutations filter
[INFO] Found plugin : Max mutations per class limit
[INFO] Found plugin : Equals shortcut equivalent mutant filter
[INFO] Found plugin : Trivial return vals equivalence filter
[INFO] Found plugin : Filters mutants with line number <= 1
[INFO] Found plugin : Division by one equivalent mutant filter
[INFO] Found plugin : Lombok junk mutations filter
[INFO] Found plugin : Filter mutations to defensive wrappers such as unmodifiableCollection on return or field write
[INFO] Found plugin : Mutant export plugin
[INFO] Found plugin : Auto add java.awt.headless=true to keep keyboard focus on Mac OS
[INFO] Found plugin : Auto set number of threads based on machine
[INFO] Found plugin : Automatically add -ea to launch args to enable assertions
[INFO] Found plugin : Default build verifier
[INFO] Found plugin : Detect missing JUnit5 plugin
[INFO] Found plugin : Detect missing TestNG plugin
```

Figure 10. *PIT command and start-up output.*

```
-7:56:28 pm PIT >> INFO : Calculated coverage in 1 seconds.
7:56:28 pm PIT >> INFO : Created 3 mutation test units
7:56:34 pm PIT >> WARNING : Minion exited abnormally due to TIMED_OUT
7:56:44 pm PIT >> WARNING : Minion exited abnormally due to TIMED_OUT
7:56:53 pm PIT >> WARNING : Minion exited abnormally due to TIMED_OUT
\7:56:53 pm PIT >> INFO : Completed in 26 seconds
================================================================================
- Mutators
================================================================================
> org.pitest.mutationtest.engine.gregor.mutators.ConditionalsBoundaryMutator
>> Generated 3 Killed 0 (0%)
> KILLED 0 SURVIVED 3 TIMED_OUT 0 NON_VIABLE 0
> MEMORY_ERROR 0 NOT_STARTED 0 STARTED 0 RUN_ERROR 0
> NO_COVERAGE 0
--------------------------------------------------------------------------------
> org.pitest.mutationtest.engine.gregor.mutators.returns.PrimitiveReturnsMutator
>> Generated 6 Killed 2 (33%)
> KILLED 2 SURVIVED 0 TIMED_OUT 0 NON_VIABLE 0
> MEMORY_ERROR 0 NOT_STARTED 0 STARTED 0 RUN_ERROR 0
> NO_COVERAGE 4
--------------------------------------------------------------------------------
> org.pitest.mutationtest.engine.gregor.mutators.VoidMethodCallMutator
>> Generated 23 Killed 2 (9%)
> KILLED 2 SURVIVED 0 TIMED_OUT 0 NON_VIABLE 0
> MEMORY_ERROR 0 NOT_STARTED 0 STARTED 0 RUN_ERROR 0
> NO_COVERAGE 21
--------------------------------------------------------------------------------
> org.pitest.mutationtest.engine.gregor.mutators.MathMutator
>> Generated 25 Killed 20 (80%)
> KILLED 17 SURVIVED 2 TIMED_OUT 3 NON_VIABLE 0
> MEMORY_ERROR 0 NOT_STARTED 0 STARTED 0 RUN_ERROR 0
> NO_COVERAGE 3
--------------------------------------------------------------------------------
> org.pitest.mutationtest.engine.gregor.mutators.returns.EmptyObjectReturnValsMutator
>> Generated 21 Killed 2 (10%)
> KILLED 2 SURVIVED 0 TIMED_OUT 0 NON_VIABLE 0
> MEMORY_ERROR 0 NOT_STARTED 0 STARTED 0 RUN_ERROR 0
> NO_COVERAGE 19
--------------------------------------------------------------------------------
> org.pitest.mutationtest.engine.gregor.mutators.NegateConditionalsMutator
>> Generated 17 Killed 10 (59%)
> KILLED 10 SURVIVED 2 TIMED_OUT 0 NON_VIABLE 0
> MEMORY_ERROR 0 NOT_STARTED 0 STARTED 0 RUN_ERROR 0
> NO_COVERAGE 5
--------------------------------------------------------------------------------
================================================================================
- Timings
================================================================================
> pre-scan for mutations : < 1 second
> scan classpath : < 1 second
> coverage and dependency analysis : 1 seconds
> build mutation tests : < 1 second
```

Figure 11. *PIT mutator summary.*

```
--------------------------------------------------------------------------------
================================================================================
- Timings
================================================================================
> pre-scan for mutations : < 1 second
> scan classpath : < 1 second
> coverage and dependency analysis : 1 seconds
> build mutation tests : < 1 second
> run mutation analysis : 24 seconds
--------------------------------------------------------------------------------
> Total  : 26 seconds
--------------------------------------------------------------------------------
================================================================================
- Statistics
================================================================================
>> Line Coverage (for mutated classes only): 36/168 (21%)
>> Generated 95 mutations Killed 36 (38%)
>> Mutations with no coverage 52. Test strength 84%
>> Ran 1231 tests (12.96 tests per mutation)
Enhanced functionality available at https://www.arcmutate.com/
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  31.591 s
[INFO] Finished at: 2026-10-06T19:56:53-04:00
[INFO] ------------------------------------------------------------------------
PS D:\sysc5105-testgen>
```

Figure 12. *PIT timings and statistics (whole project: 36 of 95 mutants killed; quadratic alone: 36 of 60, 60%).*

```
PS D:\sysc5105-testgen> java -cp target\classes quadratic.Quadratic
Welcome to Quadratic Equation Solver.
A quadratic equation can be written in the form ax^2 + bx + c = 0, where x is an unknown, a, b, and c are constants, and a is not zero.
Given values for a, b, and c, this program will produce the two roots of the equation. Both real and complex roots are supported, but not complex coefficients
Press Ctrl+C to quit at any time.
Enter a value for 'a':
1.0E154
Enter a value for 'b':
1
Enter a value for 'c':
1.0E154
```

Figure 13. *Manual check of defect D2: with a = 1.0E154, b = 1 and c = 1.0E154 the program prints no roots and never finishes (infinite loop).*

A.3 Category-partition testing (17 frames)

Method. Seventeen hand-written frames from testdata\quad_cp_frames.csv, with expected roots derived by hand (CategoryPartitionTest). The frames cover the discriminant (positive, zero, negative, equal to b squared, overflow), the sign of a, b = 0 and c = 0. Normal and strict mode, then JaCoCo and PIT.

```
mvn clean test "-Dtest=quadratic.CategoryPartitionTest"
mvn clean test "-Dtest=quadratic.CategoryPartitionTest" "-Dstrict=true"
start target\site\jacoco\index.html
mvn org.pitest:pitest-maven:mutationCoverage "-DtargetTests=quadratic.CategoryPartitionTest"
```

```
PS D:\sysc5105-testgen> mvn clean test "-Dtest=quadratic.CategoryPartitionTest"
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< sysc5105:testgen >------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24009 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ testgen ---
[INFO] Deleting D:\sysc5105-testgen\target
[INFO]
[INFO] --- jacoco:0.8.12:prepare-agent (default) @ testgen ---
[INFO] argLine set to -javaagent:C:\\Users\\anuja\\.m2\\repository\\org\\jacoco\\org.jacoco.agent\\0.8.12\\org.jacoco.agent-0.8.12-runtime.jar=destfile=D:\\sysc5105-testgen\\targe
t\\jacoco.exec
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\main\resources
[INFO]
[INFO] --- compiler:3.16.0:compile (default-compile) @ testgen ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\test\resources
[INFO]
[INFO] --- compiler:3.16.0:testCompile (default-testCompile) @ testgen ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 5 source files with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ testgen ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running quadratic.CategoryPartitionTest
[INFO] Tests run: 17, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.411 s -- in quadratic.CategoryPartitionTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 17, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jacoco:0.8.12:report (report) @ testgen ---
[INFO] Loading execution data file D:\sysc5105-testgen\target\jacoco.exec
[INFO] Analyzed bundle 'testgen' with 4 classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
```

Figure 14. *Normal mode: 17 tests, 0 failures.*

```
PS D:\sysc5105-testgen> mvn clean test "-Dtest=quadratic.CategoryPartitionTest" "-Dstrict=true"
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< sysc5105:testgen >------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24009 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ testgen ---
[INFO] Deleting D:\sysc5105-testgen\target
[INFO]
[INFO] --- jacoco:0.8.12:prepare-agent (default) @ testgen ---
[INFO] argLine set to -javaagent:C:\\Users\\anuja\\.m2\\repository\\org\\jacoco\\org.jacoco.agent\\0.8.12\\org.jacoco.agent-0.8.12-runtime.jar=destfile=D:\\sysc5105-testgen\\targe
t\\jacoco.exec
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\main\resources
[INFO]
[INFO] --- compiler:3.16.0:compile (default-compile) @ testgen ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\test\resources
[INFO]
[INFO] --- compiler:3.16.0:testCompile (default-testCompile) @ testgen ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 5 source files with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ testgen ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running quadratic.CategoryPartitionTest
[ERROR] Tests run: 17, Failures: 4, Errors: 0, Skipped: 0, Time elapsed: 0.427 s <<< FAILURE! -- in quadratic.CategoryPartitionTest
[ERROR] quadratic.CategoryPartitionTest.frame(String, String, String, String, String, String, String)[9] -- Time elapsed: 0.024 s <<< FAILURE!
org.opentest4j.AssertionFailedError: D4: F09 prints '+ -' for negative imaginary part (a=-1, b=2, c=-5) -> x1 = 1 + -2i | x2 = 1 - -2i
        at org.junit.jupiter.api.AssertionUtils.fail(AssertionUtils.java:38)
        at org.junit.jupiter.api.Assertions.fail(Assertions.java:138)
        at quadratic.CategoryPartitionTest.known(CategoryPartitionTest.java:93)
        at quadratic.CategoryPartitionTest.frame(CategoryPartitionTest.java:74)
        at java.base/java.lang.reflect.Method.invoke(Method.java:569)
        at java.base/java.util.stream.ForEachOps$ForEachOp$OfRef.accept(ForEachOps.java:183)
        at java.base/java.util.stream.ReferencePipeline$3$1.accept(ReferencePipeline.java:197)
        at java.base/java.util.stream.ReferencePipeline$2$1.accept(ReferencePipeline.java:179)
        at java.base/java.util.stream.ReferencePipeline$3$1.accept(ReferencePipeline.java:197)
        at java.base/java.util.stream.ForEachOps$ForEachOp$OfRef.accept(ForEachOps.java:183)
        at java.base/java.util.stream.ReferencePipeline$3$1.accept(ReferencePipeline.java:197)
```

Figure 15. *Strict mode: 4 of the 17 tests fail (known defects D1, D3, D4 and D5).*

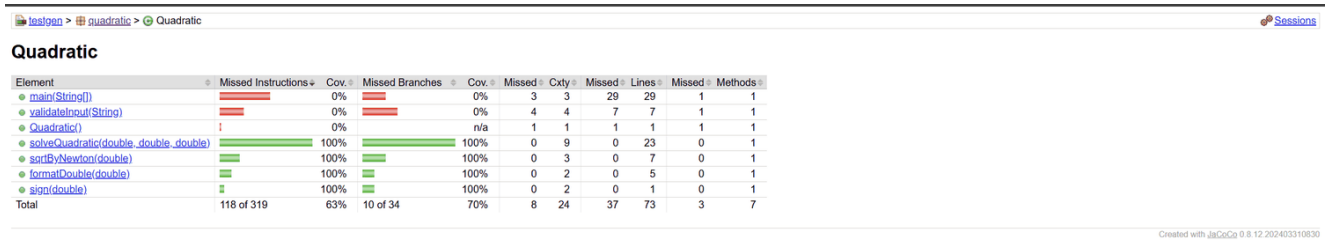


Figure 16. *JaCoCo class page for Quadratic: 63% instruction and 70% branch coverage; solveQuadratic and sqrtByNewton at 100%.*

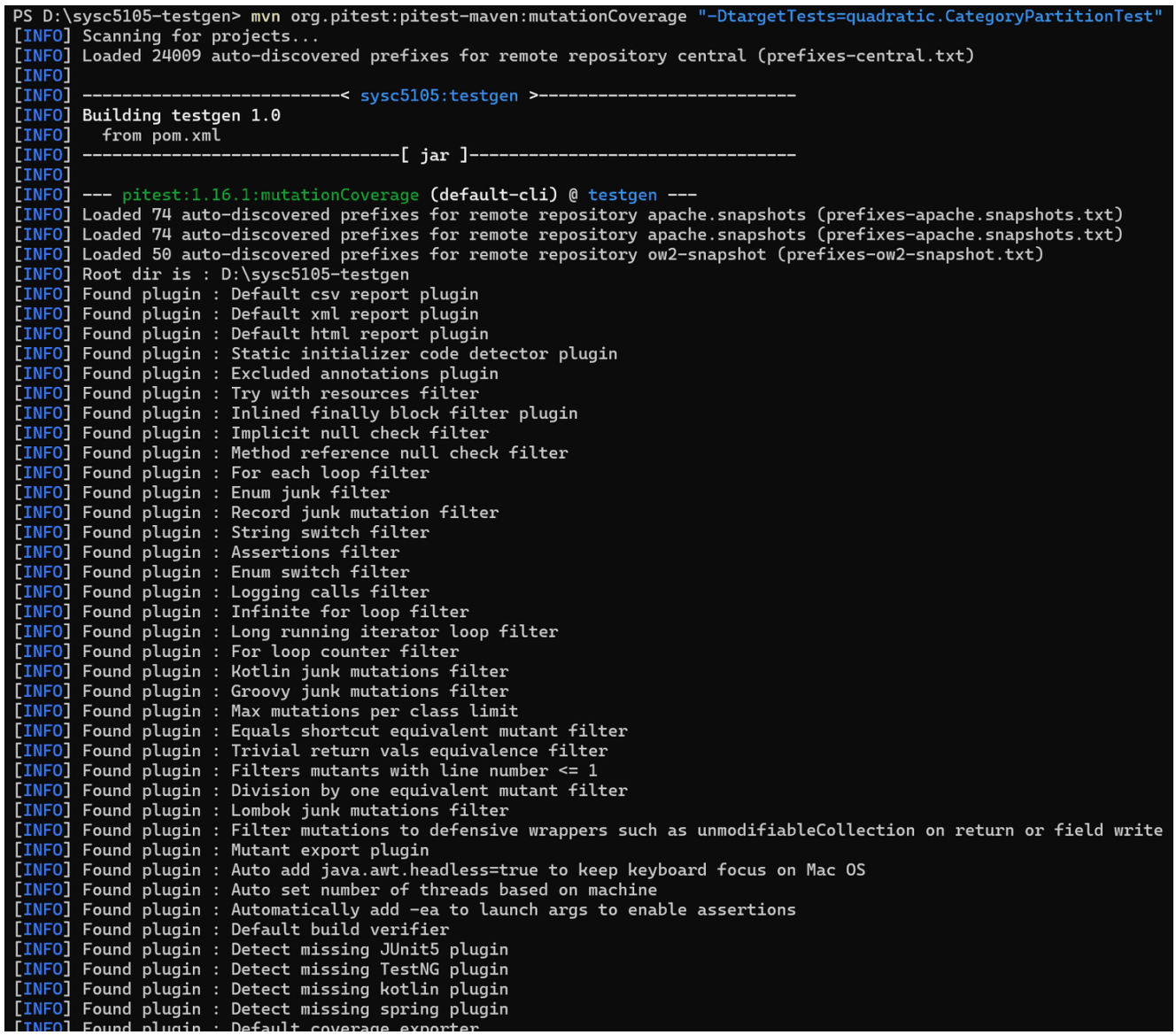


Figure 17. *PIT command and start-up output.*

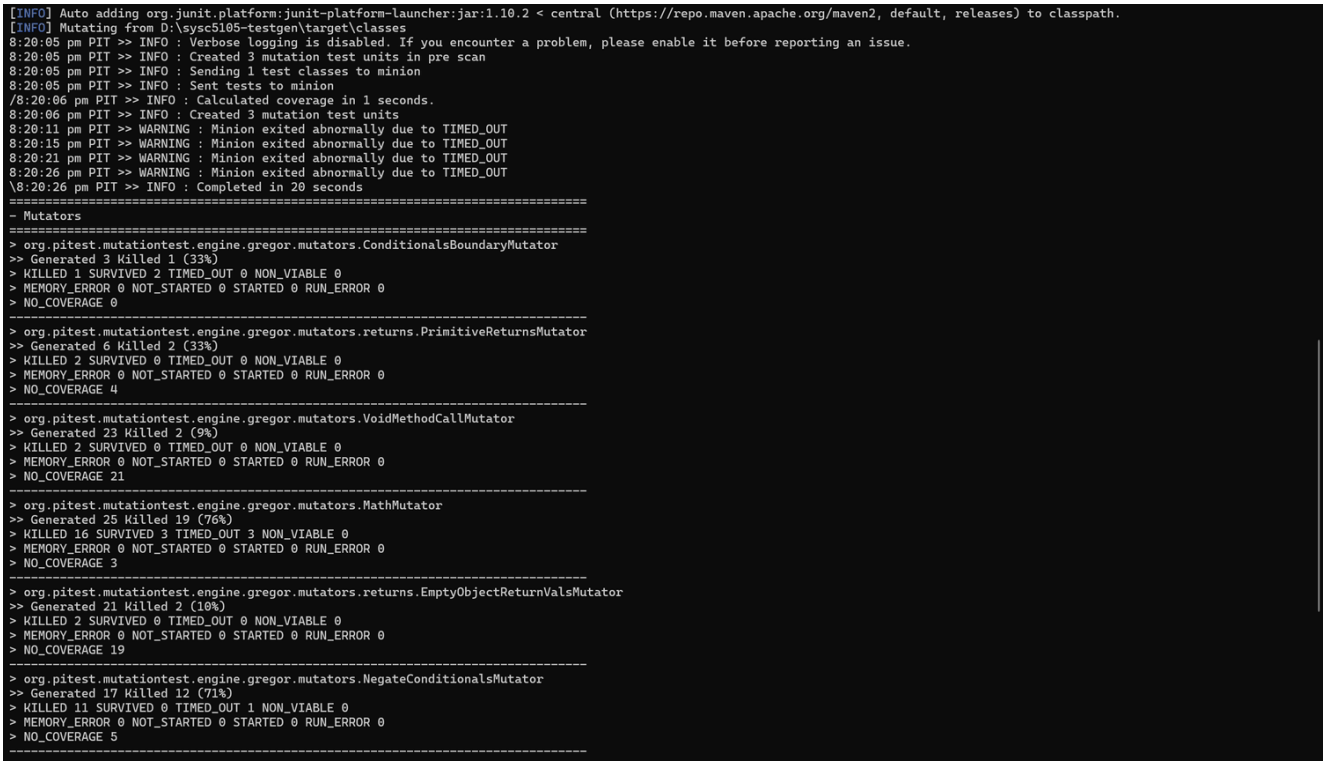


Figure 18. *PIT mutator summary.*

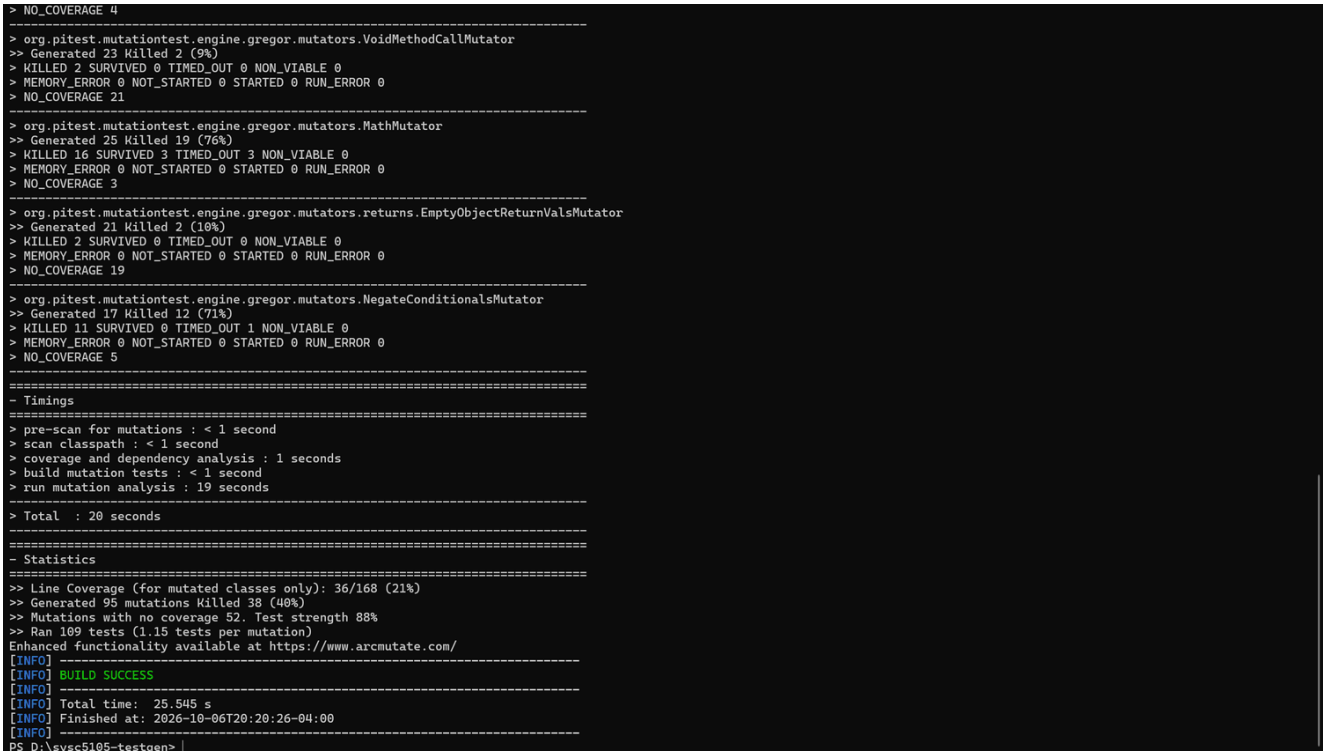


Figure 19. *PIT timings and statistics: 38 of 60 quadratic mutants killed (63%) at this point.*

A.4 Metamorphic testing

Method. MetamorphicTest uses no stored expected outputs. 216 seed equations are solved, then four relations build follow-up equations (scale all coefficients, negate b, reverse the coefficients, scale the roots); the follow-up roots must match the transformed seed roots. 216 seeds x 4 relations = 864 tests.

```
mvn clean test "-Dtest=quadratic.MetamorphicTest"
mvn clean test "-Dtest=quadratic.MetamorphicTest" "-Dstrict=true"
mvn org.pitest:pitest-maven:mutationCoverage "-DtargetTests=quadratic.MetamorphicTest"
Get-Content results\findings_metamorphic_run4.txt | Select-String '^D\d' | ForEach-Object {
$_.Line.Substring(0,2) } | Group-Object | Select-Object Name, Count
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Windows\System32> cd D:\
PS D:\> cd sysc5105-testgen
PS D:\sysc5105-testgen> notepad src\test\java\quadratic\MetamorphicTest.java
PS D:\sysc5105-testgen> mvn clean test "-Dtest=quadratic.MetamorphicTest"
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< sysc5105:testgen >------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24074 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ testgen ---
[INFO] Deleting D:\sysc5105-testgen\target
[INFO]
[INFO] --- jacoco:0.8.12:prepare-agent (default) @ testgen ---
[INFO] argLine set to -javaagent:C:\\Users\\anuja\\.m2\\repository\\org\\jacoco\\org.jacoco.agent\\0.8.12\\org.jacoco.ag
ent-0.8.12-runtime.jar=destfile=D:\\sysc5105-testgen\\target\\jacoco.exec
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\main\resources
[INFO]
[INFO] --- compiler:3.16.0:compile (default-compile) @ testgen ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[INFO]
```

Figure 20. *Normal mode run (first capture, top of the build log).*

```
PS D:\sysc5105-testgen> mvn clean test "-Dtest=quadratic.MetamorphicTest" "-Dstrict=true"
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< sysc5105:testgen >------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24074 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ testgen ---
[INFO] Deleting D:\sysc5105-testgen\target
[INFO]
[INFO] --- jacoco:0.8.12:prepare-agent (default) @ testgen ---
[INFO] argLine set to -javaagent:C:\\Users\\anuja\\.m2\\repository\\org\\jacoco\\org.jacoco.agent\\0.8.12\\org.jacoco.agent-0.8.12-runtime.jar=destfile=D:\\sysc5105-testgen\\targe
t\\jacoco.exec
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\main\resources
[INFO]
[INFO] --- compiler:3.16.0:compile (default-compile) @ testgen ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\test\resources
[INFO]
[INFO] --- compiler:3.16.0:testCompile (default-testCompile) @ testgen ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 6 source files with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ testgen ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running quadratic.MetamorphicTest
[ERROR] Tests run: 864, Failures: 208, Errors: 0, Skipped: 0, Time elapsed: 0.738 s <<< FAILURE! -- in quadratic.MetamorphicTest
[ERROR] quadratic.MetamorphicTest.mr1ScaleAllCoefficients(double, double, double)[1] -- Time elapsed: 0.111 s <<< FAILURE!
org.opentest4j.AssertionFailedError: D5: MR1(k=1.0E-9) roots changed, tiny discriminant seed(-4.0, -5.0, -4.0) -> follow-up(-4.0E-9, -5.0E-9, -4.0E-9)
        at org.junit.jupiter.api.AssertionUtils.fail(AssertionUtils.java:38)
        at org.junit.jupiter.api.Assertions.fail(Assertions.java:138)
        at quadratic.MetamorphicTest.known(MetamorphicTest.java:78)
        at quadratic.MetamorphicTest.check(MetamorphicTest.java:100)
        at quadratic.MetamorphicTest.mr1ScaleAllCoefficients(MetamorphicTest.java:45)
        at java.base/java.lang.reflect.Method.invoke(Method.java:569)
        at java.base/java.util.stream.ForEachOps$ForEachOp$OfRef.accept(ForEachOps.java:183)
        at java.base/java.util.stream.ReferencePipeline$3$1.accept(ReferencePipeline.java:197)
        at java.base/java.util.stream.ReferencePipeline$2$1.accept(ReferencePipeline.java:179)
        at java.base/java.util.stream.ReferencePipeline$3$1.accept(ReferencePipeline.java:197)
        at java.base/java.util.stream.ForEachOps$ForEachOp$OfRef.accept(ForEachOps.java:183)
```

Figure 21. *Strict mode: 208 of the 864 tests fail; all are defect D5 (tiny discriminant).*

```
PS D:\sysc5105-testgen> mvn clean test "-Dtest=quadratic.MetamorphicTest"
[INFO] Scanning for projects...
[INFO]
[INFO] ---------------------------< sysc5105:testgen >---------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24074 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ testgen ---
[INFO] Deleting D:\sysc5105-testgen\target
[INFO]
[INFO] --- jacoco:0.8.12:prepare-agent (default) @ testgen ---
[INFO] argLine set to -javaagent:C:\\Users\\anuja\\.m2\\repository\\org\\jacoco\\org.jacoco.agent\\0.8.12\\org.jacoco.agent-0.8.12-runtime.jar=destfile=D:\\sysc5105-testgen\\targ
t\\jacoco.exec
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\main\resources
[INFO]
[INFO] --- compiler:3.16.0:compile (default-compile) @ testgen ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\test\resources
[INFO]
[INFO] --- compiler:3.16.0:testCompile (default-testCompile) @ testgen ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 6 source files with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ testgen ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running quadratic.MetamorphicTest
[INFO] Tests run: 864, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.539 s -- in quadratic.MetamorphicTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 864, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jacoco:0.8.12:report (report) @ testgen ---
[INFO] Loading execution data file D:\sysc5105-testgen\target\jacoco.exec
[INFO] Analyzed bundle 'testgen' with 4 classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.480 s
```

Figure 22. *Normal mode: 864 tests, 0 failures (build log).*

```
[INFO] ---------------------------< sysc5105:testgen >---------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24074 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ testgen ---
[INFO] Deleting D:\sysc5105-testgen\target
[INFO]
[INFO] --- jacoco:0.8.12:prepare-agent (default) @ testgen ---
[INFO] argLine set to -javaagent:C:\\Users\\anuja\\.m2\\repository\\org\\jacoco\\org.jacoco.agent\\0.8.12\\org.jacoco.agent-0.8.12-runtime.jar=destfile=D:\\sysc5105-testgen\\targe
t\\jacoco.exec
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\main\resources
[INFO]
[INFO] --- compiler:3.16.0:compile (default-compile) @ testgen ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\test\resources
[INFO]
[INFO] --- compiler:3.16.0:testCompile (default-testCompile) @ testgen ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 6 source files with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ testgen ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running quadratic.MetamorphicTest
[INFO] Tests run: 864, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.539 s -- in quadratic.MetamorphicTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 864, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jacoco:0.8.12:report (report) @ testgen ---
[INFO] Loading execution data file D:\sysc5105-testgen\target\jacoco.exec
[INFO] Analyzed bundle 'testgen' with 4 classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.480 s
[INFO] Finished at: 2026-10-13T11:02:09-04:00
[INFO] ------------------------------------------------------------------------
PS D:\sysc5105-testgen>
```

Figure 23. *Normal mode, end of the build log: BUILD SUCCESS, total time 3.480 s.*

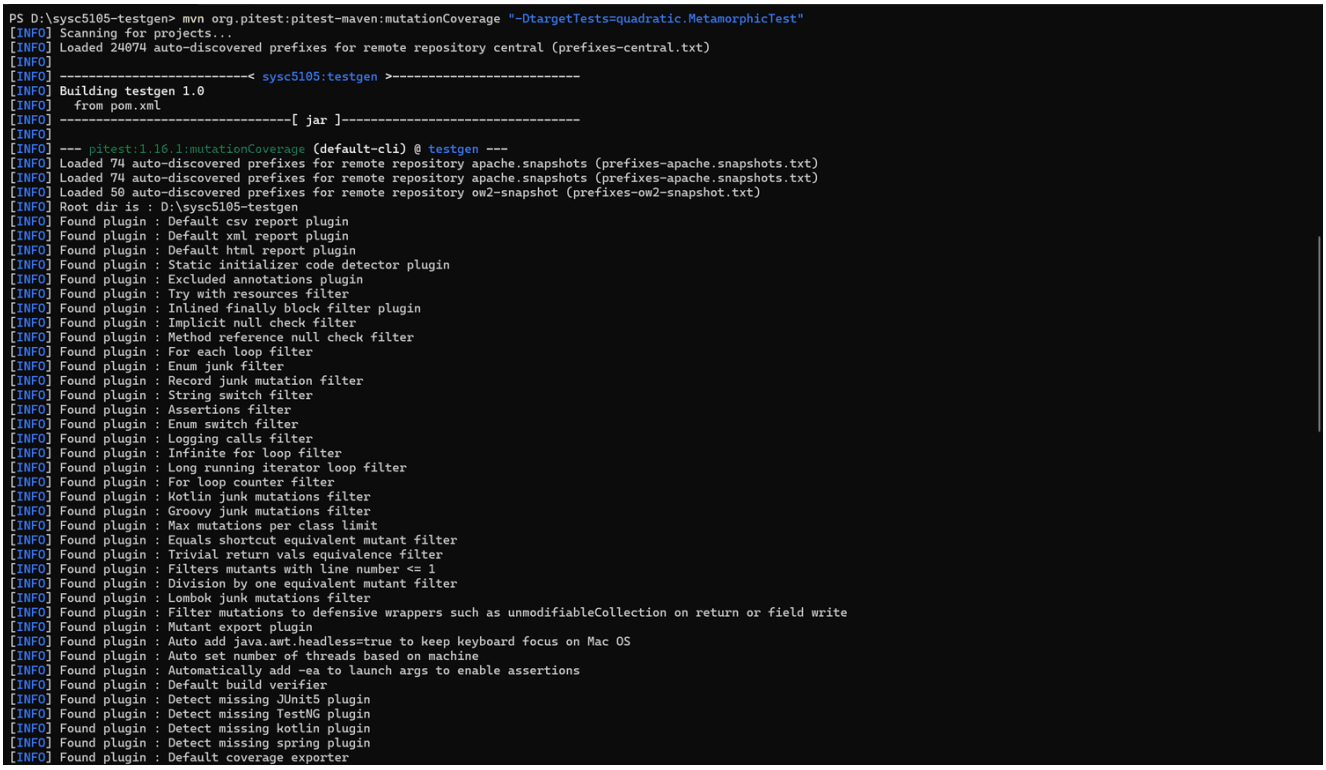


Figure 24. *PIT command and start-up output.*

```
> NO_COVERAGE 4
--------------------------------------------------------------------------------
> org.pitest.mutationtest.engine.gregor.mutators.VoidMethodCallMutator
>> Generated 23 Killed 2 (9%)
> KILLED 2 SURVIVED 0 TIMED_OUT 0 NON_VIABLE 0
> MEMORY_ERROR 0 NOT_STARTED 0 STARTED 0 RUN_ERROR 0
> NO_COVERAGE 21
--------------------------------------------------------------------------------
> org.pitest.mutationtest.engine.gregor.mutators.MathMutator
>> Generated 25 Killed 20 (80%)
> KILLED 19 SURVIVED 2 TIMED_OUT 1 NON_VIABLE 0
> MEMORY_ERROR 0 NOT_STARTED 0 STARTED 0 RUN_ERROR 0
> NO_COVERAGE 3
--------------------------------------------------------------------------------
> org.pitest.mutationtest.engine.gregor.mutators.returns.EmptyObjectReturnValsMutato
>> Generated 21 Killed 2 (10%)
> KILLED 2 SURVIVED 0 TIMED_OUT 0 NON_VIABLE 0
> MEMORY_ERROR 0 NOT_STARTED 0 STARTED 0 RUN_ERROR 0
> NO_COVERAGE 19
--------------------------------------------------------------------------------
> org.pitest.mutationtest.engine.gregor.mutators.NegateConditionalsMutator
>> Generated 17 Killed 11 (65%)
> KILLED 11 SURVIVED 1 TIMED_OUT 0 NON_VIABLE 0
> MEMORY_ERROR 0 NOT_STARTED 0 STARTED 0 RUN_ERROR 0
> NO_COVERAGE 5
--------------------------------------------------------------------------------
================================================================================
- Timings
================================================================================
> pre-scan for mutations : < 1 second
> scan classpath : < 1 second
> coverage and dependency analysis : < 1 second
> build mutation tests : < 1 second
> run mutation analysis : 9 seconds
--------------------------------------------------------------------------------
> Total  : 10 seconds
--------------------------------------------------------------------------------
================================================================================
- Statistics
================================================================================
>> Line Coverage (for mutated classes only): 35/168 (21%)
>> Generated 95 mutations Killed 37 (39%)
>> Mutations with no coverage 52. Test strength 86%
>> Ran 4581 tests (48.22 tests per mutation)
Enhanced functionality available at https://www.arcmutate.com/
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  13.704 s
[INFO] Finished at: 2026-10-13T10:56:01-04:00
[INFO] ------------------------------------------------------------------------
PS D:\sysc5105-testgen>
```

Figure 25. *PIT timings and statistics: 37 of 60 quadratic mutants killed (62%).*

```
PS D:\sysc5105-testgen> Get-Content results\findings_metamorphic_run4.txt | Select-String '^D\d' | ForEach-Object { $_.Line.Substring(0,2) } | Group-Object | Select-Object Na
ount

Name Count
---- -----
D5     208
```

Figure 26. *Defect counts from the saved findings file: D5 = 208 entries, no other defect type.*

A.5 All three techniques together

Method. The three test classes run in one Maven invocation to see what the techniques reach together.

```
mvn clean test "-
Dtest=quadratic.CombinatorialTest,quadratic.CategoryPartitionTest,quadratic.MetamorphicTest"
start target\site\jacoco\index.html
mvn org.pitest:pitest-maven:mutationCoverage "-
DtargetTests=quadratic.CombinatorialTest,quadratic.CategoryPartitionTest,quadratic.MetamorphicTest"
```

```
PS D:\sysc5105-testgen> mvn clean test "-Dtest=quadratic.CombinatorialTest,quadratic.CategoryPartitionTest,quadratic.MetamorphicTest"
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< sysc5105:testgen >------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24074 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ testgen ---
[INFO] Deleting D:\sysc5105-testgen\target
[INFO]
[INFO] --- jacoco:0.8.12:prepare-agent (default) @ testgen ---
[INFO] argLine set to -javaagent:C:\\Users\\anuja\\.m2\\repository\\org\\jacoco\\org.jacoco.agent\\0.8.12\\org.jacoco.agent-0.8.12-runtime.jar=destfile=D:\\sysc5105-testgen\\target\\jacoco.exec
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\main\resources
[INFO]
[INFO] --- compiler:3.16.0:compile (default-compile) @ testgen ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\test\resources
[INFO]
[INFO] --- compiler:3.16.0:testCompile (default-testCompile) @ testgen ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 6 source files with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ testgen ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running quadratic.CategoryPartitionTest
[INFO] Tests run: 17, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.151 s -- in quadratic.CategoryPartitionTest
[INFO] Running quadratic.CombinatorialTest
[INFO] Tests run: 242, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.172 s -- in quadratic.CombinatorialTest
[INFO] Running quadratic.MetamorphicTest
[INFO] Tests run: 864, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.331 s -- in quadratic.MetamorphicTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1123, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jacoco:0.8.12:report (report) @ testgen ---
[INFO] Loading execution data file D:\sysc5105-testgen\target\jacoco.exec
[INFO] Analyzed bundle 'testgen' with 4 classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.849 s
[INFO] Finished at: 2026-10-13T16:07:39-04:00
```

Figure 27. *Three suites together: 1123 tests (17 + 242 + 864), 0 failures.*

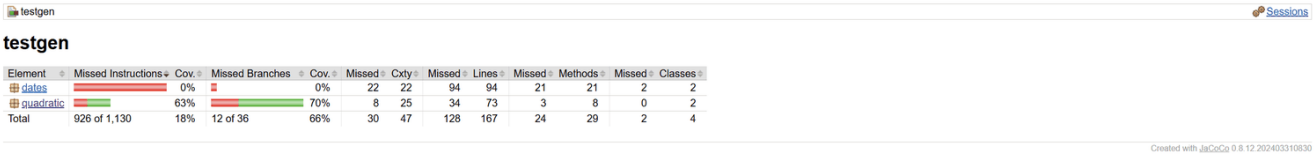


Figure 28. *JaCoCo project overview: quadratic 63% instruction and 70% branch coverage (the date code is not tested yet, 0%).*

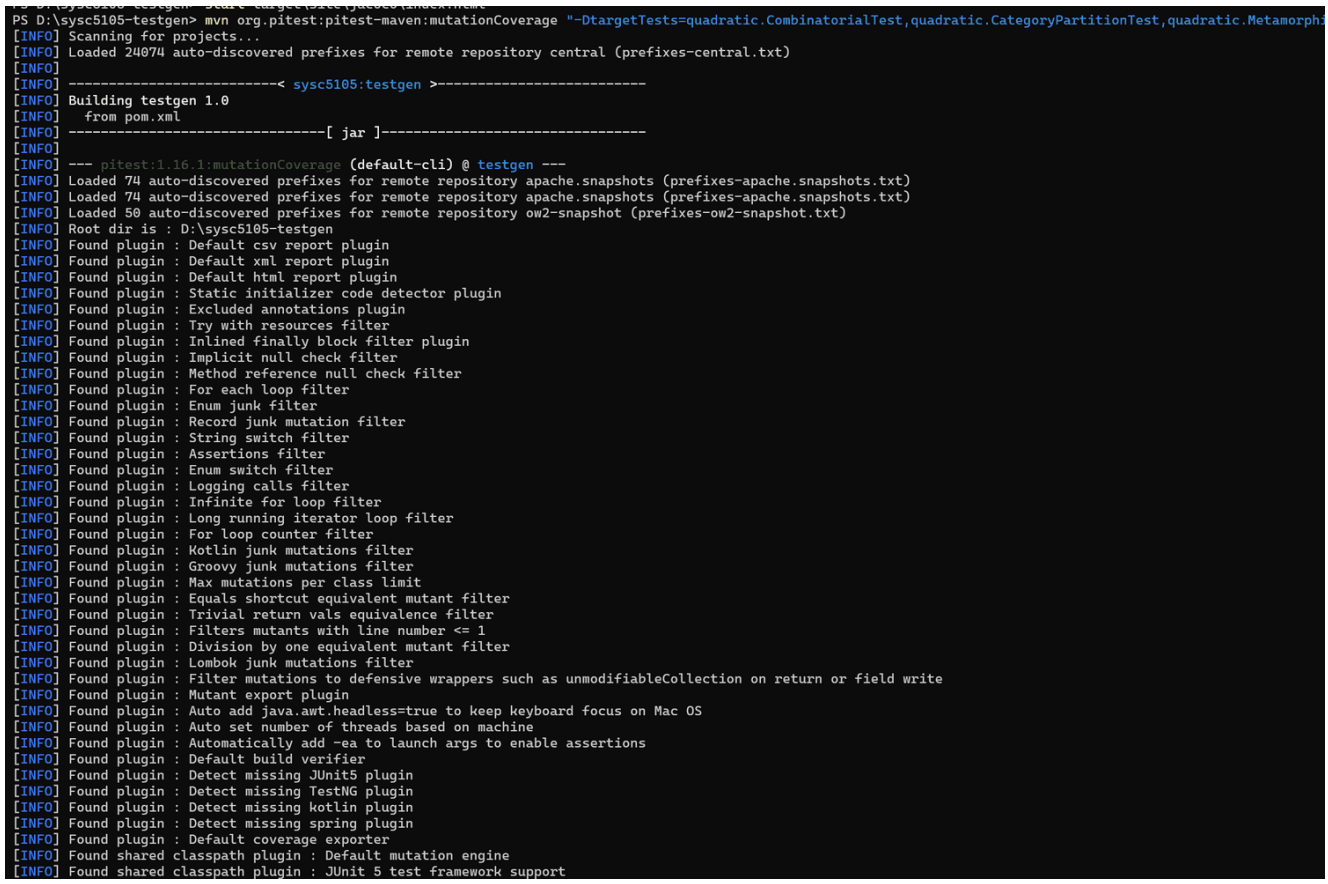


Figure 29. *PIT command and start-up output.*

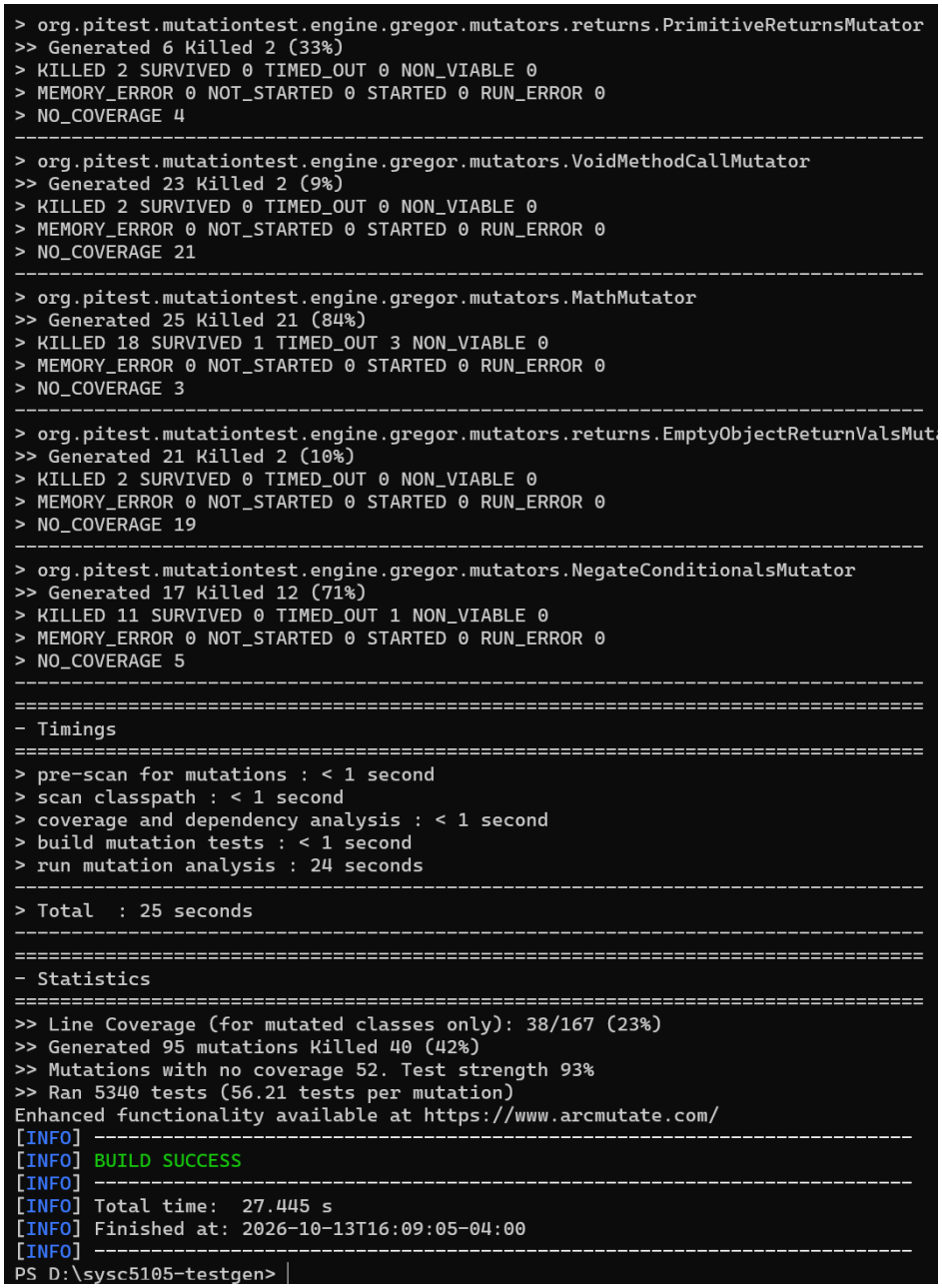


Figure 30. *PIT timings and statistics: 40 of 60 quadratic mutants killed (67%).*

A.6 Improvement round

Method. The survivor and coverage reports showed what was missing. Two fixes went into CategoryPartitionTest (the discriminant-zero exemption and a new frame F18), and a new ImprovementTest (12 tests) calls validateInput directly and drives main() with scripted input. The final run uses all four quadratic test classes.

```
mvn clean test "-Dtest=quadratic.CategoryPartitionTest"
mvn clean test "-Dtest=quadratic.ImprovementTest"
mvn clean test "-Dtest=quadratic.ImprovementTest" "-Dstrict=true"
Copy-Item results\findings_improvement.txt results\findings_improvement_run6.txt -Force
mvn clean test "-
Dtest=quadratic.CombinatorialTest,quadratic.CategoryPartitionTest,quadratic.MetamorphicTest,quadratic.I
mprovementTest"
start target\site\jacoco\index.html
mvn org.pitest:pitest-maven:mutationCoverage "-
DtargetTests=quadratic.CombinatorialTest,quadratic.CategoryPartitionTest,quadratic.MetamorphicTest,quad
ratic.ImprovementTest" "-DoutputFormats=CSV,HTML"
```

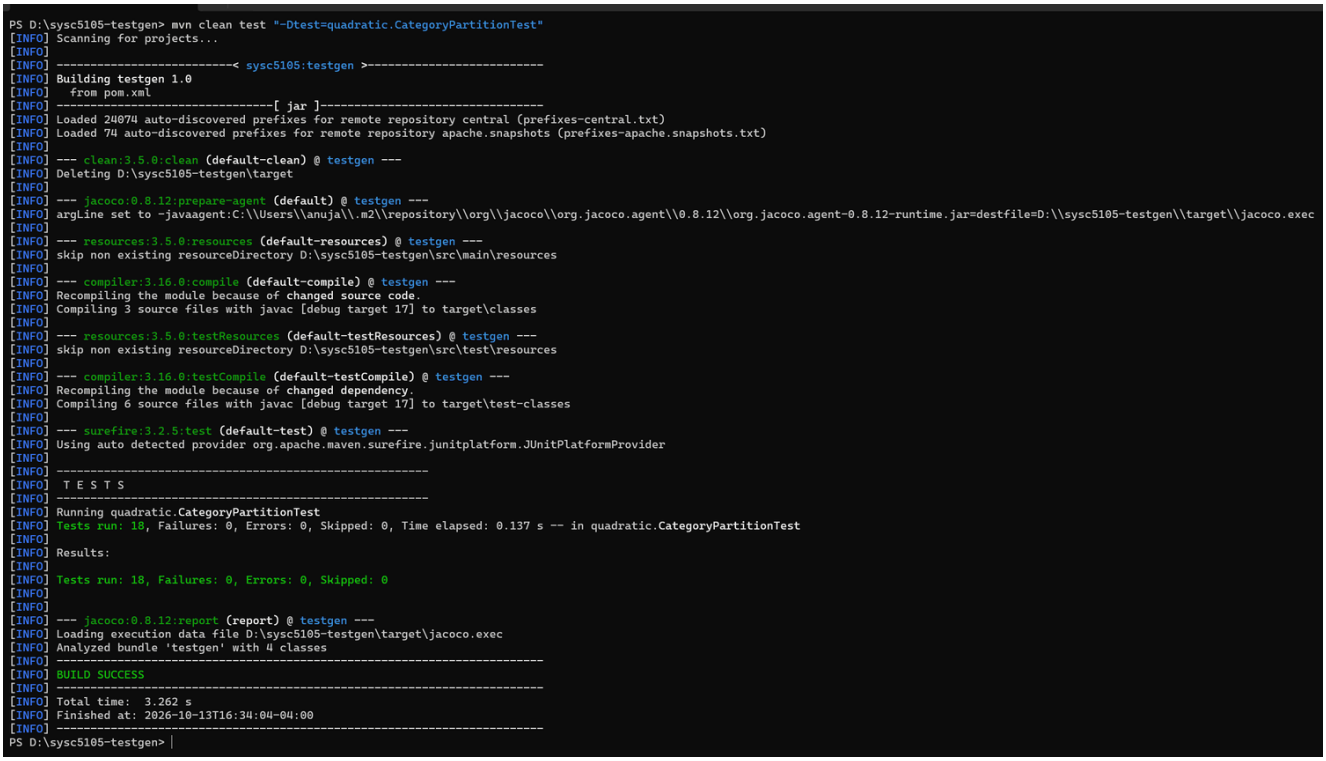


Figure 31. *CategoryPartitionTest after the fix and frame F18: 18 tests, 0 failures.*

```
PS D:\sysc5105-testgen> mvn clean test "-Dtest=quadratic.ImprovementTest"
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< sysc5105:testgen >------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24074 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ testgen ---
[INFO] Deleting D:\sysc5105-testgen\target
[INFO]
[INFO] --- jacoco:0.8.12:prepare-agent (default) @ testgen ---
[INFO] argLine set to -javaagent:C:\\Users\\anuja\\.m2\\repository\\org\\jacoco\\org.jacoco.agent\\0.8.12\\org.jacoco.agent-0.8.12-runtime.jar=destfile=D:\\sysc5105-testgen\\target\\ja
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\main\resources
[INFO]
[INFO] --- compiler:3.16.0:compile (default-compile) @ testgen ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\test\resources
[INFO]
[INFO] --- compiler:3.16.0:testCompile (default-testCompile) @ testgen ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 7 source files with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ testgen ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running quadratic.ImprovementTest
[INFO] Tests run: 12, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.083 s -- in quadratic.ImprovementTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 12, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jacoco:0.8.12:report (report) @ testgen ---
[INFO] Loading execution data file D:\sysc5105-testgen\target\jacoco.exec
[INFO] Analyzed bundle 'testgen' with 4 classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.089 s
[INFO] Finished at: 2026-10-13T16:35:53-04:00
[INFO] ------------------------------------------------------------------------
PS D:\sysc5105-testgen>
```

Figure 32. *ImprovementTest, normal mode: 12 tests, 0 failures.*

```
PS D:\sysc5105-testgen> mvn clean test "-Dtest=quadratic.ImprovementTest" "-Dstrict=true"
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< sysc5105:testgen >------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24074 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ testgen ---
[INFO] Deleting D:\sysc5105-testgen\target
[INFO]
[INFO] --- jacoco:0.8.12:prepare-agent (default) @ testgen ---
[INFO] argLine set to -javaagent:C:\\Users\\anuja\\.m2\\repository\\org\\jacoco\\org.jacoco.agent\\0.8.12\\org.jacoco.agent-0.8.12-runtime.jar=destfile=D:\\sysc5105-testgen\\target\\jacoco.exec
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\main\resources
[INFO]
[INFO] --- compiler:3.16.0:compile (default-compile) @ testgen ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\test\resources
[INFO]
[INFO] --- compiler:3.16.0:testCompile (default-testCompile) @ testgen ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 7 source files with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ testgen ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running quadratic.ImprovementTest
[ERROR] Tests run: 12, Failures: 4, Errors: 0, Skipped: 0, Time elapsed: 0.073 s <<< FAILURE! -- in quadratic.ImprovementTest
[ERROR] quadratic.ImprovementTest.harmlessFormattingVariants -- Time elapsed: 0.016 s <<< FAILURE!
org.opentest4j.AssertionFailedError: D6: validateInput rejects valid number '0.10'
        at org.junit.jupiter.api.AssertionUtils.fail(AssertionUtils.java:38)
        at org.junit.jupiter.api.Assertions.fail(Assertions.java:138)
        at quadratic.ImprovementTest.known(ImprovementTest.java:46)
        at quadratic.ImprovementTest.harmlessFormattingVariants(ImprovementTest.java:88)
        at java.base/java.lang.reflect.Method.invoke(Method.java:569)
        at java.base/java.util.ArrayList.forEach(ArrayList.java:1511)
        at java.base/java.util.ArrayList.forEach(ArrayList.java:1511)

[ERROR] quadratic.ImprovementTest.lowercaseExponentIsRejected -- Time elapsed: 0 s <<< FAILURE!
org.opentest4j.AssertionFailedError: D6: validateInput rejects valid e-notation '1e154' as too large or small
        at org.junit.jupiter.api.AssertionUtils.fail(AssertionUtils.java:38)
        at org.junit.jupiter.api.Assertions.fail(Assertions.java:138)
        at quadratic.ImprovementTest.known(ImprovementTest.java:46)
        at quadratic.ImprovementTest.lowercaseExponentIsRejected(ImprovementTest.java:79)
        at java.base/java.lang.reflect.Method.invoke(Method.java:569)
        at java.base/java.util.ArrayList.forEach(ArrayList.java:1511)
```

Figure 33. *ImprovementTest, strict mode: 4 of the 12 tests fail on recorded defects (part 1: the run and the first failure).*

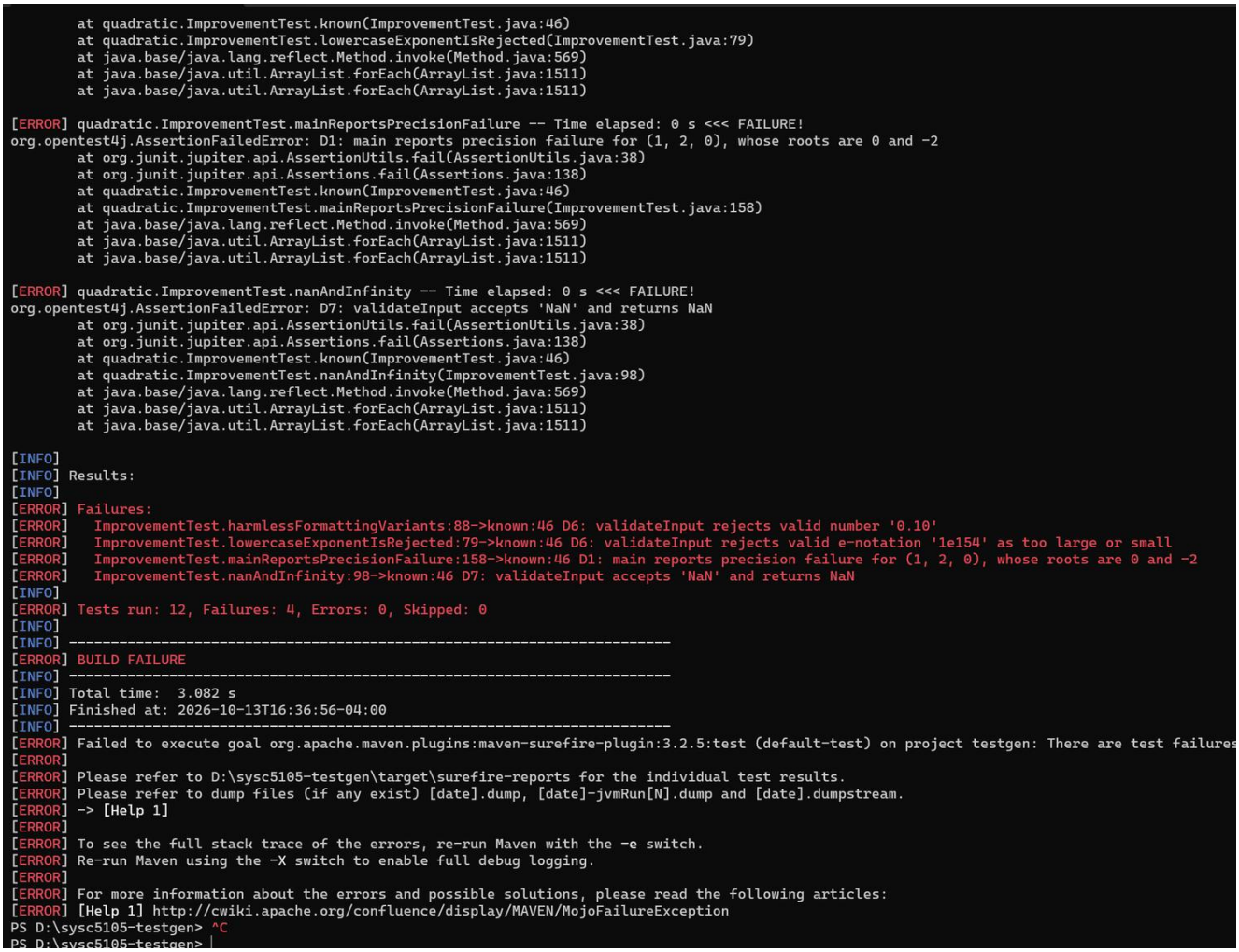


Figure 34. *Strict mode, part 2: the remaining failures (D1, D7) and the Maven summary.*

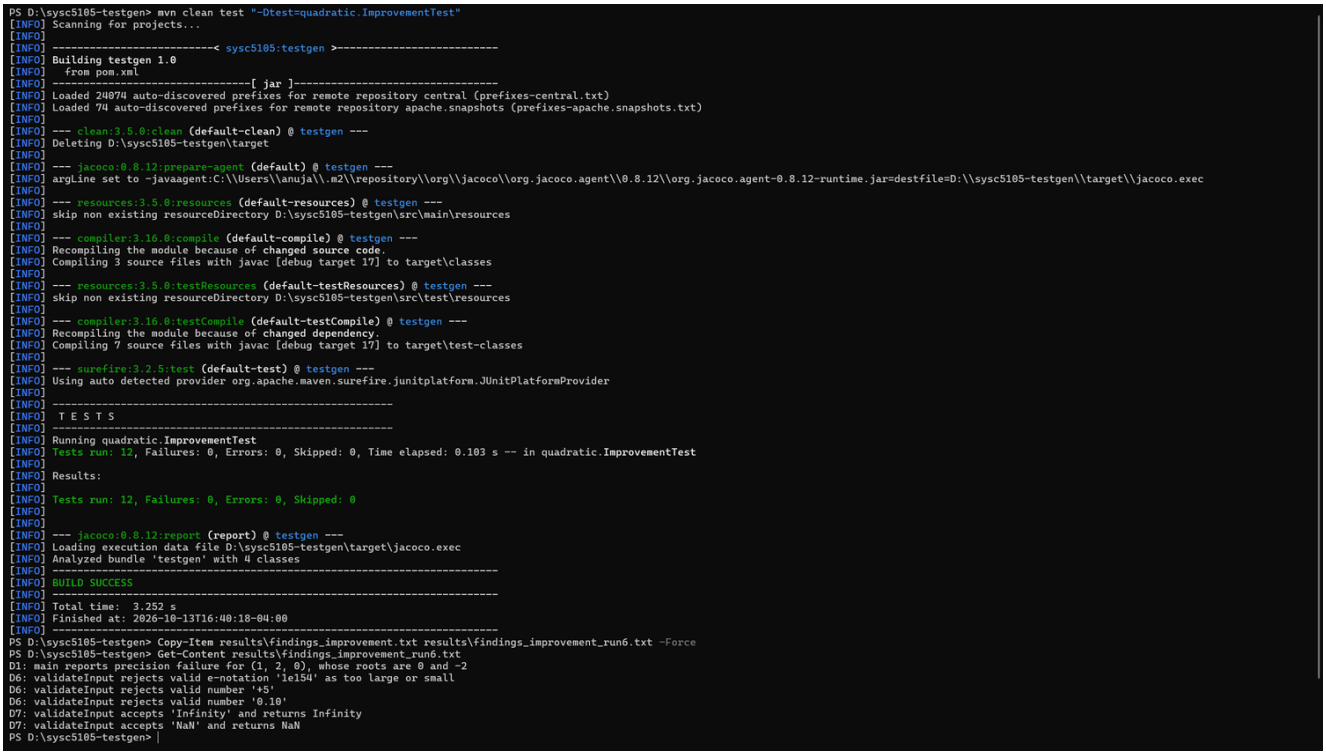


Figure 35. *Normal mode again, then the findings are copied to results\findings_improvement_run6.txt and listed (D1, D6, D7).*

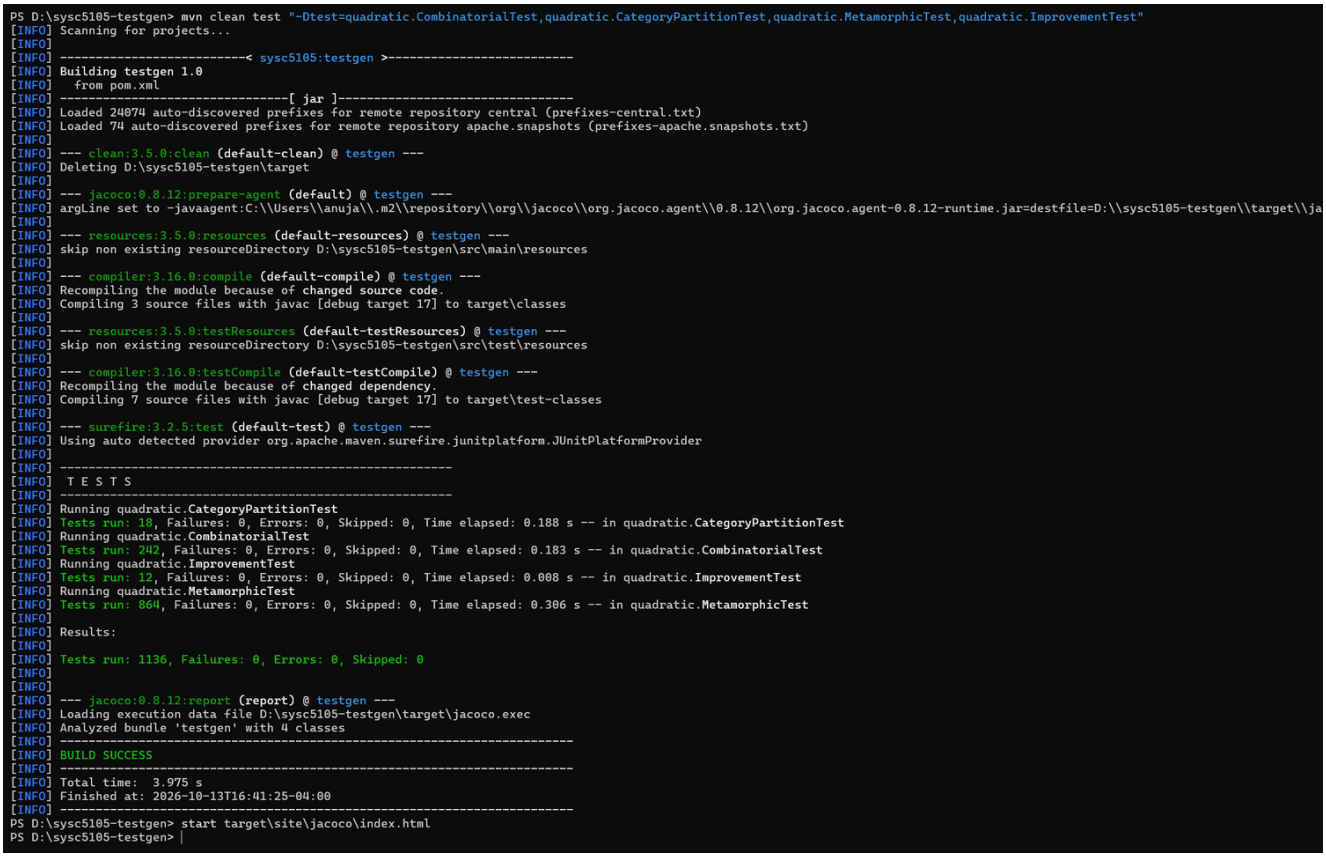
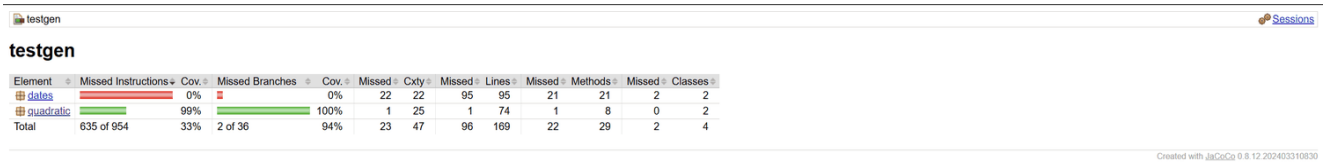


Figure 36. *Final suite, all four quadratic classes together: 1136 tests, 0 failures.*



testgen

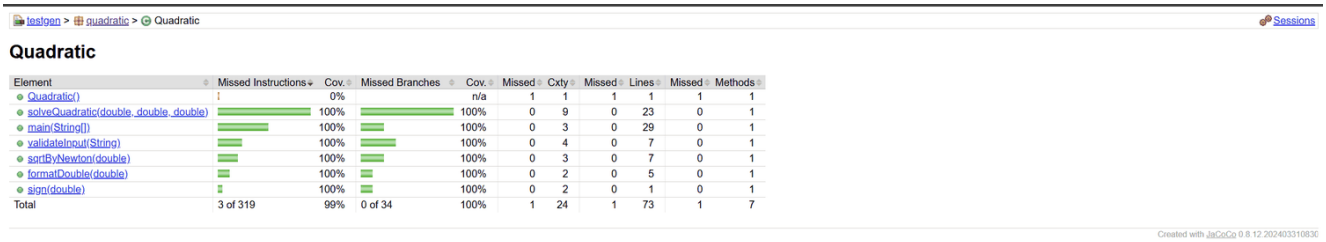
Sessions

testgen

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
dates		0%		0%	22	22	95	95	21	21	2	2
quadratic		99%		100%	1	25	1	74	1	8	0	2
Total	635 of 954	33%	2 of 36	94%	23	47	96	169	22	29	2	4

Created with JaCoCo 0.8.12.202403310830

Figure 37. *JaCoCo project overview: quadratic 99% instruction and 100% branch coverage.*



testgen > quadratic > Quadratic

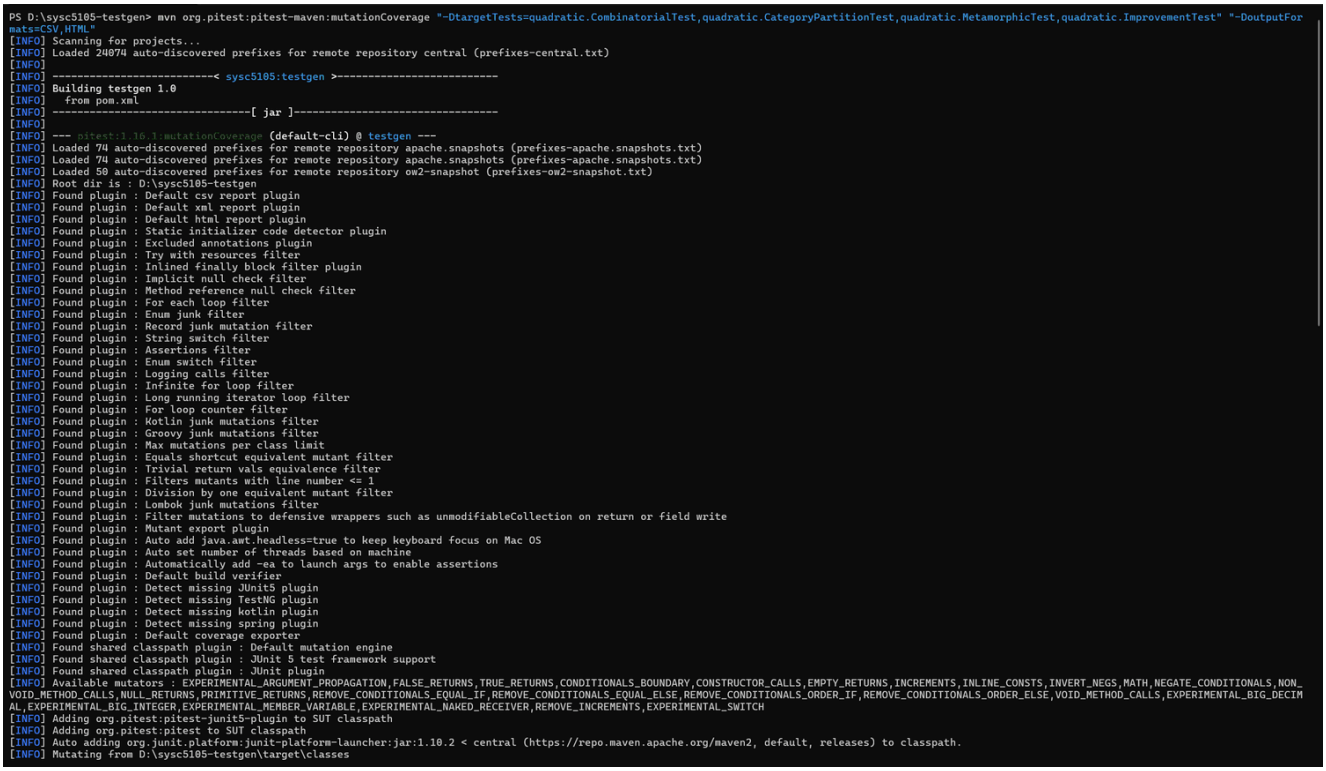
Sessions

Quadratic

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods
Quadratic()		0%		n/a	1	1	1	1	1	1
solveQuadratic(double, double, double)		100%		100%	0	9	0	23	0	1
main(String[])		100%		100%	0	3	0	29	0	1
validateInput(String)		100%		100%	0	4	0	7	0	1
sortByNewton(double)		100%		100%	0	3	0	7	0	1
formatDouble(double)		100%		100%	0	2	0	5	0	1
sign(double)		100%		100%	0	2	0	1	0	1
Total	3 of 319	99%	0 of 34	100%	1	24	1	73	1	7

Created with JaCoCo 0.8.12.202403310830

Figure 38. *JaCoCo class page for Quadratic: every method at 100%, including main and validateInput.*



```
PS D:\sysc5105-testgen> mvn org.pitest:pitest-maven:mutationCoverage "-DtargetTests=quadratic.CombinatorialTest,quadratic.CategoryPartitionTest,quadratic.MetamorphicTest,quadratic.ImprovementTest" "-DoutputFormats=CSV,HTML"
[INFO] Scanning for projects...
[INFO] Loaded 24074 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO]
[INFO] -------------------------< sysc5105:testgen >-------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- pitest:1.16.1:mutationCoverage (default-cli) @ testgen ---
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO] Loaded 50 auto-discovered prefixes for remote repository ow2-snapshot (prefixes-ow2-snapshot.txt)
[INFO] Root dir is : D:\sysc5105-testgen
[INFO] Found plugin : Default csv report plugin
[INFO] Found plugin : Default xml report plugin
[INFO] Found plugin : Default html report plugin
[INFO] Found plugin : Static initializer code detector plugin
[INFO] Found plugin : Excluded annotations plugin
[INFO] Found plugin : Try with resources filter
[INFO] Found plugin : Inlined finally block filter plugin
[INFO] Found plugin : Implicit null check filter
[INFO] Found plugin : Method reference null check filter
[INFO] Found plugin : For each loop filter
[INFO] Found plugin : Enum junk filter
[INFO] Found plugin : Record junk mutation filter
[INFO] Found plugin : String switch filter
[INFO] Found plugin : Assertions filter
[INFO] Found plugin : Enum switch filter
[INFO] Found plugin : Logging calls filter
[INFO] Found plugin : Infinite for loop filter
[INFO] Found plugin : Long running iterator loop filter
[INFO] Found plugin : For loop counter filter
[INFO] Found plugin : Kotlin junk mutations filter
[INFO] Found plugin : Groovy junk mutations filter
[INFO] Found plugin : Max mutations per class limit
[INFO] Found plugin : Equals shortcut equivalent mutant filter
[INFO] Found plugin : Trivial return vals equivalence filter
[INFO] Found plugin : Filters mutants with line number <= 1
[INFO] Found plugin : Division by one equivalent mutant filter
[INFO] Found plugin : Lombok junk mutations filter
[INFO] Found plugin : Filter mutations to defensive wrappers such as unmodifiableCollection on return or field write
[INFO] Found plugin : Mutant export plugin
[INFO] Found plugin : Auto add java.awt.headless=true to keep keyboard focus on Mac OS
[INFO] Found plugin : Auto set number of threads based on machine
[INFO] Found plugin : Automatically add -ea to launch args to enable assertions
[INFO] Found plugin : Default build verifier
[INFO] Found plugin : Detect missing JUnit5 plugin
[INFO] Found plugin : Detect missing TestNG plugin
[INFO] Found plugin : Detect missing kotlin plugin
[INFO] Found plugin : Detect missing spring plugin
[INFO] Found plugin : Default coverage exporter
[INFO] Found shared classpath plugin : Default mutation engine
[INFO] Found shared classpath plugin : JUnit 5 test framework support
[INFO] Found shared classpath plugin : JUnit plugin
[INFO] Available mutators : EXPERIMENTAL_ARGUMENT_PROPAGATION,FALSE_RETURNS,TRUE_RETURNS,CONDITIONALS_BOUNDARY,CONSTRUCTOR_CALLS,EMPTY_RETURNS,INCREMENTS,INLINE_CONSTS,INVERT_NEGS,MATH,NEGATE_CONDITIONALS,NON_VOID_METHOD_CALLS,NULL_RETURNS,PRIMITIVE_RETURNS,REMOVE_CONDITIONALS_EQUAL_IF,REMOVE_CONDITIONALS_EQUAL_ELSE,REMOVE_CONDITIONALS_ORDER_IF,REMOVE_CONDITIONALS_ORDER_ELSE,VOID_METHOD_CALLS,EXPERIMENTAL_BIG_DECIMAL,EXPERIMENTAL_BIG_INTEGER,EXPERIMENTAL_MEMBER_VARIABLE,EXPERIMENTAL_NAKED_RECEIVER,REMOVE_INCREMENTS,EXPERIMENTAL_SWITCH
[INFO] Adding org.pitest:pitest-junit5-plugin to SUT classpath
[INFO] Adding org.pitest:pitest to SUT classpath
[INFO] Auto adding org.junit.platform:junit-platform-launcher:jar:1.10.2 < central (https://repo.maven.apache.org/maven2, default, releases) to classpath.
[INFO] Mutating from D:\sysc5105-testgen\target\classes
```

Figure 39. *PIT command and start-up output.*

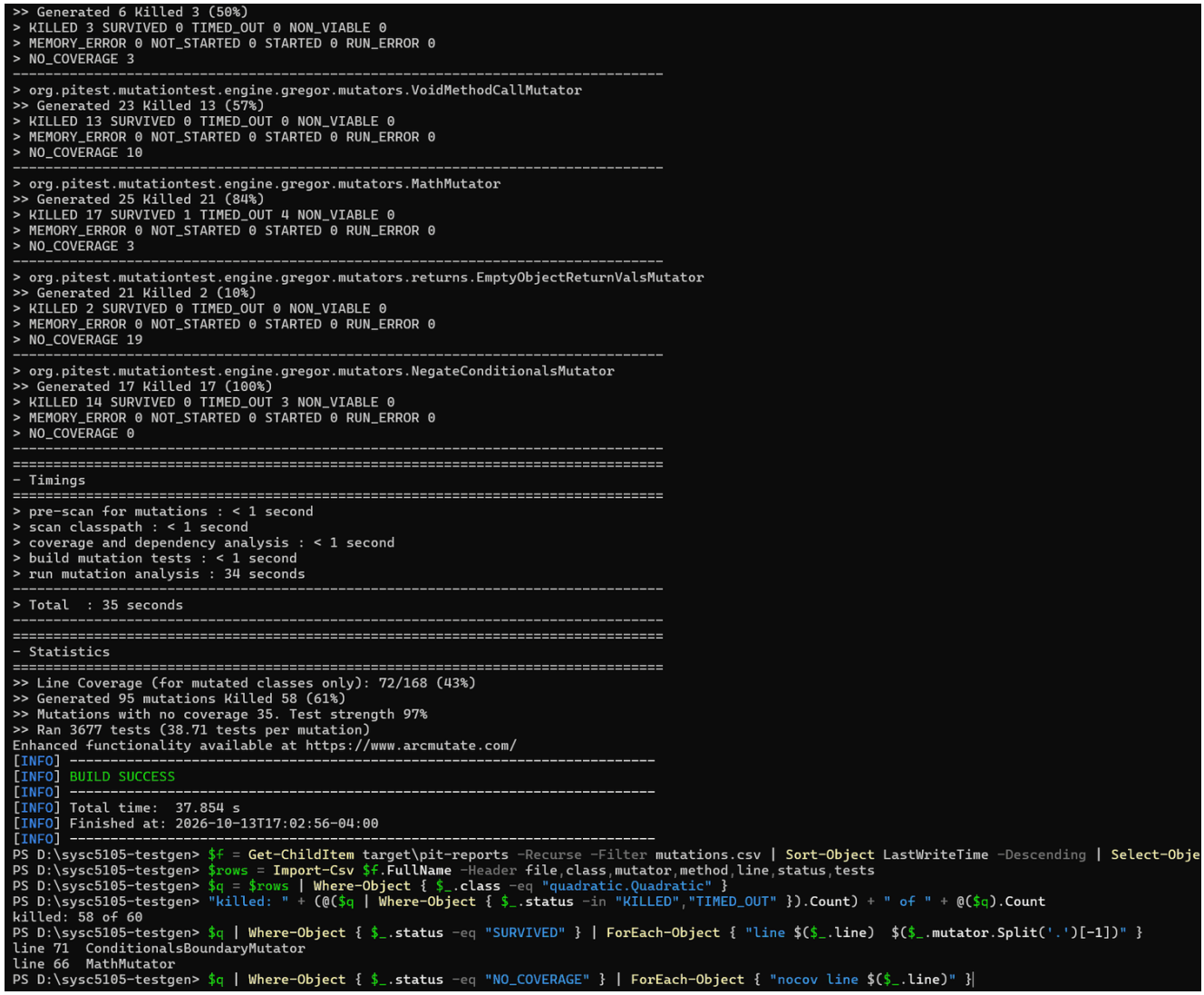


Figure 40. *PIT timings and statistics, then the PowerShell commands that list the surviving mutants.*

Pit Test Coverage Report

Project Summary

Number of Classes	Line Coverage		Mutation Coverage		Test Strength	
2	43%	72/168	61%	58/95	97%	58/60

Breakdown by Package

Name	Number of Classes	Line Coverage		Mutation Coverage		Test Strength	
dates	1	0%	0/95	0%	0/35	100%	0/0
quadratic	1	99%	72/73	97%	58/60	97%	58/60

Report generated by PIT 1.16.1

Enhanced functionality available at arcmutate.com

Figure 41. *PIT report, project summary: quadratic 99% line coverage (72 of 73) and 97% mutation score (58 of 60).*

Pit Test Coverage Report

Package Summary

quadratic

Number of Classes	Line Coverage		Mutation Coverage		Test Strength	
1	99%	72/73	97%	58/60	97%	58/60

Breakdown by Class

Name	Line Coverage		Mutation Coverage		Test Strength	
Quadratic.java	99%	72/73	97%	58/60	97%	58/60

Report generated by PIT 1.16.1

Figure 42. *PIT report, quadratic package: Quadratic.java with 58 of 60 mutants killed.*

A.7 Category-partition re-measured after the fix

Method. The earlier category-partition numbers were taken before the fix and frame F18, so CategoryPartitionTest was measured again on its own.

```
mvn clean test "-Dtest=quadratic.CategoryPartitionTest"
start target\site\jacoco\index.html
mvn org.pitest:pitest-maven:mutationCoverage "-DtargetTests=quadratic.CategoryPartitionTest" "-
DoutputFormats=CSV,HTML"
```

```
PS D:\sysc5105-testgen> mvn clean test "-Dtest=quadratic.CategoryPartitionTest"
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< sysc5105:testgen >-------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24165 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ testgen ---
[INFO] Deleting D:\sysc5105-testgen\target
[INFO]
[INFO] --- jacoco:0.8.12:prepare-agent (default) @ testgen ---
[INFO] argLine set to -javaagent:C:\\Users\\anuja\\.m2\\repository\\org\\jacoco\\org.jacoco.agent\\0.8.12\\org.jacoco.agent-0.8.12-runtime.jar=destfile=D:\\sysc5105-testgen\\target\\ja
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\main\resources
[INFO]
[INFO] --- compiler:3.16.0:compile (default-compile) @ testgen ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\test\resources
[INFO]
[INFO] --- compiler:3.16.0:testCompile (default-testCompile) @ testgen ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 11 source files with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ testgen ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running quadratic.CategoryPartitionTest
[INFO] Tests run: 18, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.390 s -- in quadratic.CategoryPartitionTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 18, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jacoco:0.8.12:report (report) @ testgen ---
[INFO] Loading execution data file D:\sysc5105-testgen\target\jacoco.exec
[INFO] Analyzed bundle 'testgen' with 4 classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  9.117 s
[INFO] Finished at: 2026-10-20T20:05:02-04:00
```

Figure 43. *CategoryPartitionTest alone after tuning: 18 tests, 0 failures.*

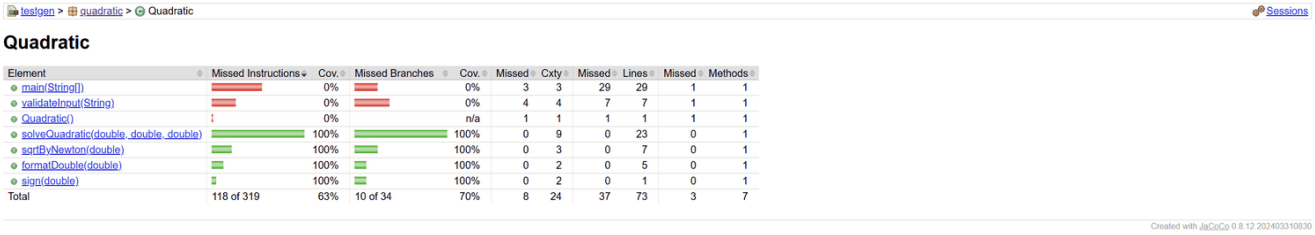


Figure 44. *JaCoCo class page for Quadratic: 63% instruction and 70% branch coverage (unchanged).*

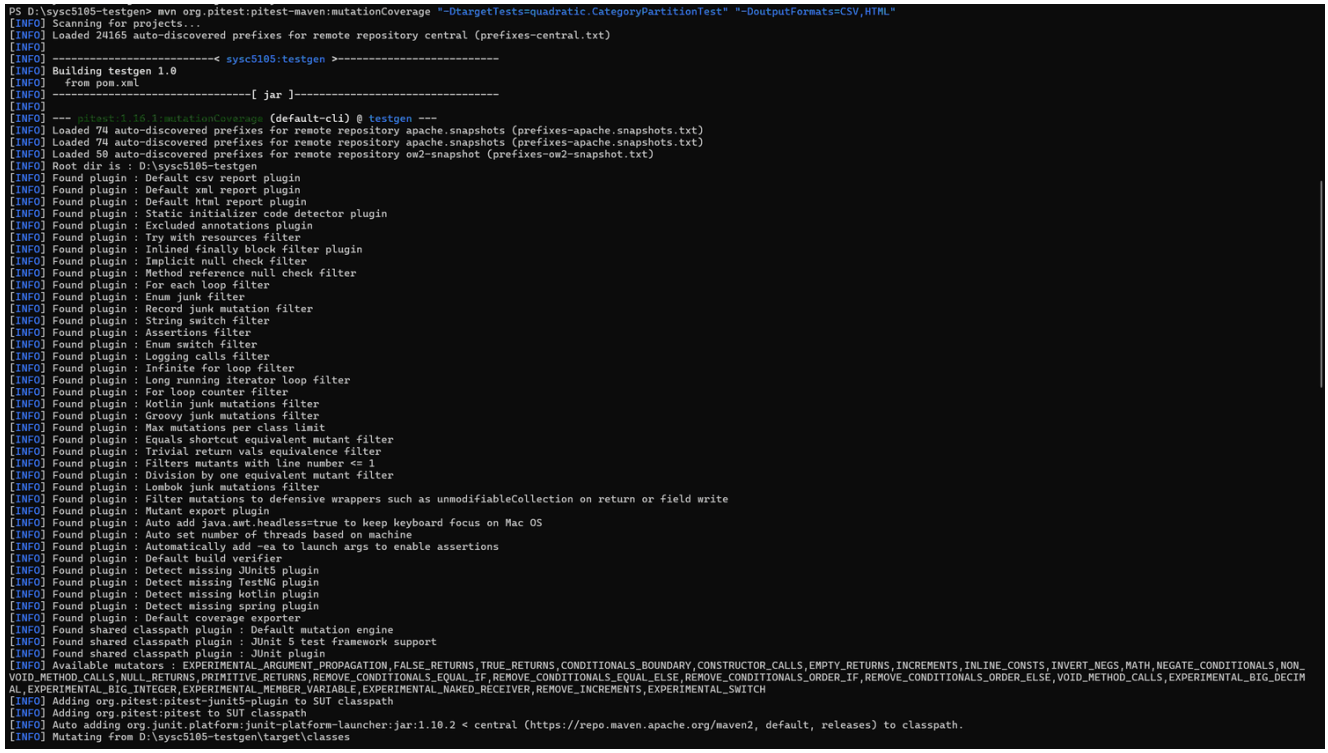


Figure 45. *PIT command and start-up output.*



Figure 46. *PIT mutator summary, timings and statistics.*

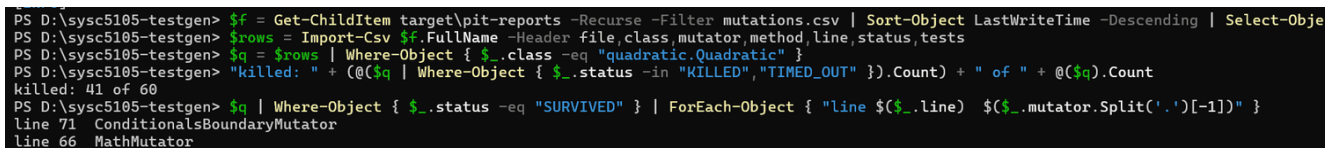


Figure 47. *Survivor script result: 41 of 60 mutants killed; survivors on lines 71 and 66.*

Part B. Date converter

Scope, confirmed by the instructor: only the conversion-related methods are measured (convert, parseDate, getDesiredFormat, getDateOnly, parseAnyDate, DateFormats.getDateFormat): 26 lines, 2 branches and 11 mutants. PIT excludes the other methods with -DexcludedMethods; the JaCoCo numbers are summed per method from jacoco.xml.

B.1 Baseline: one smoke test, scoped

Method. dates.SmokeTest converts one date. JaCoCo counts for the conversion methods are read from target\site\jacoco\jacoco.xml with a PowerShell script; PIT runs with the exclusion list.

```
mvn clean test "-Dtest=dates.SmokeTest"
$ex =
"getToday,getTomorrow,getTodayWithTime,getDaysBetweenTwoDate,getHoursBetweenTwoDate,getMinutesBetweenTw
oDates,getDateAndTime,getTimeOnly,getDateFromDays"
mvn org.pitest:pitest-maven:mutationCoverage "-DtargetTests=dates.SmokeTest" "-DexcludedMethods=$ex" "-
DoutputFormats=CSV,HTML"
```

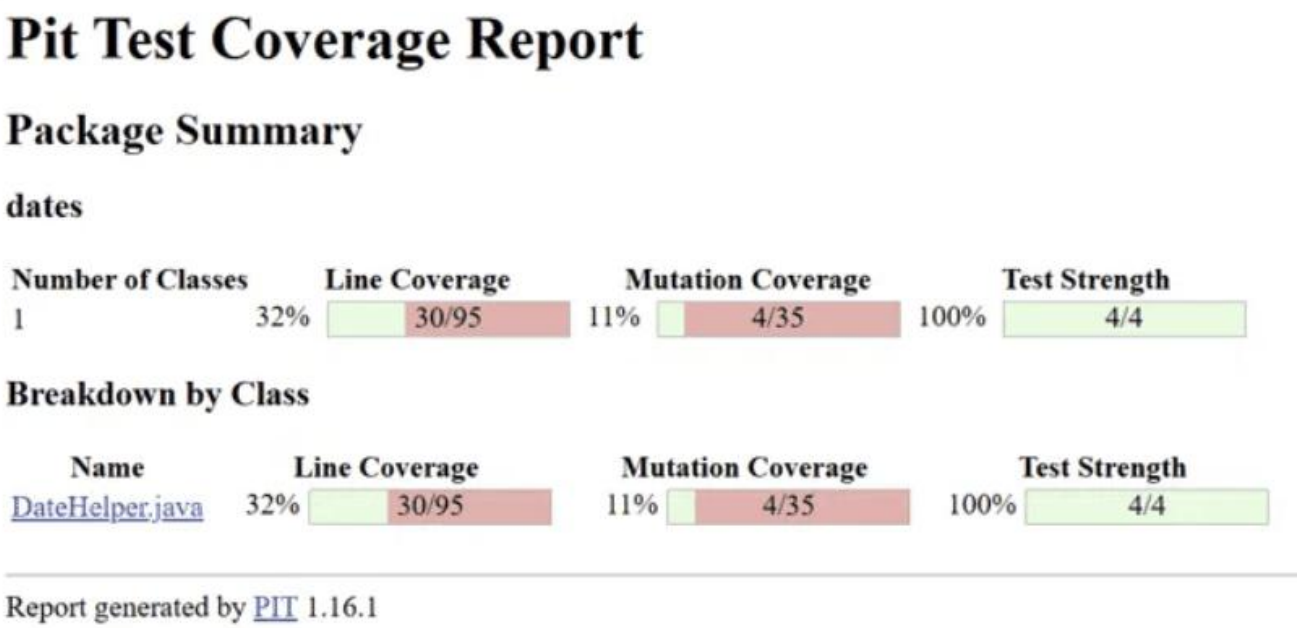
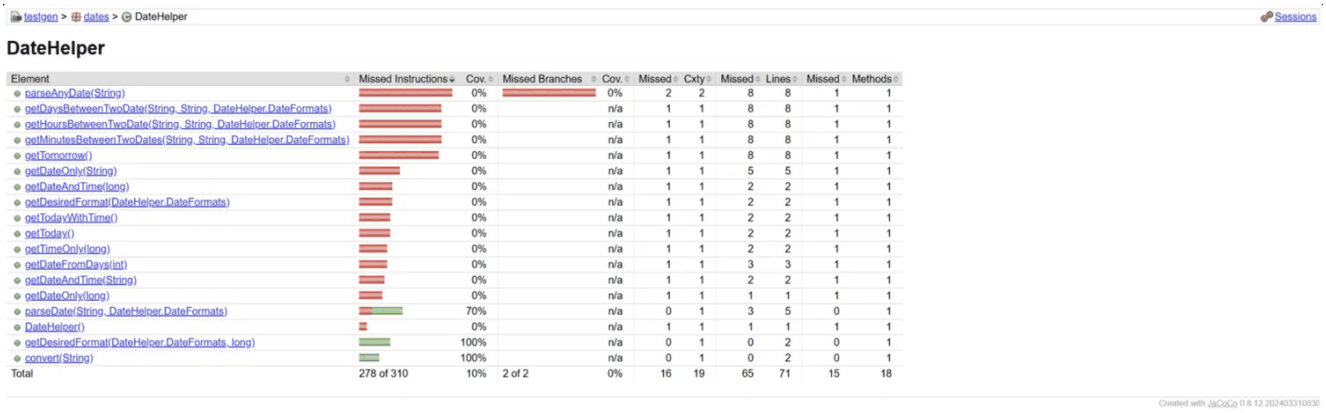


Figure 48. *PIT report, date converter baseline (whole class): 32% line coverage, 11% mutation score (4 of 35 mutants killed).*



testgen > dates > DateHelper — Sessions

DateHelper

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods
parseAnyDate(String)		0%		0%	2	2	8	8	1	1
getDaysBetweenTwoDate(String, String, DateHelper.DateFormats)		0%		n/a	1	1	8	8	1	1
getHoursBetweenTwoDate(String, String, DateHelper.DateFormats)		0%		n/a	1	1	8	8	1	1
getMinutesBetweenTwoDates(String, String, DateHelper.DateFormats)		0%		n/a	1	1	8	8	1	1
getTomorrow()		0%		n/a	1	1	8	8	1	1
getDateOnly(String)		0%		n/a	1	1	5	5	1	1
getDateAndTime(long)		0%		n/a	1	1	2	2	1	1
getDesiredFormat(DateHelper.DateFormats)		0%		n/a	1	1	2	2	1	1
getTodayWithTime()		0%		n/a	1	1	2	2	1	1
getToday()		0%		n/a	1	1	2	2	1	1
getTimeOnly(long)		0%		n/a	1	1	2	2	1	1
getDateFromDays(int)		0%		n/a	1	1	3	3	1	1
getDateAndTime(String)		0%		n/a	1	1	2	2	1	1
getDateOnly(long)		0%		n/a	1	1	1	1	1	1
parseDate(String, DateHelper.DateFormats)		70%		n/a	0	1	3	5	0	1
DateHelper()		0%		n/a	1	1	1	1	1	1
getDesiredFormat(DateHelper.DateFormats, long)		100%		n/a	0	1	0	2	0	1
convert(String)		100%		n/a	0	1	0	2	0	1
Total	278 of 310	10%	2 of 2	0%	16	19	65	71	15	18

Created with JaCoCo 0.8.12.202403310830

Figure 49. *JaCoCo class page for DateHelper, baseline: 10% instruction coverage, 0% branch coverage.*

```
PS C:\Windows\System32> cd D:\
PS D:\> cd sysc5105-testgen
PS D:\sysc5105-testgen> mvn clean test "-Dtest=dates.SmokeTest"
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< sysc5105:testgen >------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24119 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ testgen ---
[INFO] Deleting D:\sysc5105-testgen\target
[INFO]
[INFO] --- jacoco:0.8.12:prepare-agent (default) @ testgen ---
[INFO] argLine set to -javaagent:C:\\Users\\anuja\\.m2\\repository\\org\\jacoco\\org.jacoco.agent\\0.8.12\\org.jacoco.agent-0.8.12-runtime.jar=destfile=D:\\sysc5105-testgen\\target\\ja
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\main\resources
[INFO]
[INFO] --- compiler:3.16.0:compile (default-compile) @ testgen ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\test\resources
[INFO]
[INFO] --- compiler:3.16.0:testCompile (default-testCompile) @ testgen ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 7 source files with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ testgen ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running dates.SmokeTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.050 s -- in dates.SmokeTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jacoco:0.8.12:report (report) @ testgen ---
[INFO] Loading execution data file D:\sysc5105-testgen\target\jacoco.exec
[INFO] Analyzed bundle 'testgen' with 4 classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.944 s
[INFO] Finished at: 2026-10-16T20:03:50-04:00
[INFO] ------------------------------------------------------------------------
PS D:\sysc5105-testgen>
```

Figure 50. *Smoke test for the dates: 1 test, 0 failures.*

```
PS D:\sysc5105-testgen> $t = (Get-Content target\site\jacoco\jacoco.xml -Raw) -replace '<!DOCTYPE[^>]*>',''
PS D:\sysc5105-testgen> $x = [xml]$t
PS D:\sysc5105-testgen> $scope = "convert","parseDate","getDesiredFormat","getDateOnly","parseAnyDate","getDateFormat"
PS D:\sysc5105-testgen> $ms = foreach ($c in $x.report.package.class) { if ($c.name -like "dates/DateHelper*") { foreach ($m in $c.method) { if ($scope -contains $m.name) { $m } } } }
PS D:\sysc5105-testgen> foreach ($type in "LINE","BRANCH","INSTRUCTION") {
>>   $cov = 0; $mis = 0
>>   foreach ($m in $ms) { foreach ($k in $m.counter) { if ($k.type -eq $type) { $cov += [int]$k.covered; $mis += [int]$k.missed } } }
>>   "$type covered $cov missed $mis"
>> }
LINE covered 7 missed 19
BRANCH covered 0 missed 2
INSTRUCTION covered 35 missed 79
PS D:\sysc5105-testgen>
```

Figure 51. *Scoped JaCoCo numbers for the conversion methods: lines 7 covered and 19 missed, branches 0 of 2, instructions 35 covered and 79 missed.*

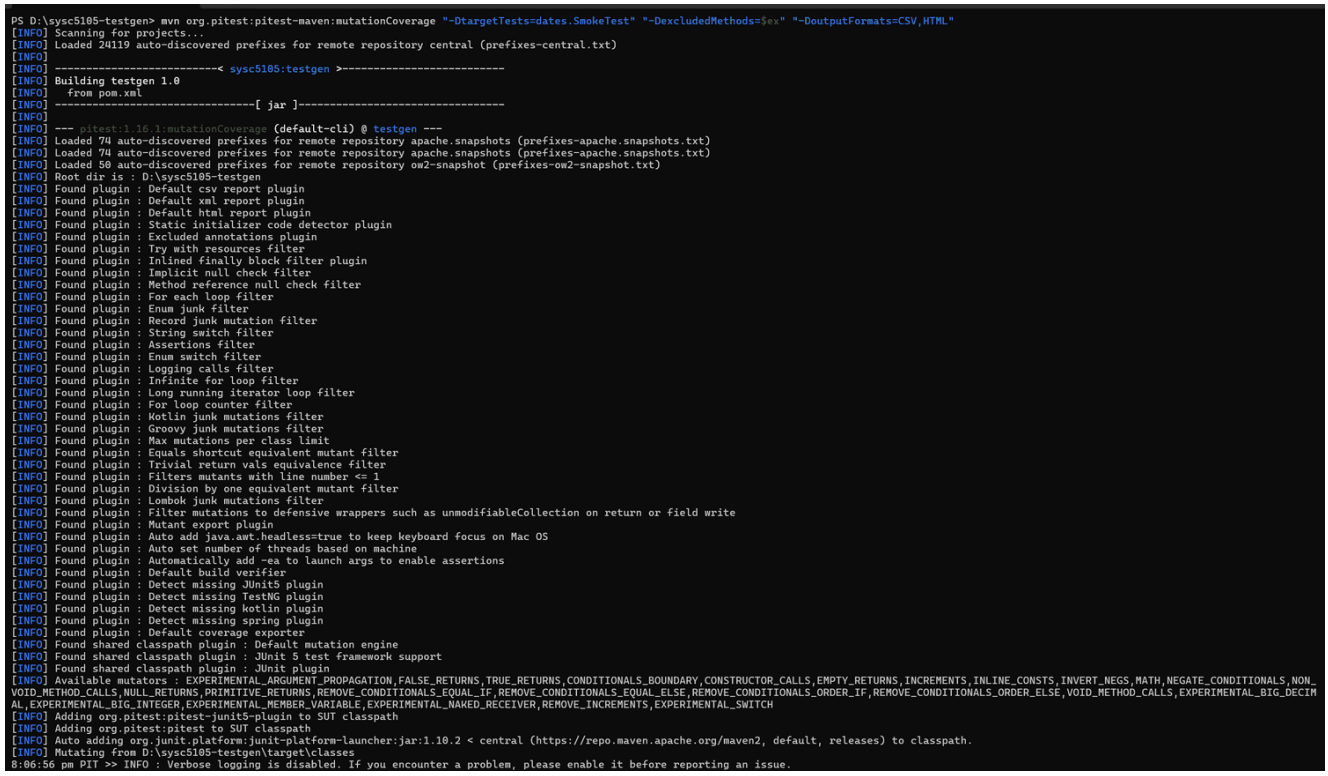


Figure 52. *PIT command with the excluded-methods list and start-up output.*



Figure 53. *PIT mutator summary, timings and statistics (scoped baseline: 4 of 11 date mutants killed).*

B.2 Combinatorial testing (PICT, pairwise and 3-way)

Method. PICT model testdata\dates.pict (year, month, day, separator) gives 50 pairwise and 294 3-way tests (exhaustive would be 588). The oracle is java.time.LocalDate: valid dates must convert exactly, invalid dates should be rejected; what DateHelper does instead is recorded as defects DT1 and DT2.

```
D:\pict\pict.exe testdata\dates.pict /o:2 /r:42 | Out-File -Encoding ascii testdata\dates_pairwise.tsv
D:\pict\pict.exe testdata\dates.pict /o:3 /r:42 | Out-File -Encoding ascii testdata\dates_3way.tsv
mvn clean test "-Dtest=dates.DatesCombinatorialTest"
mvn clean test "-Dtest=dates.DatesCombinatorialTest" "-Dstrict=true"
Copy-Item results\findings_dates_comb.txt results\findings_dates_comb_run7.txt -Force
mvn org.pitest:pitest-maven:mutationCoverage "-DtargetTests=dates.DatesCombinatorialTest" "-
DexcludedMethods=$ex" "-DoutputFormats=CSV,HTML"
```

```
PS D:\sysc5105-testgen> D:\pict\pict.exe testdata\dates.pict /o:2 /r:42 | Out-File -Encoding ascii testdata\dates_pairwise.tsv
Used seed: 42
PS D:\sysc5105-testgen> D:\pict\pict.exe testdata\dates.pict /o:3 /r:42 | Out-File -Encoding ascii testdata\dates_3way.tsv
Used seed: 42
PS D:\sysc5105-testgen> (Get-Content testdata\dates_pairwise.tsv).Count - 1
50
PS D:\sysc5105-testgen> (Get-Content testdata\dates_3way.tsv).Count - 1
294
PS D:\sysc5105-testgen> Get-Content testdata\dates_pairwise.tsv -TotalCount 6
Year    Month   Day     Sep
1970    13      28      -
1900    01      32      /
2024    13      00      /
2024    12      31      -
2100    02      30      /
PS D:\sysc5105-testgen>
```

Figure 54. *PICT generation of the date suites (pairwise and 3-way, seed 42) and the test counts: 50 pairwise and 294 3-way tests.*

```
PS D:\sysc5105-testgen> mvn clean test "-Dtest=dates.DatesCombinatorialTest"
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< sysc5105:testgen >-------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24119 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ testgen ---
[INFO] Deleting D:\sysc5105-testgen\target
[INFO]
[INFO] --- jacoco:0.8.12:prepare-agent (default) @ testgen ---
[INFO] argLine set to -javaagent:C:\\Users\\anuja\\.m2\\repository\\org\\jacoco\\org.jacoco.agent\\0.8.12\\org.jacoco.agent-0.8.12-runtime.jar=destfile=D:\\sysc5105-testgen\\target\\ja
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\main\resources
[INFO]
[INFO] --- compiler:3.16.0:compile (default-compile) @ testgen ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\test\resources
[INFO]
[INFO] --- compiler:3.16.0:testCompile (default-testCompile) @ testgen ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 8 source files with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ testgen ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running dates.DatesCombinatorialTest
[INFO] Tests run: 344, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.502 s -- in dates.DatesCombinatorialTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 344, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jacoco:0.8.12:report (report) @ testgen ---
[INFO] Loading execution data file D:\sysc5105-testgen\target\jacoco.exec
[INFO] Analyzed bundle 'testgen' with 4 classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.430 s
[INFO] Finished at: 2026-10-16T20:09:46-04:00
[INFO] ------------------------------------------------------------------------
PS D:\sysc5105-testgen>
```

Figure 55. *DatesCombinatorialTest, normal mode: 344 tests, 0 failures.*

```
PS D:\sysc5105-testgen> mvn clean test "-Dtest=dates.DatesCombinatorialTest" "-Dstrict=true"
[INFO] Scanning for projects...
[INFO]
[INFO] ---------------------------< sysc5105:testgen >---------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24119 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ testgen ---
[INFO] Deleting D:\sysc5105-testgen\target
[INFO]
[INFO] --- jacoco:0.8.12:prepare-agent (default) @ testgen ---
[INFO] argLine set to -javaagent:C:\\Users\\anuja\\.m2\\repository\\org\\jacoco\\org.jacoco.agent\\0.8.12\\org.jacoco.agent-0.8.12-runtime.jar=destfile=D:\\sysc5105-testgen\\target\\ja
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\main\resources
[INFO]
[INFO] --- compiler:3.16.0:compile (default-compile) @ testgen ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\test\resources
[INFO]
[INFO] --- compiler:3.16.0:testCompile (default-testCompile) @ testgen ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 8 source files with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ testgen ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running dates.DatesCombinatorialTest
[ERROR] Tests run: 344, Failures: 272, Errors: 0, Skipped: 0, Time elapsed: 0.699 s <<< FAILURE! -- in dates.DatesCombinatorialTest
[ERROR] dates.DatesCombinatorialTest.threeWaySuite(String, String, String, String)[1] -- Time elapsed: 0.089 s <<< FAILURE!
org.opentest4j.AssertionFailedError: DT2: unparsable input '1900/02/31' silently becomes the epoch date 31/12/1969
        at org.junit.jupiter.api.AssertionUtils.fail(AssertionUtils.java:38)
        at org.junit.jupiter.api.Assertions.fail(Assertions.java:138)
        at dates.DatesCombinatorialTest.known(DatesCombinatorialTest.java:72)
        at dates.DatesCombinatorialTest.runCase(DatesCombinatorialTest.java:96)
        at dates.DatesCombinatorialTest.threeWaySuite(DatesCombinatorialTest.java:67)
        at java.base/java.lang.reflect.Method.invoke(Method.java:569)
        at java.base/java.util.stream.ForEachOps$ForEachOp$OfRef.accept(ForEachOps.java:183)
        at java.base/java.util.stream.ReferencePipeline$3$1.accept(ReferencePipeline.java:197)
        at java.base/java.util.stream.ReferencePipeline$2$1.accept(ReferencePipeline.java:179)
        at java.base/java.util.stream.ReferencePipeline$3$1.accept(ReferencePipeline.java:197)
        at java.base/java.util.stream.ForEachOps$ForEachOp$OfRef.accept(ForEachOps.java:183)
        at java.base/java.util.stream.ReferencePipeline$3$1.accept(ReferencePipeline.java:197)
        at java.base/java.util.stream.ReferencePipeline$3$1.accept(ReferencePipeline.java:197)
        at java.base/java.util.stream.ReferencePipeline$3$1.accept(ReferencePipeline.java:197)
        at java.base/java.util.stream.ForEachOps$ForEachOp$OfRef.accept(ForEachOps.java:183)
        at java.base/java.util.stream.ReferencePipeline$3$1.accept(ReferencePipeline.java:197)
        at java.base/java.util.stream.ForEachOps$ForEachOp$OfRef.accept(ForEachOps.java:183)
        at java.base/java.util.stream.ReferencePipeline$3$1.accept(ReferencePipeline.java:197)
        at java.base/java.util.stream.ReferencePipeline$3$1.accept(ReferencePipeline.java:197)
        at java.base/java.util.stream.ReferencePipeline$2$1.accept(ReferencePipeline.java:179)
        at java.base/java.util.stream.SliceOps$1$1.accept(SliceOps.java:200)
        at java.base/java.util.ArrayList$ArrayListSpliterator.forEachRemaining(ArrayList.java:1625)
        at java.base/java.util.stream.AbstractPipeline.copyInto(AbstractPipeline.java:509)
```

Figure 56. *Strict mode: 272 of the 344 tests fail (invalid dates silently converted).*

```
PS D:\sysc5105-testgen> mvn org.pitest:pitest-maven:mutationCoverage "-DtargetTests=dates.DatesCombinatorialTest" "-DexcludedMethods=$ex" "-DoutputFormats=CSV,HTML"
[INFO] Scanning for projects...
[INFO] Loaded 24119 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO]
[INFO] ---------------------------< sysc5105:testgen >---------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- pitest:1.16.1:mutationCoverage (default-cli) @ testgen ---
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO] Loaded 50 auto-discovered prefixes for remote repository ow2-snapshot (prefixes-ow2-snapshot.txt)
[INFO] Root dir is : D:\sysc5105-testgen
[INFO] Found plugin : Default csv report plugin
[INFO] Found plugin : Default xml report plugin
[INFO] Found plugin : Default html report plugin
[INFO] Found plugin : Static initializer code detector plugin
[INFO] Found plugin : Excluded annotations plugin
[INFO] Found plugin : Try with resources filter
[INFO] Found plugin : Inlined finally block filter plugin
[INFO] Found plugin : Implicit null check filter
[INFO] Found plugin : Method reference null check filter
[INFO] Found plugin : For each loop filter
[INFO] Found plugin : Enum junk filter
[INFO] Found plugin : Record junk mutation filter
[INFO] Found plugin : String switch filter
[INFO] Found plugin : Assertions filter
[INFO] Found plugin : Enum switch filter
[INFO] Found plugin : Logging calls filter
[INFO] Found plugin : Infinite for loop filter
[INFO] Found plugin : Long running iterator loop filter
[INFO] Found plugin : For loop counter filter
[INFO] Found plugin : Kotlin junk mutations filter
[INFO] Found plugin : Groovy junk mutations filter
[INFO] Found plugin : Max mutations per class limit
[INFO] Found plugin : Equals shortcut equivalent mutant filter
[INFO] Found plugin : Trivial return vals equivalence filter
[INFO] Found plugin : Filters mutants with line number <= 1
[INFO] Found plugin : Division by one equivalent mutant filter
[INFO] Found plugin : Lombok junk mutations filter
[INFO] Found plugin : Filter mutations to defensive wrappers such as unmodifiableCollection on return or field write
[INFO] Found plugin : Mutant export plugin
[INFO] Found plugin : Auto add java.awt.headless=true to keep keyboard focus on Mac OS
[INFO] Found plugin : Auto set number of threads based on machine
[INFO] Found plugin : Automatically add -ea to launch args to enable assertions
[INFO] Found plugin : Default build verifier
[INFO] Found plugin : Detect missing JUnit5 plugin
[INFO] Found plugin : Detect missing TestNG plugin
[INFO] Found plugin : Detect missing kotlin plugin
[INFO] Found plugin : Detect missing spring plugin
[INFO] Found plugin : Default coverage exporter
[INFO] Found shared classpath plugin : Default mutation engine
[INFO] Found shared classpath plugin : JUnit 5 test framework support
[INFO] Found shared classpath plugin : JUnit plugin
[INFO] Available mutators : EXPERIMENTAL_ARGUMENT_PROPAGATION,FALSE_RETURNS,TRUE_RETURNS,CONDITIONALS_BOUNDARY,CONSTRUCTOR_CALLS,EMPTY_RETURNS,INCREMENTS,INLINE_CONSTS,INVERT_NEGS,MATH,NEGATE_CONDITIONALS,NON_
VOID_METHOD_CALLS,NULL_RETURNS,PRIMITIVE_RETURNS,REMOVE_CONDITIONALS_EQUAL_IF,REMOVE_CONDITIONALS_EQUAL_ELSE,REMOVE_CONDITIONALS_ORDER_IF,REMOVE_CONDITIONALS_ORDER_ELSE,VOID_METHOD_CALLS,EXPERIMENTAL_BIG_DECIM
AL,EXPERIMENTAL_BIG_INTEGER,EXPERIMENTAL_MEMBER_VARIABLE,EXPERIMENTAL_NAKED_RECEIVER,REMOVE_INCREMENTS,EXPERIMENTAL_SWITCH
[INFO] Adding org.pitest:pitest-junit5-plugin to SUT classpath
[INFO] Adding org.pitest:pitest to SUT classpath
[INFO] Auto adding org.junit.platform:junit-platform-launcher:jar:1.10.2 < central (https://repo.maven.apache.org/maven2, default, releases) to classpath.
[INFO] Mutating from D:\sysc5105-testgen\target\classes
8:11:22 pm PIT >> INFO : Verbose logging is disabled. If you encounter a problem, please enable it before reporting an issue.
```

Figure 57. *PIT command and start-up output. (first PIT run)*

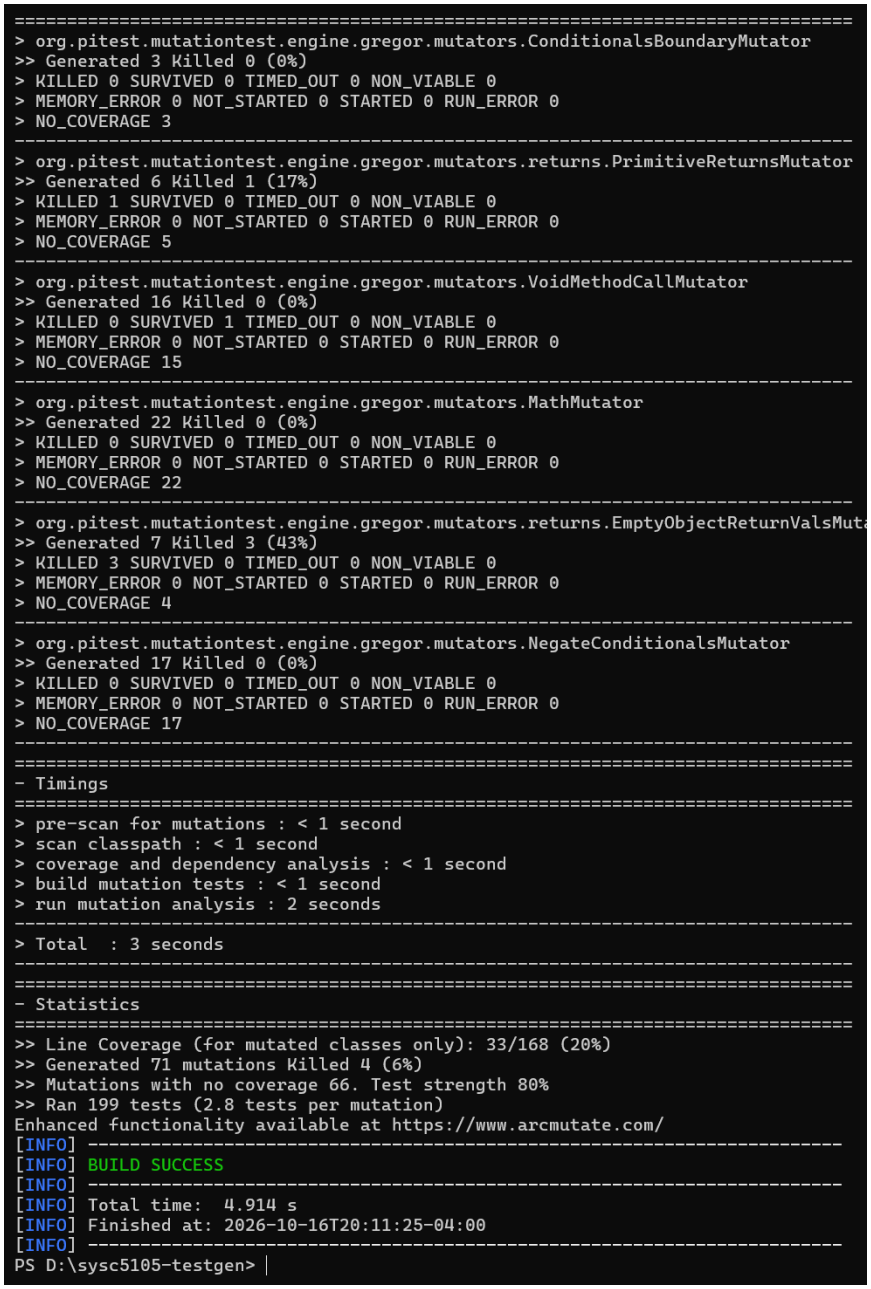


Figure 58. *PIT mutator summary, timings and statistics (first PIT run).*

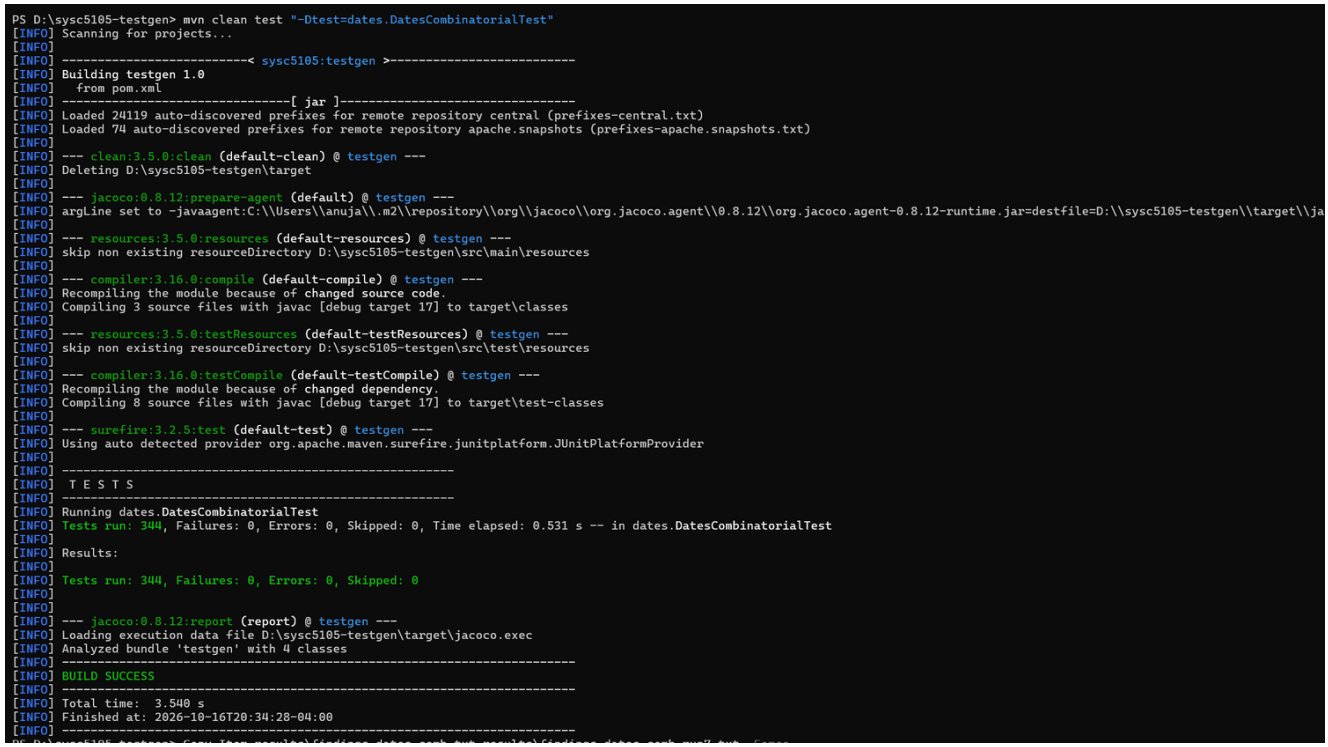


Figure 59. *Normal run repeated before the findings were saved.*

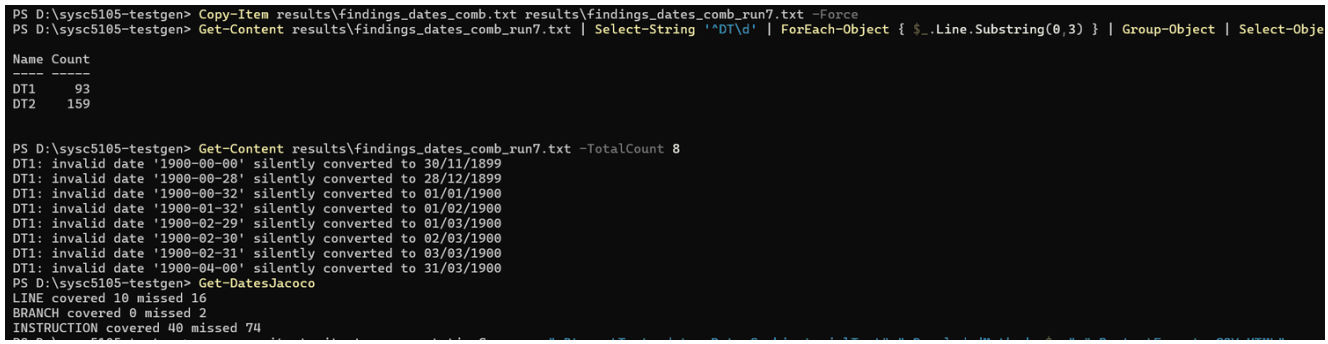


Figure 60. *Findings saved and counted: DT1 = 93, DT2 = 159, with the first example lines and the scoped JaCoCo numbers (10 of 26 lines).*

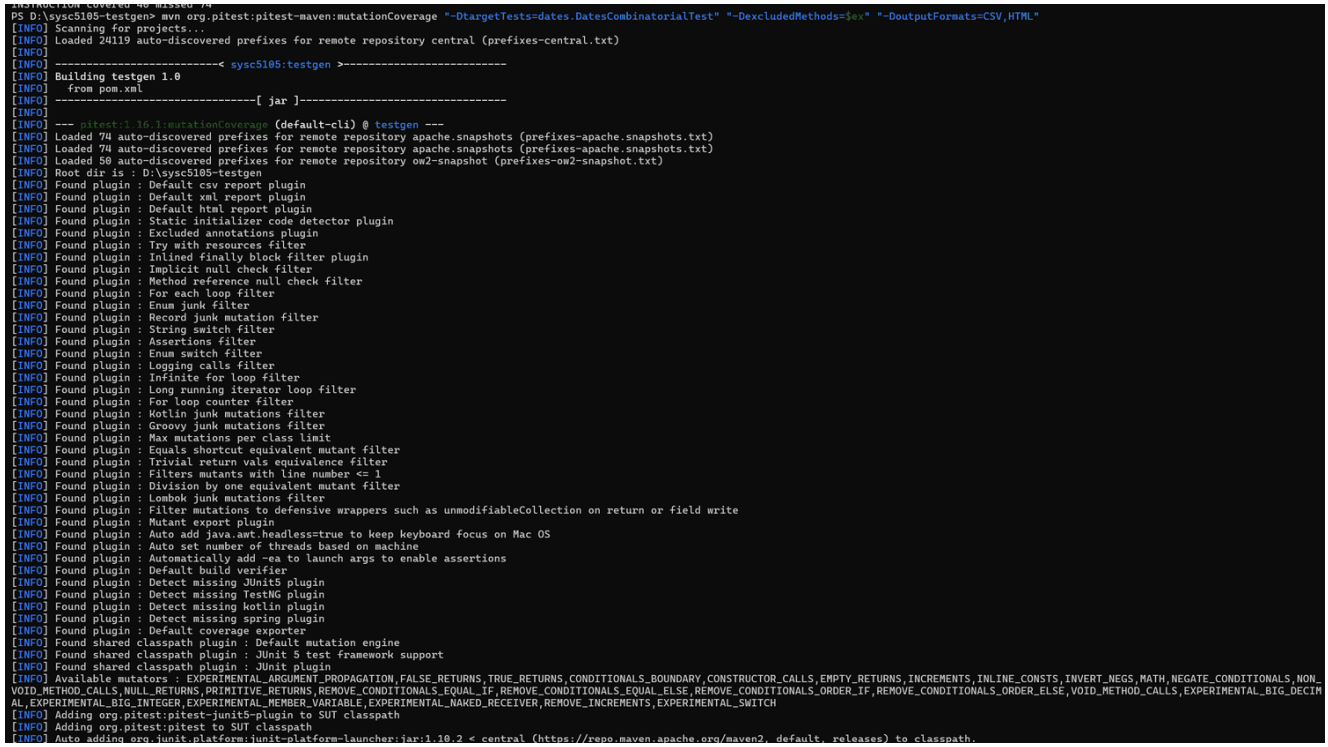


Figure 61. *PIT command and start-up output. (repeat run)*

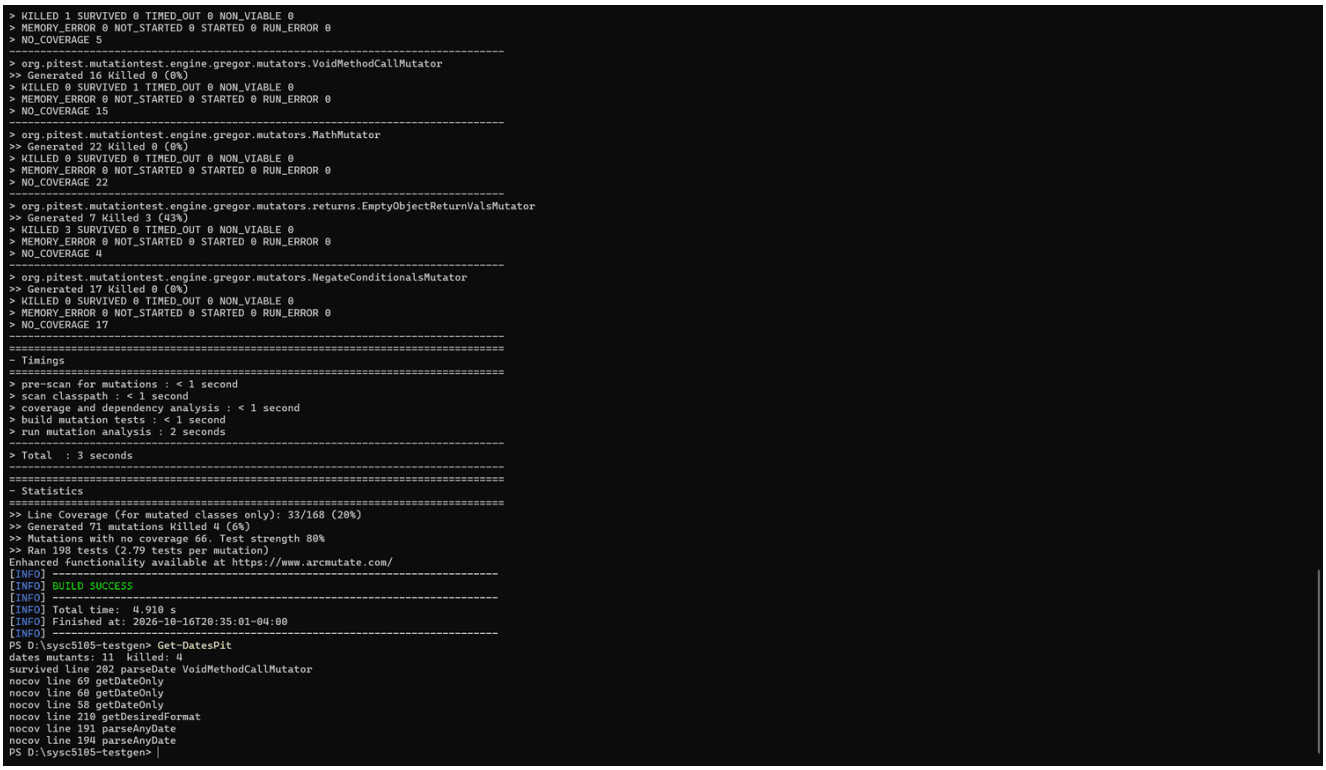


Figure 62. *PIT timings and statistics, then the scoped result: 4 of 11 mutants killed; the survivor is line 202 of parseDate.*

B.3 Category-partition testing (28 frames and a null test)

Method. Frames from testdata\dates_cp_frames.csv: 12 valid dates (leap days, month ends, boundaries) with hand-derived output, checked against java.time, and 16 invalid inputs (impossible dates, bad months and days, malformed strings), plus a convert(null) test. 29 tests.

```
mvn clean test "-Dtest=dates.DatesCategoryPartitionTest"
mvn clean test "-Dtest=dates.DatesCategoryPartitionTest" "-Dstrict=true"
Copy-Item results\findings_dates_catpart.txt results\findings_dates_catpart_run9.txt -Force
mvn org.pitest:pitest-maven:mutationCoverage "-DtargetTests=dates.DatesCategoryPartitionTest" "-
DexcludedMethods=$ex" "-DoutputFormats=CSV,HTML"
```

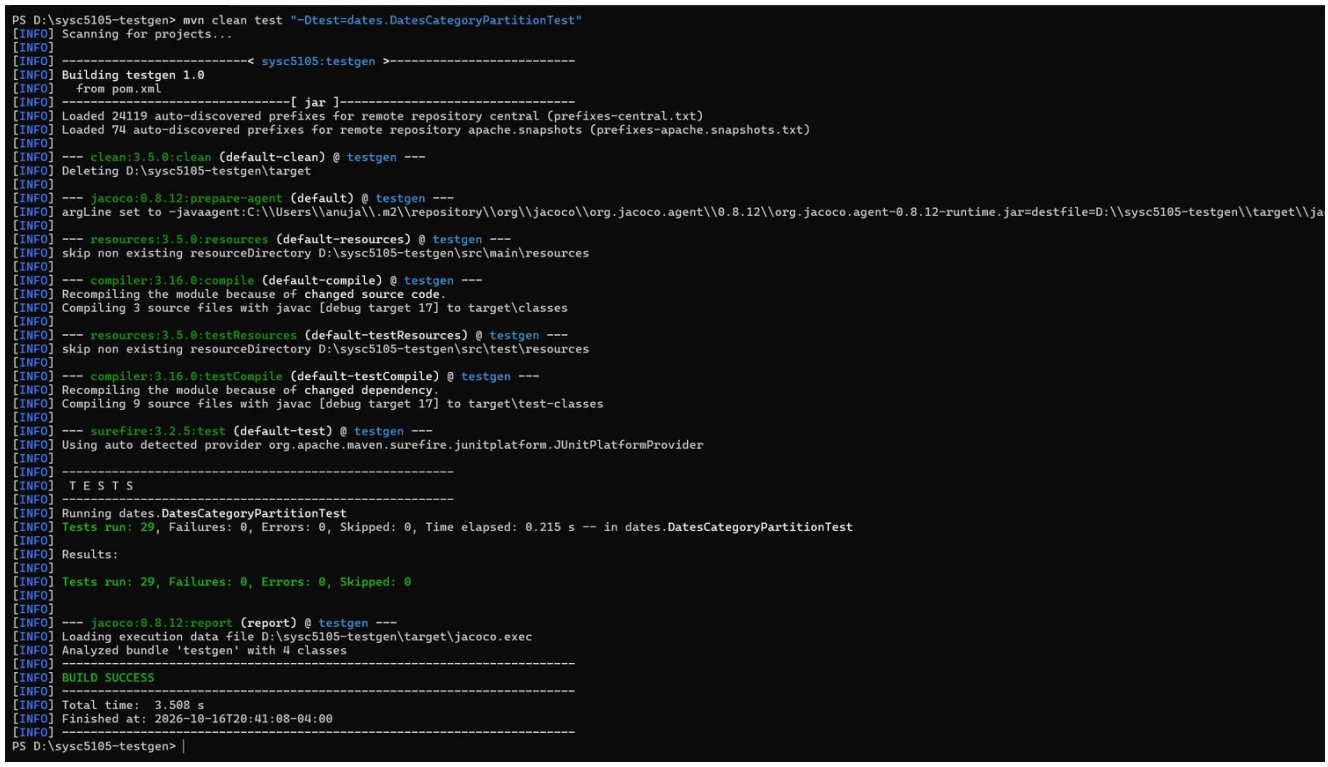


Figure 63. *DatesCategoryPartitionTest, normal mode: 29 tests, 0 failures.*

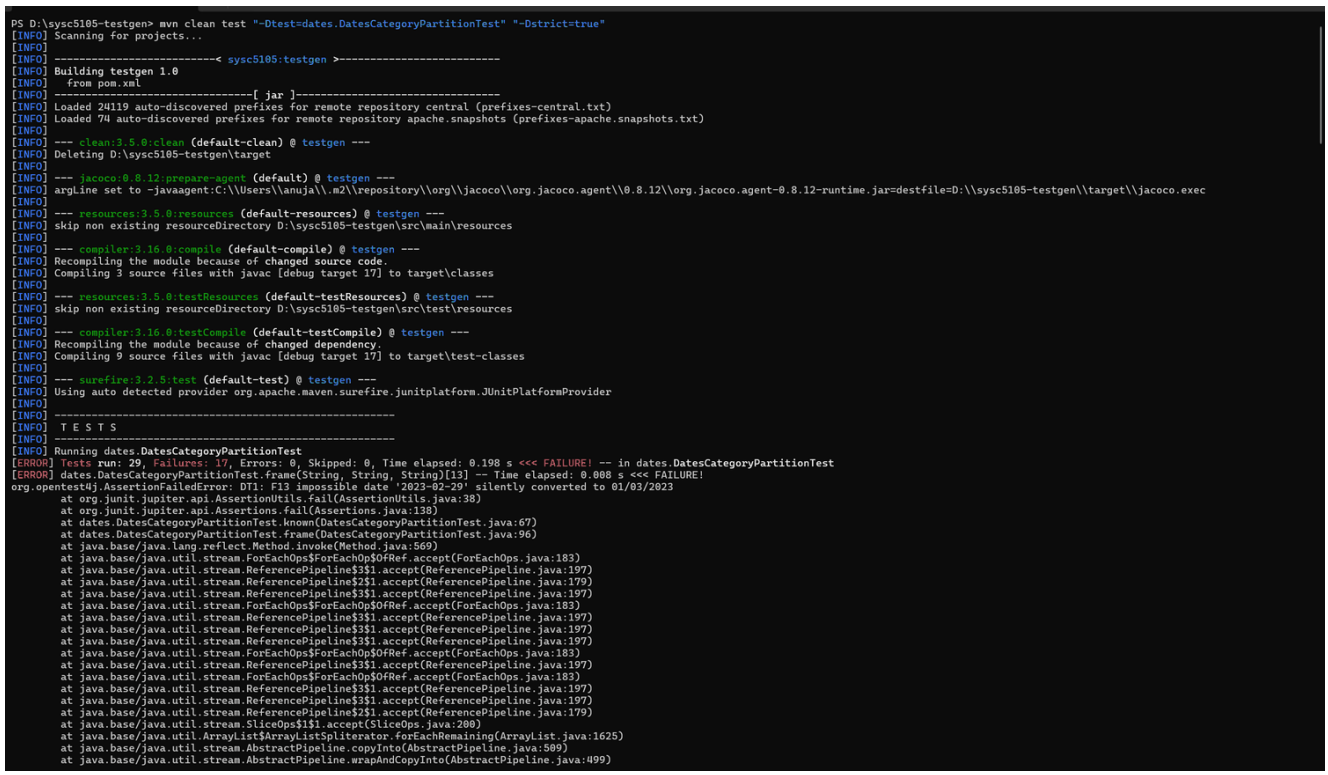


Figure 64. *Strict mode: 17 of the 29 tests fail.*

```
PS D:\sysc5105-testgen> Copy-Item results\findings_dates_catpart.txt results\findings_dates_catpart_run9.txt -Force
PS D:\sysc5105-testgen> Get-Content results\findings_dates_catpart_run9.txt | Select-String '^DT\d' | ForEach-Object { $_.Line.Substring(0,3) } | Group-Object | Select-Obje

Name Count
---- -----
DT1      9
DT2      3
DT3      4
DT4      1


PS D:\sysc5105-testgen> Get-Content results\findings_dates_catpart_run9.txt
DT1: F13 impossible date '2023-02-29' silently converted to 01/03/2023
DT1: F14 impossible date '1900-02-29' silently converted to 01/03/1900
DT1: F15 impossible date '2100-02-29' silently converted to 01/03/2100
DT1: F16 impossible date '2023-02-30' silently converted to 02/03/2023
DT1: F17 impossible date '2023-04-31' silently converted to 01/05/2023
DT1: F18 impossible date '2023-13-01' silently converted to 01/01/2024
DT1: F19 impossible date '2023-00-10' silently converted to 10/12/2022
DT1: F20 impossible date '2023-01-00' silently converted to 31/12/2022
DT1: F21 impossible date '2023-01-32' silently converted to 01/02/2023
DT2: F22 unparsable input '2023/12/25' silently becomes the epoch date 31/12/1969
DT2: F23 unparsable input '' silently becomes the epoch date 31/12/1969
DT2: F24 unparsable input 'abc' silently becomes the epoch date 31/12/1969
DT3: F25 malformed input '2023-1-5' accepted and converted to 05/01/2023
DT3: F26 malformed input '2023-12-25abc' accepted and converted to 25/12/2023
DT3: F27 malformed input '23-12-25' accepted and converted to 25/12/0023
DT3: F28 malformed input '2023-12-25T10:00' accepted and converted to 25/12/2023
DT4: convert(null) throws NullPointerException
PS D:\sysc5105-testgen> Get-DatesJacoco
LINE covered 10 missed 16
BRANCH covered 0 missed 2
INSTRUCTION covered 40 missed 74
```

Figure 65. *Findings saved and counted: DT1 = 9, DT2 = 3, DT3 = 4, DT4 = 1, with the full list and the scoped JaCoCo numbers.*

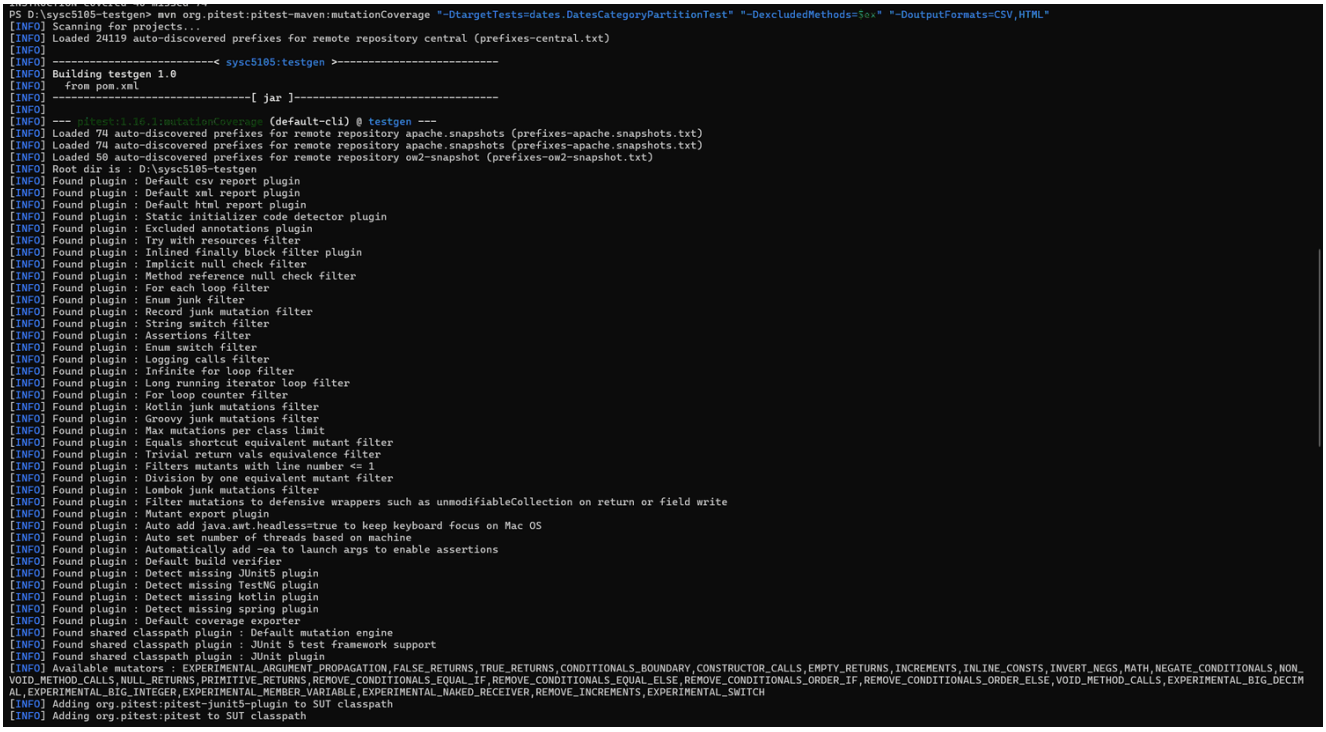


Figure 66. *PIT command and start-up output.*

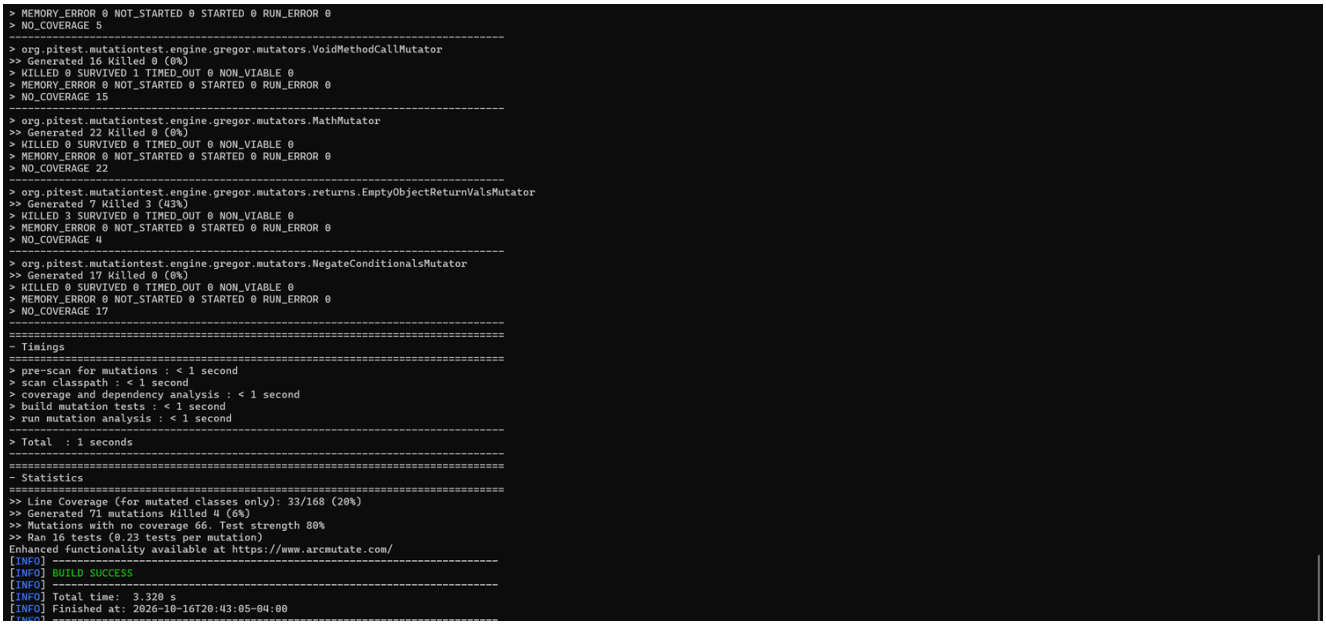


Figure 67. *PIT mutator summary, timings and statistics.*

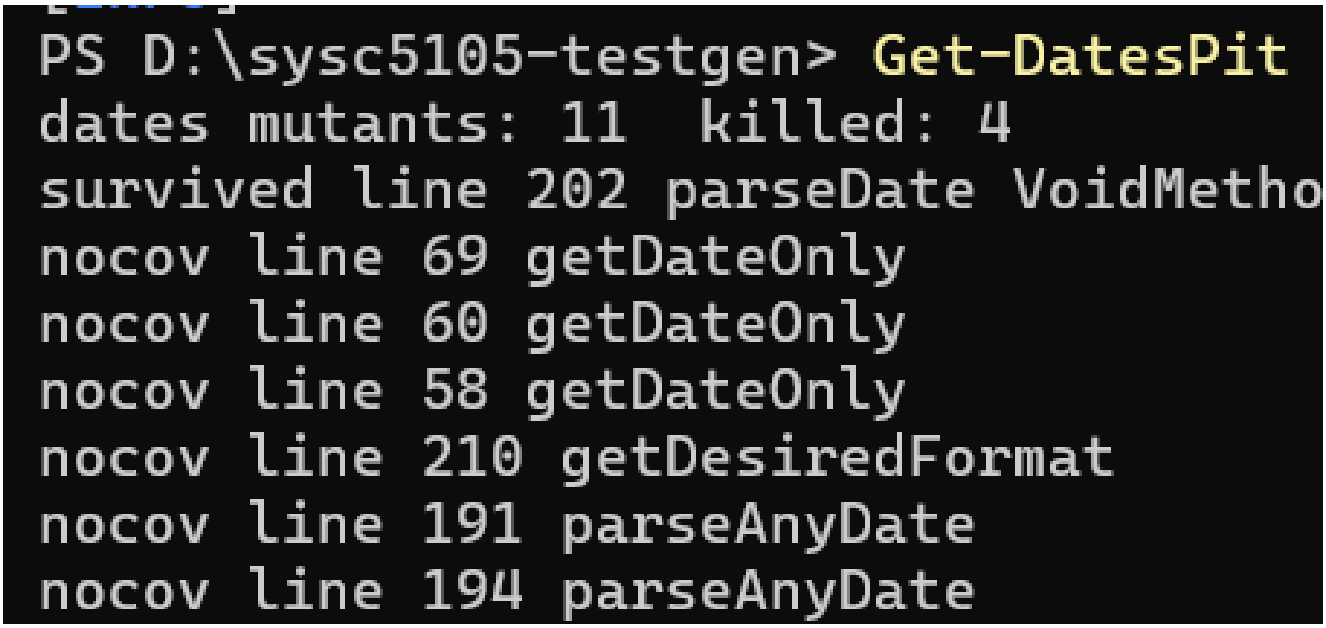


Figure 68. *Scoped result: 4 of 11 mutants killed; survivor on line 202, six mutants uncovered.*

B.4 Metamorphic testing

Method. DatesMetamorphicTest has no stored expected outputs. 284 valid dates are converted and five relations are checked: round trip, day-shift consistency, order preservation, agreement between two code paths, and invariance under time zone and locale changes. 856 tests.

```
mvn clean test "-Dtest=dates.DatesMetamorphicTest"
mvn clean test "-Dtest=dates.DatesMetamorphicTest" "-Dstrict=true"
Copy-Item results\findings_dates_meta.txt results\findings_dates_meta_run10.txt -Force
mvn org.pitest:pitest-maven:mutationCoverage "-DtargetTests=dates.DatesMetamorphicTest" "-
DexcludedMethods=$ex" "-DoutputFormats=CSV,HTML"
```

```
PS D:\sysc5105-testgen> mvn clean test "-Dtest=dates.DatesMetamorphicTest"
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------------< sysc5105:testgen >----------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24119 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ testgen ---
[INFO] Deleting D:\sysc5105-testgen\target
[INFO]
[INFO] --- jacoco:0.8.12:prepare-agent (default) @ testgen ---
[INFO] argLine set to -javaagent:C:\\Users\\anuja\\.m2\\repository\\org\\jacoco\\org.jacoco.agent\\0.8.12\\org.jacoco.agent-0.8.12-runtime.jar=destfile=D:\\sysc5105-testgen\\target\\ja
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\main\resources
[INFO]
[INFO] --- compiler:3.16.0:compile (default-compile) @ testgen ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\test\resources
[INFO]
[INFO] --- compiler:3.16.0:testCompile (default-testCompile) @ testgen ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 10 source files with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ testgen ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running dates.DatesMetamorphicTest
[INFO] Tests run: 856, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.675 s -- in dates.DatesMetamorphicTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 856, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jacoco:0.8.12:report (report) @ testgen ---
[INFO] Loading execution data file D:\sysc5105-testgen\target\jacoco.exec
[INFO] Analyzed bundle 'testgen' with 4 classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.759 s
[INFO] Finished at: 2026-10-16T21:31:42-04:00
[INFO] ------------------------------------------------------------------------
PS D:\sysc5105-testgen>
```

Figure 69. *DatesMetamorphicTest, normal mode: 856 tests, 0 failures.*

```
PS D:\sysc5105-testgen> mvn clean test "-Dtest=dates.DatesMetamorphicTest" "-Dstrict=true"
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------------< sysc5105:testgen >----------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24119 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ testgen ---
[INFO] Deleting D:\sysc5105-testgen\target
[INFO]
[INFO] --- jacoco:0.8.12:prepare-agent (default) @ testgen ---
[INFO] argLine set to -javaagent:C:\\Users\\anuja\\.m2\\repository\\org\\jacoco\\org.jacoco.agent\\0.8.12\\org.jacoco.agent-0.8.12-runtime.jar=destfile=D:\\sysc5105-testgen\\target\\jacoco.exec
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\main\resources
[INFO]
[INFO] --- compiler:3.16.0:compile (default-compile) @ testgen ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\test\resources
[INFO]
[INFO] --- compiler:3.16.0:testCompile (default-testCompile) @ testgen ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 10 source files with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ testgen ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running dates.DatesMetamorphicTest
[ERROR] Tests run: 856, Failures: 2, Errors: 0, Skipped: 0, Time elapsed: 0.584 s <<< FAILURE! -- in dates.DatesMetamorphicTest
[ERROR] dates.DatesMetamorphicTest.mr4TimeZoneInvalidInput -- Time elapsed: 0.029 s <<< FAILURE!
org.opentest4j.AssertionFailedError: DT5: unparsable input 'abc' gives 31/12/1969 in America/Toronto but 01/01/1970 in UTC
        at org.junit.jupiter.api.AssertionUtils.fail(AssertionUtils.java:38)
        at org.junit.jupiter.api.Assertions.fail(Assertions.java:138)
        at dates.DatesMetamorphicTest.known(DatesMetamorphicTest.java:66)
        at dates.DatesMetamorphicTest.mr4TimeZoneInvalidInput(DatesMetamorphicTest.java:158)
        at java.base/java.lang.reflect.Method.invoke(Method.java:569)
        at java.base/java.util.ArrayList.forEach(ArrayList.java:1511)
        at java.base/java.util.ArrayList.forEach(ArrayList.java:1511)

[ERROR] dates.DatesMetamorphicTest.mr4LocaleValidDates -- Time elapsed: 0.081 s <<< FAILURE!
org.opentest4j.AssertionFailedError: DT7: default locale ar-EG changes the output for valid date 1900-01-01: ??/??/????
        at org.junit.jupiter.api.AssertionUtils.fail(AssertionUtils.java:38)
        at org.junit.jupiter.api.Assertions.fail(Assertions.java:138)
        at dates.DatesMetamorphicTest.known(DatesMetamorphicTest.java:66)
        at dates.DatesMetamorphicTest.mr4LocaleValidDates(DatesMetamorphicTest.java:170)
        at java.base/java.lang.reflect.Method.invoke(Method.java:569)
        at java.base/java.util.ArrayList.forEach(ArrayList.java:1511)
        at java.base/java.util.ArrayList.forEach(ArrayList.java:1511)

[INFO]
[INFO] Results:
[INFO]
[ERROR] Failures:
[ERROR]   DatesMetamorphicTest.mr4LocaleValidDates:170->known:66 DT7: default locale ar-EG changes the output for valid date 1900-01-01: ??/??/????
```

Figure 70. *Strict mode: 2 of the 856 tests fail (known defects DT5, time zone, and DT7, locale).*

```
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\test\resources
[INFO]
[INFO] --- compiler:3.16.0:testCompile (default-testCompile) @ testgen ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 10 source files with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ testgen ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running dates.DatesMetamorphicTest
[ERROR] Tests run: 856, Failures: 2, Errors: 0, Skipped: 0, Time elapsed: 0.584 s <<< FAILURE! -- in dates.DatesMetamorphicTest
[ERROR] dates.DatesMetamorphicTest.mr4TimeZoneInvalidInput -- Time elapsed: 0.029 s <<< FAILURE!
org.opentest4j.AssertionFailedError: DT5: unparsable input 'abc' gives 31/12/1969 in America/Toronto but 01/01/1970 in UTC
        at org.junit.jupiter.api.AssertionUtils.fail(AssertionUtils.java:38)
        at org.junit.jupiter.api.Assertions.fail(Assertions.java:138)
        at dates.DatesMetamorphicTest.known(DatesMetamorphicTest.java:66)
        at dates.DatesMetamorphicTest.mr4TimeZoneInvalidInput(DatesMetamorphicTest.java:158)
        at java.base/java.lang.reflect.Method.invoke(Method.java:569)
        at java.base/java.util.ArrayList.forEach(ArrayList.java:1511)
        at java.base/java.util.ArrayList.forEach(ArrayList.java:1511)

[ERROR] dates.DatesMetamorphicTest.mr4LocaleValidDates -- Time elapsed: 0.081 s <<< FAILURE!
org.opentest4j.AssertionFailedError: DT7: default locale ar-EG changes the output for valid date 1900-01-01: ??/??/????
        at org.junit.jupiter.api.AssertionUtils.fail(AssertionUtils.java:38)
        at org.junit.jupiter.api.Assertions.fail(Assertions.java:138)
        at dates.DatesMetamorphicTest.known(DatesMetamorphicTest.java:66)
        at dates.DatesMetamorphicTest.mr4LocaleValidDates(DatesMetamorphicTest.java:170)
        at java.base/java.lang.reflect.Method.invoke(Method.java:569)
        at java.base/java.util.ArrayList.forEach(ArrayList.java:1511)
        at java.base/java.util.ArrayList.forEach(ArrayList.java:1511)

[INFO]
[INFO] Results:
[INFO]
[ERROR] Failures:
[ERROR]   DatesMetamorphicTest.mr4LocaleValidDates:170->known:66 DT7: default locale ar-EG changes the output for valid date 1900-01-01: ??/??/????
[ERROR]   DatesMetamorphicTest.mr4TimeZoneInvalidInput:158->known:66 DT5: unparsable input 'abc' gives 31/12/1969 in America/Toronto but 01/01/1970
[INFO]
[ERROR] Tests run: 856, Failures: 2, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[ERROR] BUILD FAILURE
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.614 s
[INFO] Finished at: 2026-10-16T21:32:34-04:00
[INFO] ------------------------------------------------------------------------
[ERROR] Failed to execute goal org.apache.maven.plugins:maven-surefire-plugin:3.2.5:test (default-test) on project testgen: There are test failures.
[ERROR]
[ERROR] Please refer to D:\sysc5105-testgen\target\surefire-reports for the individual test results.
[ERROR] Please refer to dump files (if any exist) [date].dump, [date]-jvmRun[N].dump and [date].dumpstream.
[ERROR] -> [Help 1]
[ERROR]
[ERROR] To see the full stack trace of the errors, re-run Maven with the -e switch.
[ERROR] Re-run Maven using the -X switch to enable full debug logging.
[ERROR]
[ERROR] For more information about the errors and possible solutions, please read the following articles:
[ERROR] [Help 1] http://cwiki.apache.org/confluence/display/MAVEN/MojoFailureException
PS D:\sysc5105-testgen>
```

Figure 71. *Strict mode (continued): failure details and the Maven summary.*

```
PS D:\sysc5105-testgen> Copy-Item results\findings_dates_meta.txt results\findings_dates_meta_run10.txt -Force
PS D:\sysc5105-testgen> Get-Content results\findings_dates_meta_run10.txt -Encoding UTF8 | Select-String '^DT\d' | ForEach-Object { $_.Line.Substring(0,3) } | Group-Object | Sele
ct-Object Name, Count

Name Count
---- -----
DT5      3
DT7      4


PS D:\sysc5105-testgen> Get-Content results\findings_dates_meta_run10.txt -Encoding UTF8
DT5: unparsable input 'abc' gives 31/12/1969 in America/Toronto but 01/01/1970 in UTC
DT5: unparsable input 'abc' gives 31/12/1969 in Pacific/Kiritimati but 01/01/1970 in UTC
DT5: unparsable input 'abc' gives 31/12/1969 in Pacific/Pago_Pago but 01/01/1970 in UTC
DT7: default locale ar-EG changes the output for valid date 1900-01-01: ٠١/٠١/١٩٠٠
DT7: default locale ja-JP-u-ca-japanese changes the output for valid date 2000-02-29: 01/03/2000
DT7: default locale th-TH changes the output for valid date 2000-02-29: 01/03/2000
DT7: default locale th-TH-u-nu-thai changes the output for valid date 1900-01-01: ๐๑/๐๑/๑๙๐๐
PS D:\sysc5105-testgen> Get-DatesJacoco
LINE covered 13 missed 13
BRANCH covered 0 missed 2
INSTRUCTION covered 60 missed 54
```

Figure 72. *Findings saved and counted: DT5 = 3, DT7 = 4, with the list and the scoped JaCoCo numbers (13 of 26 lines).*

```
PS D:\sysc5105-testgen> mvn org.pitest:pitest-maven:mutationCoverage "-DtargetTests=dates.DatesMetamorphicTest" "-DexcludedMethods=$ex" "-DoutputFormats=CSV,HTML"
[INFO] Scanning for projects...
[INFO] Loaded 24119 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO]
[INFO] ----------------------------< sysc5105:testgen >----------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- pitest:1.16.1:mutationCoverage (default-cli) @ testgen ---
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO] Loaded 50 auto-discovered prefixes for remote repository ow2-snapshot (prefixes-ow2-snapshot.txt)
[INFO] Root dir is : D:\sysc5105-testgen
[INFO] Found plugin : Default csv report plugin
[INFO] Found plugin : Default xml report plugin
[INFO] Found plugin : Default html report plugin
[INFO] Found plugin : Static initializer code detector plugin
[INFO] Found plugin : Excluded annotations plugin
[INFO] Found plugin : Try with resources filter
[INFO] Found plugin : Inlined finally block filter plugin
[INFO] Found plugin : Implicit null check filter
[INFO] Found plugin : Method reference null check filter
[INFO] Found plugin : For each loop filter
[INFO] Found plugin : Enum junk filter
[INFO] Found plugin : Record junk mutation filter
[INFO] Found plugin : String switch filter
[INFO] Found plugin : Assertions filter
[INFO] Found plugin : Enum switch filter
[INFO] Found plugin : Logging calls filter
[INFO] Found plugin : Infinite for loop filter
[INFO] Found plugin : Long running iterator loop filter
[INFO] Found plugin : For loop counter filter
[INFO] Found plugin : Kotlin junk mutations filter
[INFO] Found plugin : Groovy junk mutations filter
[INFO] Found plugin : Max mutations per class limit
[INFO] Found plugin : Equals shortcut equivalent mutant filter
[INFO] Found plugin : Trivial return vals equivalence filter
[INFO] Found plugin : Filters mutants with line number <= 1
[INFO] Found plugin : Division by one equivalent mutant filter
[INFO] Found plugin : Lombok junk mutations filter
[INFO] Found plugin : Filter mutations to defensive wrappers such as unmodifiableCollection on return or field write
[INFO] Found plugin : Mutant export plugin
[INFO] Found plugin : Auto add java.awt.headless=true to keep keyboard focus on Mac OS
[INFO] Found plugin : Auto set number of threads based on machine
[INFO] Found plugin : Automatically add -ea to launch args to enable assertions
[INFO] Found plugin : Default build verifier
[INFO] Found plugin : Detect missing JUnit5 plugin
[INFO] Found plugin : Detect missing TestNG plugin
[INFO] Found plugin : Detect missing kotlin plugin
[INFO] Found plugin : Detect missing spring plugin
[INFO] Found plugin : Default coverage exporter
[INFO] Found shared classpath plugin : Default mutation engine
[INFO] Found shared classpath plugin : JUnit 5 test framework support
[INFO] Found shared classpath plugin : JUnit plugin
[INFO] Available mutators : EXPERIMENTAL_ARGUMENT_PROPAGATION,FALSE_RETURNS,TRUE_RETURNS,CONDITIONALS_BOUNDARY,CONSTRUCTOR_CALLS,EMPTY_RETURNS,INCREMENTS,INLINE_CONSTS,INVERT_NEGS,MATH,NEGATE_CONDITIONALS,NON_
VOID_METHOD_CALLS,NULL_RETURNS,PRIMITIVE_RETURNS,REMOVE_CONDITIONALS_EQUAL_IF,REMOVE_CONDITIONALS_EQUAL_ELSE,REMOVE_CONDITIONALS_ORDER_IF,REMOVE_CONDITIONALS_ORDER_ELSE,VOID_METHOD_CALLS,EXPERIMENTAL_BIG_DECIM
AL,EXPERIMENTAL_BIG_INTEGER,EXPERIMENTAL_MEMBER_VARIABLE,EXPERIMENTAL_NAKED_RECEIVER,REMOVE_INCREMENTS,EXPERIMENTAL_SWITCH
[INFO] Adding org.pitest:pitest-junit5-plugin to SUT classpath
[INFO] Adding org.pitest:pitest to SUT classpath
[INFO] Auto adding org.junit.platform:junit-platform-launcher:jar:1.10.2 < central (https://repo.maven.apache.org/maven2, default, releases) to classpath.
```

Figure 73. *PIT command and start-up output.*

```
>> Generated 3 Killed 0 (0%)
> KILLED 0 SURVIVED 0 TIMED_OUT 0 NON_VIABLE 0
> MEMORY_ERROR 0 NOT_STARTED 0 STARTED 0 RUN_ERROR 0
> NO_COVERAGE 3
------------------------------------------------------------------------------
> org.pitest.mutationtest.engine.gregor.mutators.returns.PrimitiveReturnsMutator
>> Generated 6 Killed 2 (33%)
> KILLED 2 SURVIVED 0 TIMED_OUT 0 NON_VIABLE 0
> MEMORY_ERROR 0 NOT_STARTED 0 STARTED 0 RUN_ERROR 0
> NO_COVERAGE 4
------------------------------------------------------------------------------
> org.pitest.mutationtest.engine.gregor.mutators.VoidMethodCallMutator
>> Generated 16 Killed 0 (0%)
> KILLED 0 SURVIVED 1 TIMED_OUT 0 NON_VIABLE 0
> MEMORY_ERROR 0 NOT_STARTED 0 STARTED 0 RUN_ERROR 0
> NO_COVERAGE 15
------------------------------------------------------------------------------
> org.pitest.mutationtest.engine.gregor.mutators.MathMutator
>> Generated 22 Killed 0 (0%)
> KILLED 0 SURVIVED 0 TIMED_OUT 0 NON_VIABLE 0
> MEMORY_ERROR 0 NOT_STARTED 0 STARTED 0 RUN_ERROR 0
> NO_COVERAGE 22
------------------------------------------------------------------------------
> org.pitest.mutationtest.engine.gregor.mutators.returns.EmptyObjectReturnValsMuta
>> Generated 7 Killed 4 (57%)
> KILLED 4 SURVIVED 0 TIMED_OUT 0 NON_VIABLE 0
> MEMORY_ERROR 0 NOT_STARTED 0 STARTED 0 RUN_ERROR 0
> NO_COVERAGE 3
------------------------------------------------------------------------------
> org.pitest.mutationtest.engine.gregor.mutators.NegateConditionalsMutator
>> Generated 17 Killed 0 (0%)
> KILLED 0 SURVIVED 0 TIMED_OUT 0 NON_VIABLE 0
> MEMORY_ERROR 0 NOT_STARTED 0 STARTED 0 RUN_ERROR 0
> NO_COVERAGE 17
------------------------------------------------------------------------------
==============================================================================
- Timings
==============================================================================
> pre-scan for mutations : < 1 second
> scan classpath : < 1 second
> coverage and dependency analysis : < 1 second
> build mutation tests : < 1 second
> run mutation analysis : 2 seconds
------------------------------------------------------------------------------
> Total  : 3 seconds
------------------------------------------------------------------------------
==============================================================================
- Statistics
==============================================================================
>> Line Coverage (for mutated classes only): 36/168 (21%)
>> Generated 71 mutations Killed 6 (8%)
>> Mutations with no coverage 64. Test strength 86%
>> Ran 7 tests (0.1 tests per mutation)
Enhanced functionality available at https://www.arcmutate.com/
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.029 s
[INFO] Finished at: 2026-10-16T21:34:09-04:00
[INFO] ------------------------------------------------------------------------
```

Figure 74. *PIT mutator summary, timings and statistics.*

```
PS D:\sysc5105-testgen> Get-DatesPit
dates mutants: 11  killed: 6
survived line 202 parseDate VoidMetho
nocov line 60 getDateOnly
nocov line 210 getDesiredFormat
nocov line 191 parseAnyDate
nocov line 194 parseAnyDate
PS D:\sysc5105-testgen>
```

Figure 75. *Scoped result: 6 of 11 mutants killed.*

B.5 All three techniques together

Method. The three date test classes run in one Maven invocation.

```
mvn clean test "-
Dtest=dates.DatesCombinatorialTest,dates.DatesCategoryPartitionTest,dates.DatesMetamorphicTest"
mvn org.pitest:pitest-maven:mutationCoverage "-
DtargetTests=dates.DatesCombinatorialTest,dates.DatesCategoryPartitionTest,dates.DatesMetamorphicTest"
"-DexcludedMethods=$ex" "-DoutputFormats=CSV,HTML"
```

```
PS D:\sysc5105-testgen> mvn clean test "-Dtest=dates.DatesCombinatorialTest,dates.DatesCategoryPartitionTest,dates.DatesMetamorphicTest"
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< sysc5105:testgen >------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24165 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ testgen ---
[INFO] Deleting D:\sysc5105-testgen\target
[INFO]
[INFO] --- jacoco:0.8.12:prepare-agent (default) @ testgen ---
[INFO] argLine set to -javaagent:C:\\Users\\anuja\\.m2\\repository\\org\\jacoco\\org.jacoco.agent\\0.8.12\\org.jacoco.agent-0.8.12-runtime.jar=destfile=D:\\sysc5105-testgen\\target\\jacoco.exec
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\main\resources
[INFO]
[INFO] --- compiler:3.16.0:compile (default-compile) @ testgen ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\test\resources
[INFO]
[INFO] --- compiler:3.16.0:testCompile (default-testCompile) @ testgen ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 11 source files with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ testgen ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running dates.DatesCategoryPartitionTest
[INFO] Tests run: 29, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.504 s -- in dates.DatesCategoryPartitionTest
[INFO] Running dates.DatesCombinatorialTest
[INFO] Tests run: 344, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.204 s -- in dates.DatesCombinatorialTest
[INFO] Running dates.DatesMetamorphicTest
[INFO] Tests run: 856, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.176 s -- in dates.DatesMetamorphicTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1229, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jacoco:0.8.12:report (report) @ testgen ---
[INFO] Loading execution data file D:\sysc5105-testgen\target\jacoco.exec
[INFO] Analyzed bundle 'testgen' with 4 classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  11.685 s
[INFO] Finished at: 2026-10-20T19:13:54-04:00
```

Figure 76. *Three suites together: 1229 tests (344 + 29 + 856), 0 failures.*

```
PS D:\sysc5105-testgen> Get-DatesJacoco
LINE covered 13 missed 13
BRANCH covered 0 missed 2
INSTRUCTION covered 60 missed 54
PS D:\sysc5105-testgen>
```

Figure 77. *Scoped JaCoCo: 13 of 26 lines covered, 0 of 2 branches.*

```
PS D:\sysc5105-testgen> mvn org.pitest:pitest-maven:mutationCoverage "-DtargetTests=dates.DatesCombinatorialTest,dates.DatesCategoryPartitionTest,dates.DatesMetamorphicTest" "-DexcludedMethods=
$ex" "-DoutputFormats=CSV,HTML"
[INFO] Scanning for projects...
[INFO] Loaded 24165 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO]
[INFO] ---------------------------< sysc5105:testgen >---------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- pitest:1.16.1:mutationCoverage (default-cli) @ testgen ---
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[WARNING] Could not transfer metadata /.meta/prefixes.txt from/to ow2-snapshot (https://repository.ow2.org/nexus/content/repositories/snapshots): HTTP Status: 503
[INFO] Loaded 50 auto-discovered prefixes for remote repository ow2-snapshot (prefixes-ow2-snapshot.txt)
[INFO] Root dir is : D:\sysc5105-testgen
[INFO] Found plugin : Default csv report plugin
[INFO] Found plugin : Default xml report plugin
[INFO] Found plugin : Default html report plugin
[INFO] Found plugin : Static initializer code detector plugin
[INFO] Found plugin : Excluded annotations plugin
[INFO] Found plugin : Try with resources filter
[INFO] Found plugin : Inlined finally block filter plugin
[INFO] Found plugin : Implicit null check filter
[INFO] Found plugin : Method reference null check filter
[INFO] Found plugin : For each loop filter
[INFO] Found plugin : Enum junk filter
[INFO] Found plugin : Record junk mutation filter
[INFO] Found plugin : String switch filter
[INFO] Found plugin : Assertions filter
[INFO] Found plugin : Enum switch filter
[INFO] Found plugin : Logging calls filter
[INFO] Found plugin : Infinite for loop filter
[INFO] Found plugin : Long running iterator loop filter
[INFO] Found plugin : For loop counter filter
[INFO] Found plugin : Kotlin junk mutations filter
[INFO] Found plugin : Groovy junk mutations filter
[INFO] Found plugin : Max mutations per class limit
[INFO] Found plugin : Equals shortcut equivalent mutant filter
[INFO] Found plugin : Trivial return vals equivalence filter
[INFO] Found plugin : Filters mutants with line number <= 1
[INFO] Found plugin : Division by one equivalent mutant filter
[INFO] Found plugin : Lombok junk mutations filter
[INFO] Found plugin : Filter mutations to defensive wrappers such as unmodifiableCollection on return or field write
[INFO] Found plugin : Mutant export plugin
[INFO] Found plugin : Auto add java.awt.headless=true to keep keyboard focus on Mac OS
[INFO] Found plugin : Auto set number of threads based on machine
[INFO] Found plugin : Automatically add -ea to launch args to enable assertions
[INFO] Found plugin : Default build verifier
[INFO] Found plugin : Detect missing JUnit5 plugin
[INFO] Found plugin : Detect missing TestNG plugin
[INFO] Found plugin : Detect missing kotlin plugin
[INFO] Found plugin : Detect missing spring plugin
[INFO] Found plugin : Default coverage exporter
```

Figure 78. *PIT command and start-up output.*

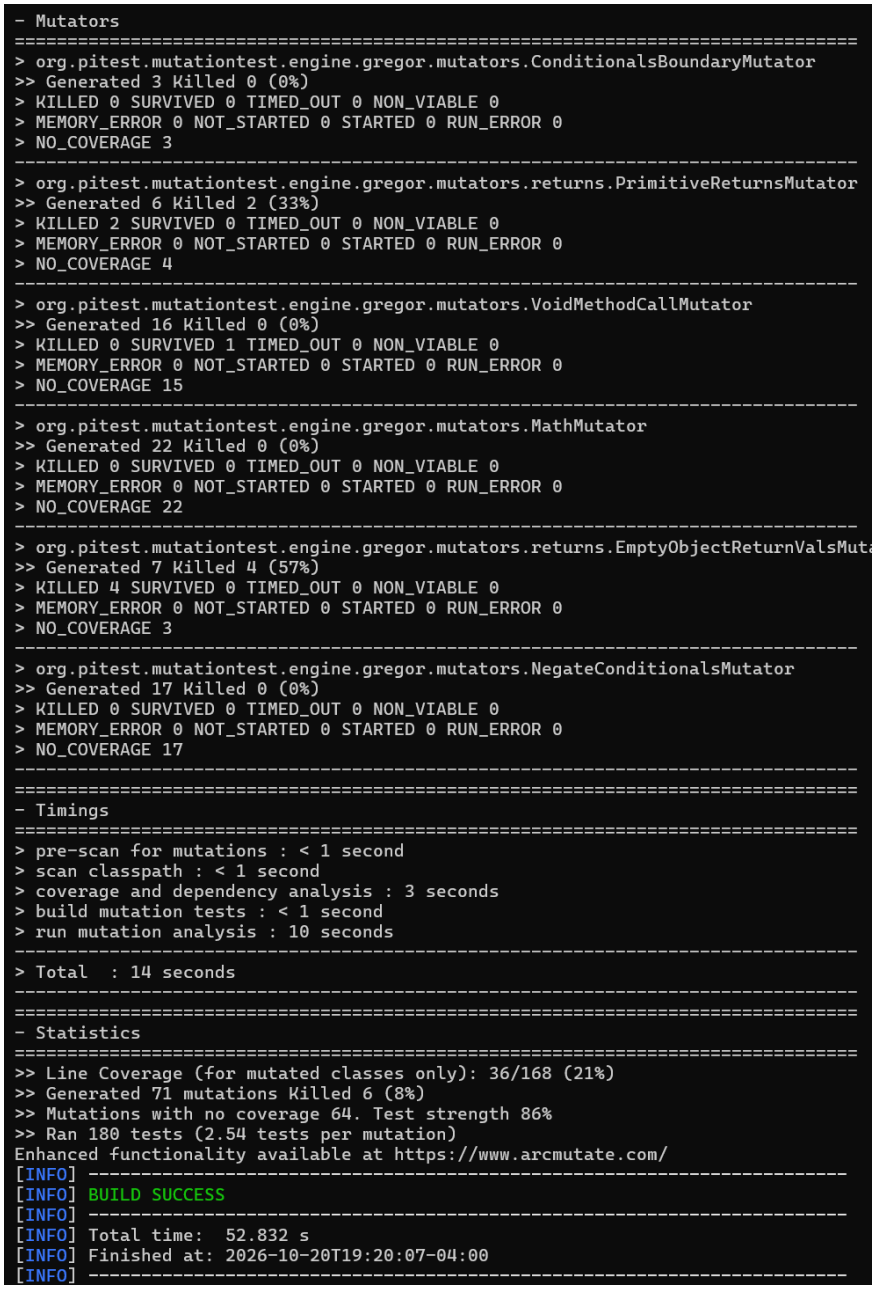


Figure 79. *PIT mutator summary, timings and statistics.*

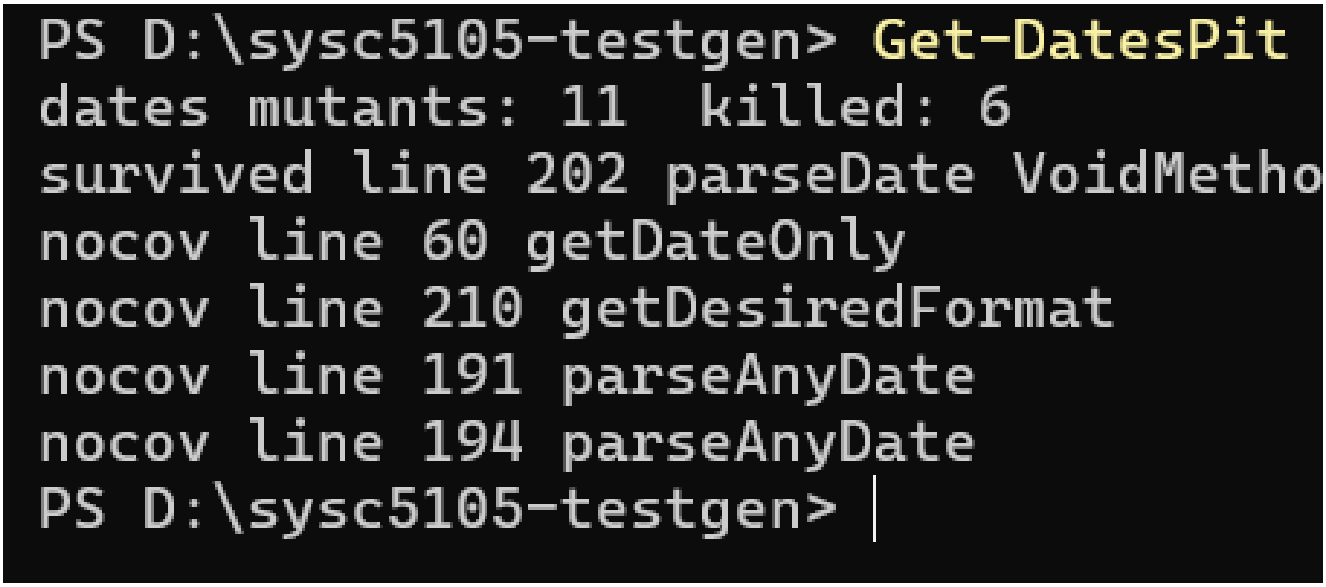


Figure 80. *Scoped result: 6 of 11 mutants killed; survivor on line 202.*

B.6 Improvement round

Method. DatesImprovementTest (7 tests) calls getDateOnly, parseDate, parseAnyDate and getDesiredFormat directly, which convert() never reaches, and captures System.err to observe the printed stack traces. The final run uses all four date test classes.

```
mvn clean test "-Dtest=dates.DatesImprovementTest"
mvn clean test "-Dtest=dates.DatesImprovementTest" "-Dstrict=true"
Copy-Item results\findings_dates_improvement.txt results\findings_dates_improvement_run11.txt -Force
mvn clean test "-
Dtest=dates.DatesCombinatorialTest,dates.DatesCategoryPartitionTest,dates.DatesMetamorphicTest,dates.Da
tesImprovementTest"
mvn org.pitest:pitest-maven:mutationCoverage "-
DtargetTests=dates.DatesCombinatorialTest,dates.DatesCategoryPartitionTest,dates.DatesMetamorphicTest,d
ates.DatesImprovementTest" "-DexcludedMethods=$ex" "-DoutputFormats=CSV,HTML"
```

```
PS D:\sysc5105-testgen> mvn clean test "-Dtest=dates.DatesImprovementTest"
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< sysc5105:testgen >------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24165 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ testgen ---
[INFO] Deleting D:\sysc5105-testgen\target
[INFO]
[INFO] --- jacoco:0.8.12:prepare-agent (default) @ testgen ---
[INFO] argLine set to -javaagent:C:\\Users\\anuja\\.m2\\repository\\org\\jacoco\\org.jacoco.agent\\0.8.12\\org.jacoco.agent-0.8.12-runtime.jar=destfile=D:\\sysc5105-testgen\\target\\jacoco.exec
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\main\resources
[INFO]
[INFO] --- compiler:3.16.0:compile (default-compile) @ testgen ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\test\resources
[INFO]
[INFO] --- compiler:3.16.0:testCompile (default-testCompile) @ testgen ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 11 source files with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ testgen ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running dates.DatesImprovementTest
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.465 s -- in dates.DatesImprovementTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jacoco:0.8.12:report (report) @ testgen ---
[INFO] Loading execution data file D:\sysc5105-testgen\target\jacoco.exec
[INFO] Analyzed bundle 'testgen' with 4 classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  12.086 s
[INFO] Finished at: 2026-10-20T19:03:34-04:00
[INFO] ------------------------------------------------------------------------
PS D:\sysc5105-testgen>
```

Figure 81. *DatesImprovementTest, normal mode: 7 tests, 0 failures.*

```
PS D:\sysc5105-testgen> mvn clean test "-Dtest=dates.DatesImprovementTest" "-Dstrict=true"
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< sysc5105:testgen >------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24165 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ testgen ---
[INFO] Deleting D:\sysc5105-testgen\target
[INFO]
[INFO] --- jacoco:0.8.12:prepare-agent (default) @ testgen ---
[INFO] argLine set to -javaagent:C:\\Users\\anuja\\.m2\\repository\\org\\jacoco\\org.jacoco.agent\\0.8.12\\org.jacoco.agent-0.8.12-runtime.jar=destfile=D:\\sysc5105-testgen\\target\\jacoco.exec
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\main\resources
[INFO]
[INFO] --- compiler:3.16.0:compile (default-compile) @ testgen ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\test\resources
[INFO]
[INFO] --- compiler:3.16.0:testCompile (default-testCompile) @ testgen ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 11 source files with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ testgen ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running dates.DatesImprovementTest
[ERROR] Tests run: 7, Failures: 2, Errors: 0, Skipped: 0, Time elapsed: 0.403 s <<< FAILURE! -- in dates.DatesImprovementTest
[ERROR] dates.DatesImprovementTest.parseAnyDateIsoString -- Time elapsed: 0.068 s <<< FAILURE!
org.opentest4j.AssertionFailedError: DT8: parseAnyDate('2023-12-25') returns the date 0031-06-15 (the last matching format wins)
        at org.junit.jupiter.api.AssertionUtils.fail(AssertionUtils.java:38)
        at org.junit.jupiter.api.Assertions.fail(Assertions.java:138)
        at dates.DatesImprovementTest.known(DatesImprovementTest.java:49)
        at dates.DatesImprovementTest.checkParseAny(DatesImprovementTest.java:119)
        at dates.DatesImprovementTest.parseAnyDateIsoString(DatesImprovementTest.java:128)
        at java.base/java.lang.reflect.Method.invoke(Method.java:569)
        at java.base/java.util.ArrayList.forEach(ArrayList.java:1511)
        at java.base/java.util.ArrayList.forEach(ArrayList.java:1511)

[ERROR] dates.DatesImprovementTest.parseAnyDateSlashString -- Time elapsed: 0.058 s <<< FAILURE!
org.opentest4j.AssertionFailedError: DT9: parseAnyDate('25/12/2023') prints 40 stack traces although it succeeds
        at org.junit.jupiter.api.AssertionUtils.fail(AssertionUtils.java:38)
        at org.junit.jupiter.api.Assertions.fail(Assertions.java:138)
        at dates.DatesImprovementTest.known(DatesImprovementTest.java:49)
        at dates.DatesImprovementTest.checkParseAny(DatesImprovementTest.java:122)
        at dates.DatesImprovementTest.parseAnyDateSlashString(DatesImprovementTest.java:133)
```

Figure 82. *Strict mode: 2 of the 7 tests fail (parseAnyDate defects), part 1.*

```
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running dates.DatesImprovementTest
[ERROR] Tests run: 7, Failures: 2, Errors: 0, Skipped: 0, Time elapsed: 0.403 s <<< FAILURE! -- in dates.DatesImprovementTest
[ERROR] dates.DatesImprovementTest.parseAnyDateIsoString -- Time elapsed: 0.068 s <<< FAILURE!
org.opentest4j.AssertionFailedError: DT8: parseAnyDate('2023-12-25') returns the date 0031-06-15 (the last matching format wins)
        at org.junit.jupiter.api.AssertionUtils.fail(AssertionUtils.java:38)
        at org.junit.jupiter.api.Assertions.fail(Assertions.java:138)
        at dates.DatesImprovementTest.known(DatesImprovementTest.java:49)
        at dates.DatesImprovementTest.checkParseAny(DatesImprovementTest.java:119)
        at dates.DatesImprovementTest.parseAnyDateIsoString(DatesImprovementTest.java:128)
        at java.base/java.lang.reflect.Method.invoke(Method.java:569)
        at java.base/java.util.ArrayList.forEach(ArrayList.java:1511)
        at java.base/java.util.ArrayList.forEach(ArrayList.java:1511)

[ERROR] dates.DatesImprovementTest.parseAnyDateSlashString -- Time elapsed: 0.058 s <<< FAILURE!
org.opentest4j.AssertionFailedError: DT9: parseAnyDate('25/12/2023') prints 40 stack traces although it succeeds
        at org.junit.jupiter.api.AssertionUtils.fail(AssertionUtils.java:38)
        at org.junit.jupiter.api.Assertions.fail(Assertions.java:138)
        at dates.DatesImprovementTest.known(DatesImprovementTest.java:49)
        at dates.DatesImprovementTest.checkParseAny(DatesImprovementTest.java:122)
        at dates.DatesImprovementTest.parseAnyDateSlashString(DatesImprovementTest.java:133)
        at java.base/java.lang.reflect.Method.invoke(Method.java:569)
        at java.base/java.util.ArrayList.forEach(ArrayList.java:1511)
        at java.base/java.util.ArrayList.forEach(ArrayList.java:1511)

[INFO]
[INFO] Results:
[INFO]
[ERROR] Failures:
[ERROR]   DatesImprovementTest.parseAnyDateIsoString:128->checkParseAny:119->known:49 DT8: parseAnyDate('2023-12-25') returns the date 0031-06-15 (the last matching format wins)
[ERROR]   DatesImprovementTest.parseAnyDateSlashString:133->checkParseAny:122->known:49 DT9: parseAnyDate('25/12/2023') prints 40 stack traces although it succeeds
[INFO]
[ERROR] Tests run: 7, Failures: 2, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[ERROR] BUILD FAILURE
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  8.890 s
[INFO] Finished at: 2026-10-20T19:04:51-04:00
[INFO] ------------------------------------------------------------------------
[ERROR] Failed to execute goal org.apache.maven.plugins:maven-surefire-plugin:3.2.5:test (default-test) on project testgen: There are test failures.
[ERROR]
[ERROR] Please refer to D:\sysc5105-testgen\target\surefire-reports for the individual test results.
[ERROR] Please refer to dump files (if any exist) [date].dump, [date]-jvmRun[N].dump and [date].dumpstream.
[ERROR] -> [Help 1]
[ERROR]
[ERROR] To see the full stack trace of the errors, re-run Maven with the -e switch.
[ERROR] Re-run Maven using the -X switch to enable full debug logging.
[ERROR]
[ERROR] For more information about the errors and possible solutions, please read the following articles:
[ERROR] [Help 1] http://cwiki.apache.org/confluence/display/MAVEN/MojoFailureException
PS D:\sysc5105-testgen>
```

Figure 83. *Strict mode, part 2: DT8 (parseAnyDate returns the date 0031-06-15 for 2023-12-25) and DT9 (40 stack traces per call).*

```
PS D:\sysc5105-testgen> Copy-Item results\findings_dates_improvement.txt results\findings_dates_impr
PS D:\sysc5105-testgen> Get-Content results\findings_dates_improvement_run11.txt -Encoding UTF8
DT8: parseAnyDate('2023-12-25') returns the date 0031-06-15 (the last matching format wins)
DT9: parseAnyDate('2023-12-25') prints 40 stack traces although it succeeds
DT9: parseAnyDate('25/12/2023') prints 40 stack traces although it succeeds
PS D:\sysc5105-testgen>
```

Figure 84. *Findings saved and listed: DT8 once and DT9 twice.*

```
PS D:\sysc5105-testgen> mvn clean test "-Dtest=dates.DatesCombinatorialTest,dates.DatesCategoryPartitionTest,dates.DatesMetamorphicTest,dates.DatesImprovementTest"
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< sysc5105:testgen >-------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24165 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ testgen ---
[INFO] Deleting D:\sysc5105-testgen\target
[INFO]
[INFO] --- jacoco:0.8.12:prepare-agent (default) @ testgen ---
[INFO] argLine set to -javaagent:C:\\Users\\anuja\\.m2\\repository\\org\\jacoco\\org.jacoco.agent\\0.8.12\\org.jacoco.agent-0.8.12-runtime.jar=destfile=D:\\sysc5105-testgen\\target\\ja
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\main\resources
[INFO]
[INFO] --- compiler:3.16.0:compile (default-compile) @ testgen ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\test\resources
[INFO]
[INFO] --- compiler:3.16.0:testCompile (default-testCompile) @ testgen ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 11 source files with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ testgen ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running dates.DatesCategoryPartitionTest
[INFO] Tests run: 29, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.528 s -- in dates.DatesCategoryPartitionTest
[INFO] Running dates.DatesCombinatorialTest
[INFO] Tests run: 344, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.084 s -- in dates.DatesCombinatorialTest
[INFO] Running dates.DatesImprovementTest
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.094 s -- in dates.DatesImprovementTest
[INFO] Running dates.DatesMetamorphicTest
[INFO] Tests run: 856, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.019 s -- in dates.DatesMetamorphicTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1236, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jacoco:0.8.12:report (report) @ testgen ---
[INFO] Loading execution data file D:\sysc5105-testgen\target\jacoco.exec
[INFO] Analyzed bundle 'testgen' with 4 classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  11.289 s
[INFO] Finished at: 2026-10-20T19:22:35-04:00
[INFO] ------------------------------------------------------------------------
PS D:\sysc5105-testgen>
```

Figure 85. *Final suite, four date classes together: 1236 tests, 0 failures.*

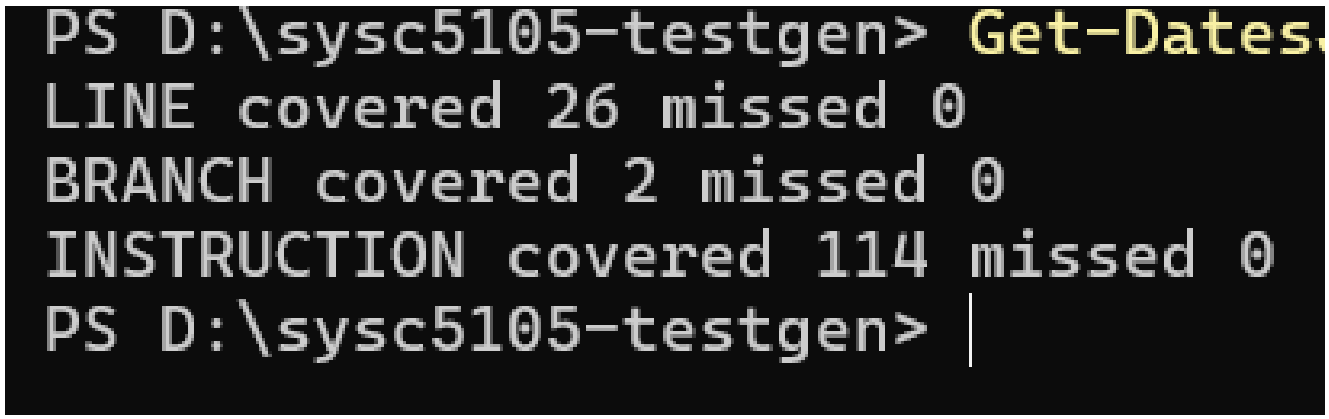


Figure 86. *Scoped JaCoCo: 26 of 26 lines and 2 of 2 branches covered.*

```
PS D:\sysc5105-testgen> mvn org.pitest:pitest-maven:mutationCoverage "-DtargetTests=dates.DatesCombinatorialTest,dates.DatesCategoryPartitionTest,dates.DatesMetamorphicTest,dates.DatesImprovementTest" "-Dexclu
dedMethods=$ex" "-DoutputFormats=CSV,HTML"
[INFO] Scanning for projects...
[INFO] Loaded 24165 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO]
[INFO] ------------------------< sysc5105:testgen >-------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- pitest:1.16.1:mutationCoverage (default-cli) @ testgen ---
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots.txt)
[INFO] Loaded 50 auto-discovered prefixes for remote repository ow2-snapshot (prefixes-ow2-snapshot.txt)
[INFO] Root dir is : D:\sysc5105-testgen
[INFO] Found plugin : Default csv report plugin
[INFO] Found plugin : Default xml report plugin
[INFO] Found plugin : Default html report plugin
[INFO] Found plugin : Static initializer code detector plugin
[INFO] Found plugin : Excluded annotations plugin
[INFO] Found plugin : Try with resources filter
[INFO] Found plugin : Inlined finally block filter plugin
[INFO] Found plugin : Implicit null check filter
[INFO] Found plugin : Method reference null check filter
[INFO] Found plugin : For each loop filter
[INFO] Found plugin : Enum junk filter
[INFO] Found plugin : Record junk mutation filter
[INFO] Found plugin : String switch filter
[INFO] Found plugin : Assertions filter
[INFO] Found plugin : Enum switch filter
[INFO] Found plugin : Logging calls filter
[INFO] Found plugin : Infinite for loop filter
[INFO] Found plugin : Long running iterator loop filter
[INFO] Found plugin : For loop counter filter
[INFO] Found plugin : Kotlin junk mutations filter
[INFO] Found plugin : Groovy junk mutations filter
[INFO] Found plugin : Max mutations per class limit
[INFO] Found plugin : Equals shortcut equivalent mutant filter
[INFO] Found plugin : Trivial return vals equivalence filter
[INFO] Found plugin : Filters mutants with line number <= 1
[INFO] Found plugin : Division by one equivalent mutant filter
[INFO] Found plugin : Lombok junk mutations filter
[INFO] Found plugin : Filter mutations to defensive wrappers such as unmodifiableCollection on return or field write
[INFO] Found plugin : Mutant export plugin
[INFO] Found plugin : Auto add java.awt.headless=true to keep keyboard focus on Mac OS
[INFO] Found plugin : Auto set number of threads based on machine
[INFO] Found plugin : Automatically add -ea to launch args to enable assertions
[INFO] Found plugin : Default build verifier
[INFO] Found plugin : Detect missing JUnit5 plugin
[INFO] Found plugin : Detect missing TestNG plugin
[INFO] Found plugin : Detect missing kotlin plugin
[INFO] Found plugin : Detect missing spring plugin
[INFO] Found plugin : Default coverage exporter
[INFO] Found shared classpath plugin : Default mutation engine
[INFO] Found shared classpath plugin : JUnit 5 test framework support
[INFO] Found shared classpath plugin : JUnit plugin
[INFO] Available mutators : EXPERIMENTAL_ARGUMENT_PROPAGATION,FALSE_RETURNS,TRUE_RETURNS,CONDITIONALS_BOUNDARY,CONSTRUCTOR_CALLS,EMPTY_RETURNS,INCREMENTS,INLINE_CONSTS,INVERT_NEGS,MATH,NEGATE_CONDITIONALS,NON_
VOID_METHOD_CALLS,NULL_RETURNS,PRIMITIVE_RETURNS,REMOVE_CONDITIONALS_EQUAL_IF,REMOVE_CONDITIONALS_EQUAL_ELSE,REMOVE_CONDITIONALS_ORDER_IF,REMOVE_CONDITIONALS_ORDER_ELSE,VOID_METHOD_CALLS,EXPERIMENTAL_BIG_DECIM
AL,EXPERIMENTAL_BIG_INTEGER,EXPERIMENTAL_MEMBER_VARIABLE,EXPERIMENTAL_NAKED_RECEIVER,REMOVE_INCREMENTS,EXPERIMENTAL_SWITCH
[INFO] Adding org.pitest:pitest-junit5-plugin to SUT classpath
[INFO] Adding org.pitest:pitest to SUT classpath
[INFO] Auto adding org.junit.platform:junit-platform-launcher:jar:1.10.2 < central (https://repo.maven.apache.org/maven2, default, releases) to classpath.
```

Figure 87. *PIT command and start-up output.*

```
================================================================================
- Mutators
================================================================================
> org.pitest.mutationtest.engine.gregor.mutators.ConditionalsBoundaryMutator
>> Generated 3 Killed 0 (0%)
> KILLED 0 SURVIVED 0 TIMED_OUT 0 NON_VIABLE 0
> MEMORY_ERROR 0 NOT_STARTED 0 STARTED 0 RUN_ERROR 0
> NO_COVERAGE 3
--------------------------------------------------------------------------------
> org.pitest.mutationtest.engine.gregor.mutators.returns.PrimitiveReturnsMutator
>> Generated 6 Killed 3 (50%)
> KILLED 3 SURVIVED 0 TIMED_OUT 0 NON_VIABLE 0
> MEMORY_ERROR 0 NOT_STARTED 0 STARTED 0 RUN_ERROR 0
> NO_COVERAGE 3
--------------------------------------------------------------------------------
> org.pitest.mutationtest.engine.gregor.mutators.VoidMethodCallMutator
>> Generated 16 Killed 3 (19%)
> KILLED 3 SURVIVED 0 TIMED_OUT 0 NON_VIABLE 0
> MEMORY_ERROR 0 NOT_STARTED 0 STARTED 0 RUN_ERROR 0
> NO_COVERAGE 13
--------------------------------------------------------------------------------
> org.pitest.mutationtest.engine.gregor.mutators.MathMutator
>> Generated 22 Killed 0 (0%)
> KILLED 0 SURVIVED 0 TIMED_OUT 0 NON_VIABLE 0
> MEMORY_ERROR 0 NOT_STARTED 0 STARTED 0 RUN_ERROR 0
> NO_COVERAGE 22
--------------------------------------------------------------------------------
> org.pitest.mutationtest.engine.gregor.mutators.returns.EmptyObjectReturnValsMutator
>> Generated 7 Killed 5 (71%)
> KILLED 5 SURVIVED 0 TIMED_OUT 0 NON_VIABLE 0
> MEMORY_ERROR 0 NOT_STARTED 0 STARTED 0 RUN_ERROR 0
> NO_COVERAGE 2
--------------------------------------------------------------------------------
> org.pitest.mutationtest.engine.gregor.mutators.NegateConditionalsMutator
>> Generated 17 Killed 0 (0%)
> KILLED 0 SURVIVED 0 TIMED_OUT 0 NON_VIABLE 0
> MEMORY_ERROR 0 NOT_STARTED 0 STARTED 0 RUN_ERROR 0
> NO_COVERAGE 17
--------------------------------------------------------------------------------
================================================================================
- Timings
================================================================================
> pre-scan for mutations : < 1 second
> scan classpath : < 1 second
> coverage and dependency analysis : 3 seconds
> build mutation tests : < 1 second
> run mutation analysis : 9 seconds
--------------------------------------------------------------------------------
> Total  : 13 seconds
--------------------------------------------------------------------------------
================================================================================
- Statistics
================================================================================
>> Line Coverage (for mutated classes only): 49/168 (29%)
>> Generated 71 mutations Killed 11 (15%)
>> Mutations with no coverage 60. Test strength 100%
>> Ran 180 tests (2.54 tests per mutation)
Enhanced functionality available at https://www.arcmutate.com/
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  17.986 s
[INFO] Finished at: 2026-10-20T19:23:41-04:00
```

Figure 88. *PIT mutator summary, timings and statistics.*

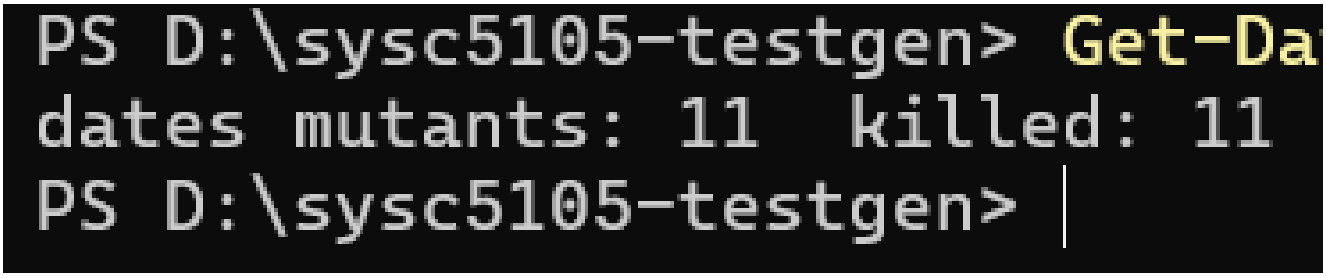


Figure 89. *Scoped result: all 11 of 11 mutants killed.*

Part C. Execution time

C.1 Timing of all eight suites

Method. The project is compiled once with JaCoCo skipped, so the coverage agent does not slow the tests. scripts\timing.ps1 then runs each suite five times and averages the class time from the surefire XML report.

```
mvn clean test-compile "-Djacoco.skip=true"
. .\scripts\timing.ps1
```

```
PS D:\sysc5105-testgen> mvn clean test-compile "-Djacoco.skip=true"
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------------< sysc5105:testgen >-------------------------
[INFO] Building testgen 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] Loaded 24165 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ testgen ---
[INFO] Deleting D:\sysc5105-testgen\target
[INFO]
[INFO] --- jacoco:0.8.12:prepare-agent (default) @ testgen ---
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.
[INFO] Skipping JaCoCo execution because property jacoco.skip is set.
[INFO] argLine set to empty
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\main\resources
[INFO]
[INFO] --- compiler:3.16.0:compile (default-compile) @ testgen ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ testgen ---
[INFO] skip non existing resourceDirectory D:\sysc5105-testgen\src\test\resources
[INFO]
[INFO] --- compiler:3.16.0:testCompile (default-testCompile) @ testgen ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 11 source files with javac [debug target 17] to target\test-classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.016 s
[INFO] Finished at: 2026-10-20T19:53:43-04:00
[INFO] ------------------------------------------------------------------------
PS D:\sysc5105-testgen>
```

Figure 90. *Compile step with JaCoCo skipped.*

```
PS D:\sysc5105-testgen> . .\scripts\timing.ps1
quadratic.CombinatorialTest: mean 0.695 s (runs: 0.713, 0.695, 0.659, 0.681, 0.727)
quadratic.CategoryPartitionTest: mean 0.241 s (runs: 0.288, 0.236, 0.238, 0.226, 0.219)
quadratic.MetamorphicTest: mean 1.391 s (runs: 1.369, 1.414, 1.381, 1.419, 1.372)
quadratic.ImprovementTest: mean 0.143 s (runs: 0.127, 0.127, 0.153, 0.159, 0.149)
dates.DatesCombinatorialTest: mean 1.182 s (runs: 1.205, 1.095, 1.234, 1.167, 1.208)
dates.DatesCategoryPartitionTest: mean 0.260 s (runs: 0.311, 0.281, 0.295, 0.301, 0.112)
dates.DatesMetamorphicTest: mean 1.694 s (runs: 0.873, 1.991, 1.961, 1.804, 1.839)
dates.DatesImprovementTest: mean 0.257 s (runs: 0.259, 0.261, 0.265, 0.268, 0.233)
PS D:\sysc5105-testgen>
```

Figure 91. *Timing results: mean class time over 5 runs for each suite (quadratic and dates).*